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Thesis for the Doctor of Philosophy

Brain to artificial agents: Insect brain-inspired algorithms for vision-based navigation

Angel Canelo

Graduate School of Hanyang University

August 2024

Thesis for the Doctor of Philosophy

Brain to artificial agents: Insect brain-inspired algorithms for vision-based navigation

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A Thesis submitted to the graduate school of Hanyang University in partial fulfillment of the requirements for the degree of Doctor of Philosophy

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Abstract

Brain to artificial agents:
Insect brain-inspired algorithms
for vision-based navigation

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The brain of the fruit fly *Drosophila melanogaster* has been extensively studied in recent years, for both its structure and function, at the level of single neurons. This, along with its sophisticated visual behaviors, makes this model animal an ideal platform for developing mathematical frameworks for artificial agents performing vision-based navigation. In this study, we have developed mathematical models based on the fly's visual processing, enabling artificial agents to detect complex visual features in their surroundings and navigate accordingly.

We have developed a neuromorphic convolutional neural network (CNN), FlyVisNet, that is based on the fly's visual system architecture. FlyVisNet implements neural pathways that allow pattern recognition, enabling the detection of objects in different shapes and looming objects. We found that FlyVisNet outperformed other architectures of the same size segment in the classification accuracy of objects of different complexities. With its small size, it allows to be deployed in the most memory and power demanding embedded systems such as nano drones. FlyVisNet allowed visually guided navigation and collision avoidance of virtual and real agents.

The fly encodes these visual inputs into visual reflexive behaviors. For that purpose, the different neural pathways need to be coordinated when they share a motor actuator such as the fly's wing muscles. In visual reflexive behaviors, it has been found that the fly uses an internal signal, the efference copy (EC), to suppress self-generated visual optic flow. This mechanism allows the fly to stabilize its flights and prioritize specific visual inputs. We provide an integrative model for reflexive visual behaviors in which an EC mechanism permits to coordinate fly's different pathways in complex visual environment with multiple visual patterns. We found that our virtual fly responded faster to the visual patterns when using EC. We also found behavioral evidence of EC when the flying fly navigates in cluttered visual environments.

In goal-driven behaviors, the fly can find specific places via visual cues. Through a preliminary study, we have built a model-based reinforcement learning (RL) framework in which an autoencoder (AE) maps the visual inputs into a fly's central brain-like latent representation. By using FlyVisNet as encoder, an agent could navigate to find a target in a virtual environment.

Overall, we have demonstrated that *Drosophila*'s brain can be used as a baseline to develop vision-based navigation algorithms. Our algorithms outperformed other architectures of similar size and allowed artificial agents to perform visually guided navigation in complex visual environments. Our results highlight the importance of brain-inspired computations for a deeper understanding of their biological counterparts, with potential applicability to other domains.

Chapter 1. Introduction

Organisms perceive the external world through a multitude of sensory mechanisms, each finely tuned to detect specific stimuli. These stimuli can range from light and sound to touch, taste, and smell. The process of perception involves the reception of these stimuli by sensory organs, their transmission to the brain, and the interpretation of these signals to form a coherent representation of the surrounding environment.

Once the sensory perception of the external world is integrated organisms generate behavior. For example, a plant might grow towards a source of light, a predator might stalk its prey using visual and olfactory cues, or a human might withdraw their hand from a hot surface upon sensing pain. These behaviors are crucial for survival and reproduction, as they enable organisms to respond appropriately to environmental stimuli, avoid danger, and seek out resources.

Likewise other reverse engineering examples in history such as the Soviet Tupolev Tu-4 based on the American B-29 bomber or the space rocket Explorer 1 based on the German V-2 rocket that provided such good results, mimicking these biological processes in artificial systems can lead to more adaptive and intelligent behavior for machines. By incorporating sensory perception-like mathematical models, artificial agents can interact more effectively with their environment and perform tasks with greater autonomy and versatility.

Sensory perception in artificial agents is possible through the use of sensors that detect various environmental stimuli. For example, cameras can capture visual information, microphones can record sound, and pressure sensors can detect touch. These sensors provide data similarly to biological counterparts and by analyzing them provide insights about the surroundings.

Once sensory information is obtained, artificial systems can employ brain-

inspired algorithms to process and interpret this data and produce intelligent decision-making. Such is the case of artificial neural networks (ANNs) that can be trained to recognize patterns in sensory input and make predictions or decisions based on this information. For instance, a self-driving car can use visual input from cameras to detect obstacles and make decisions about steering and braking.

Among all modalities of sensory, vision surpasses them in its ability to provide rich, multidimensional data about the environment, offering insights into spatial relationships, object recognition, and temporal dynamics. Implementing vision-based algorithms inspired by biological counterparts for artificial agents would allow processing information and produce versatile behavior for a wide range of tasks. This would improve agents' autonomy and adaptability in dynamic and complex environments.

1.1 Motivation and basic approaches

Reverse engineering the human brain is an endeavor that sits at the pinnacle of scientific ambition and curiosity. However, the brain's complexity is staggering, with its billions of neurons interconnected in intricate networks, orchestrating human thought, emotion, and behavior. Attempts to mimic the brain's neural processing led to the birth of artificial intelligence (AI). Despite significant advancements in neuroscience, understanding the brain in its entirety remains elusive.

In light of these challenges, scientists have turned to simpler organisms to unravel the mysteries of neural circuitry. One such organism is the fruit fly, *Drosophila melanogaster*, renowned for its relatively simple yet function-

ally sophisticated brain. While vastly different from the human brain in scale and complexity, the fruit fly brain has been studied extensively and dissected at a cell level with the recently released connectomes [1, 2]. This makes the fly a perfect baseline and opportunity to develop new AI. Specifically, its visual system of about 80,000 neurons allows it to navigate, either flying or walking, through complex environments with stunning precision. Their ability to process visual information in real-time, often with limited computational resources, is a prove of the efficiency of evolutionary design.

In the present thesis, the fly's visual system is used as a baseline to develop ANNs and flight control algorithms that are used as the "brain" of artificial flying agents, both virtual and real. We derive algorithms from quantitative experimental analysis as well as mathematical modeling that allows the agents to fly in cluttered environments performing visual reflexive behaviors and collision avoidance.

The *Drosophila* visual system involves compound eyes with numerous ommatidia, each housing photoreceptor neurons (PRs) [3]. Among its visual remarkable features, motion detection [4] is facilitated by specialized neurons such as the lobula plate tangential cells (LPTCs) and the horizontal system (HS) cells [5]. These neurons enable the fly to perceive moving objects, optic flow, etc. allowing a quick response to changes in its visual environment. Pattern recognition is another key function of its visual system. Through the integration of signals from multiple ommatidia and subsequent processing in the deeper layers such as the Lobula and optic glomeruli [6], *Drosophila* is capable of perceiving shapes and patterns in their surroundings [7, 8].

Particularly, Convolutional Neural Networks (CNNs), inspired by the structure of the visual cortex in animals, are exceptionally adept at learn-

ing hierarchical representations of images. This hierarchical approach enables them to capture intricate patterns and features at different levels of abstraction, from edges and textures to complex objects and scenes. CNNs have demonstrated unparalleled performance across various visual recognition tasks. Through layers of convolutional, pooling, and activation operations [9], CNNs can efficiently extract and process visual information, making them exceptionally powerful tools for computer vision tasks. CNNs can be trained by using supervised learning, which permits the network to learn from labeled examples, where each input image is associated with a corresponding label or class. During the training process, the network adjusts its parameters iteratively to minimize the distance between its predictions and the ground truth labels, typically measured using a loss function. This training is often accomplished using gradient-based techniques like backpropagation, where gradients of the loss function with respect to the network parameters are computed and used to update the weights. Thus, the intrinsic characteristics of CNNs make them a perfect approach to build a comprehensive model of the visual system of *Drosophila*. Previous studies have shown that CNNs can be applied for such purpose in collision detection tasks [10, 11], classification [12], and optic flow detection [13, 14], but they mainly implemented partial circuits or early to medium visual layers. In this thesis, we have built the FlyVisNet, a CNN that includes neurons from the retina to deep higher visual centers such as lobula and optic glomeruli, that extract high-level visual features. We show that the artificial neurons' responses are in line with the real ones and that embedded in artificial agents allows visually guided navigation.

Once the fly extracts and processes the visual features, it will generate visual reflexive behaviors. These behaviors represent the interplay between

sensory input and motor output and are adaptive strategies employed by flies to navigate their environment effectively. Among these behaviors are the optomotor response, bar fixation, and spot aversion. The optomotor response is a reflexive behavior observed in flies where they adjust their body orientation in response to visual motion. When exposed to a rotating visual stimulus, such as stripes or bars moving in a specific direction, flies will reflexively orient themselves to maintain their position relative to the moving pattern [15, 16, 17]. Bar fixation is another visual reflexive behavior observed in flies, where they fixate on and follow a narrow, elongated visual stimulus, such as a dark bar or stripe [18, 19]. This behavior is thought to aid in maintaining stability during flight and may also play a role in visual navigation. Spot aversion refers to the avoidance behavior exhibited by flies when they tend to move away from short object like spots, possibly as a mechanism to avoid potential threats or disturbances [20, 19].

Dynamical systems theory presents a mathematical approach for capturing the temporal evolution of these behaviors, considering how visual sensory stimuli and internal states dynamically interact to produce observable motor-related actions. Particularly, control theory techniques allow to implement feedback mechanisms and design control strategies to regulate the fly's behavior in response to environmental stimuli. Previous works have shown dynamical system models of the fly for visuomotor behaviors in which visual input generated gaze angle of the fly [17, 21]. A similar approach was used to model proprioceptive-based feedback control [22]. In these studies, only one single input was provided at a time. In this thesis, we provide an integrative dynamical system model of the visuomotor reflexive behaviors of the fly for complex visual environments where multiple visual patterns take place simultaneously.

We have implemented an efference copy (EC) mechanism to coordinate different visual pathways that can activate a common actuator, the wing muscles. EC is a copy of motor-related signals that suppress self-generated visual stimulus [23], allowing the fly to suppress undesired visual inputs [24, 25, 26].

Besides the visual reflexive behaviors, the processed visual features can also be encoded into internal representations for goal-driven behaviors such as spatial navigation. There exist neural pathways that connect the visual system with the central complex's ellipsoid body projection gall (EPG) neurons [27]. By using this pathway the fly can associate visual cues with reward in a navigation-based visual learning task [28].

Reinforcement learning (RL) offers a convenient approach to model such behaviors in the fly. The reward prediction error signal central to RL algorithms bears similarities to dopamine-mediated reinforcement signals observed in the brain of flies and other animals. Specifically, the central complex is involved in processing and integrating visual information to create a representation of the surrounding space [29, 30]. This representation forms the basis for the internal model used in model-based RL. Agents would use its internal model to plan sequences of actions that lead to desired outcomes [31]. Visual cues related to goal locations or relevant landmarks help specify these goals within the internal representation. The central complex, involved in encoding spatial maps and representations of the environment, contributes to the formation and maintenance of goal representations based on visual cues [32, 33]. In silico, this can be achieved by using autoencoders (AEs) that allow to map visual inputs to a lower dimensional latent space [34, 35, 36]. In this work, we provide preliminary model-based framework in which we use FlyVisNet as the encoder part of an AE to map a virtual environment to a central complex-like

internal representation. This allows a virtual agent, trained through RL, to navigate and find a target in the environment.

1.2 Organization of the dissertation

This thesis presents two main storylines and one preliminary study in which by maintaining a common baseline, the *Drosophila*'s visual system, we have adopted two different principles for modeling. Thus, mimicking fly's visual system architecture via CNN models with supervised learning and modeling visual reflexive behaviors via dynamical systems (see Figure 1.1).

Within this first chapter, we cover the background and literature review of the fields that are the foundations of the work presented in this thesis. In particular, we will give an overview of the following four subjects: 1) Brain-inspired AI 2) *Drosophila*'s vision and models 3) Vision-based artificial agents.

In chapter 2, we will build FlyVisNet, a CNN that follows the same architecture as the fly's visual system. The network will include neurons from Retina to deeper layers up to optic glomeruli. It will implement each neuron as a convolution layer whose kernel will be initialized with positive or negative values depending on whether the corresponding real neuron provides excitatory or inhibitory synapse respectively. The CNN will classify the images in three categories corresponding to loom, bar, and spot corresponding to the implemented feature-detecting neurons. It will also implement motion-detecting neurons. The network's inference will allow an agent to navigate in a virtual environment. It will also be embedded in a drone to allow it to navigate in a real arena.

In chapter 3, We will develop an integrative model of the fly visual reflexive

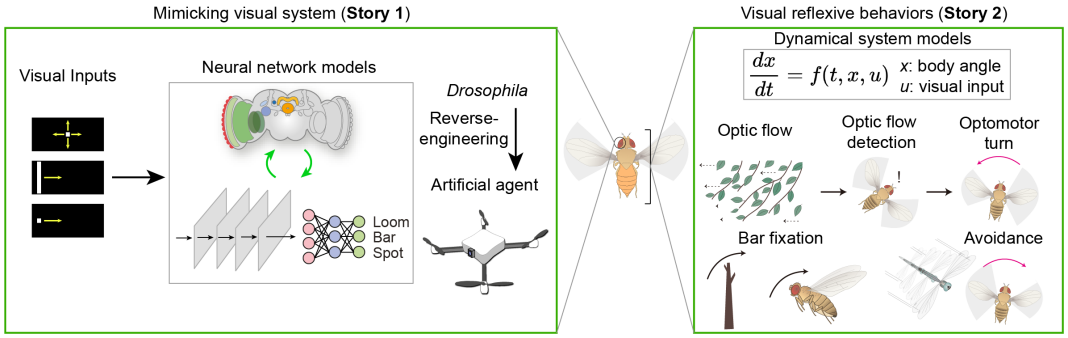


Figure 1.1: Graphical summary of the thesis. This work aims to reverse-engineering the *Drosophila*'s visual system and its application for the vision-guided navigation of artificial agents. In the first story, we mimic the architecture of the visual system via CNN models. In the second story, we build an integrative model for visual reflexive behaviors by using dynamical system models.

behaviors for complex visual environments in which an EC mechanism will be used to coordinate concurrent visual pathways that share a common actuator. Our model will simulate a virtual fly that can process multiple visual patterns and generate an appropriate action for steering control. We will present three different strategies of integration of visual inputs and will show behavioral evidence of EC through experimental analysis.

We will conclude this thesis in chapter 4 by discussing the significance of the main results and potential future directions. Among these, we will present a preliminary model-based RL framework where by using an AE we will map visual inputs from a virtual environment to lower dimensional internal representations. Via RL, a virtual agent will be trained to maximize reward by arriving at a target. Thus, the internal representation will be based on the encoded visual cues in *Drosophila*'s central complex and will allow the agent to navigate and find the target.

1.3 Background and literature review

1.3.1 Brain-inspired AI

With its 86 billion neurons interconnecting in a complex web of electrical and chemical signals, the human brain is responsible for our thoughts, emotions, and actions. This unparalleled capability has long fascinated scientists and philosophers alike. What makes the brain truly remarkable is its efficiency. Despite its vast complexity, the brain operates on about 20 watts of power, roughly equivalent to a dim light bulb. With this efficiency, the brain integrates sensory inputs, learns from experience, and solves complex problems in real-time.

AI began trying to unlock the secrets of the brain's intelligence and cognition. One of its early steps was when Norbert Wiener introduced the term "cybernetics" in the 1940s [37], which laid foundational concepts for understanding feedback systems and their application to intelligent machines. He explored how all intelligent behavior was the result of feedback mechanisms and how that could be applied to create self-regulating systems and machines that can learn from their environment.

Another important milestone was the introduction of neural networks, computational models designed to mimic the brain's neural architecture. The McCulloch-Pitts neural model, developed by Warren McCulloch and Walter Pitts in 1943, simplified the biological neuron (see Figure 1.2 A, B) into a mathematical abstraction, showcasing how neurons could be computationally modeled [38]. Their model was based on the idea of binary threshold logic, where neurons either "fire" (output a signal) or remain inactive based on the weighted sum of their inputs. These networks, composed of interconnected

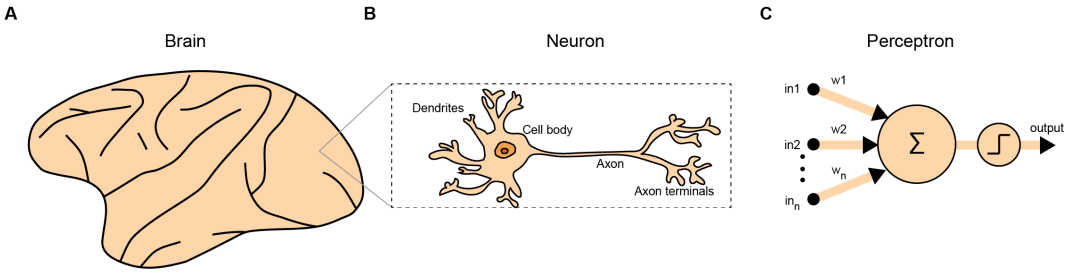


Figure 1.2: The simplest processing unit of the brain, the neuron, can be modeled as a perceptron. (A). A mammalian brain. (B). A neuron. (C). The perceptron.

nodes akin to artificial neurons, showed promise in learning from data and making predictions—a rudimentary reflection of the brain's ability to recognize patterns and adapt over time.

In the 1950s, Frank Rosenblatt introduced perceptrons (see Figure 1.2 C), early prototypes of neural networks capable of learning from input data through weighted connections [39]. Rosenblatt's perceptron, while limited to linearly separable tasks, sparked enthusiasm for neural network research and laid the groundwork for future developments in AI and machine learning

In the late 1950s and early 1960s, Wiesel and Hubel performed experiments that provided groundbreaking insights into the organization and function of the visual cortex in mammals, particularly cats [40, 41]. Specifically, they identified distinct types of neurons in the visual cortex with specific receptive fields—areas of the visual field that trigger neural responses. They classified neurons into "simple cells" and "complex cells." Simple cells responded to oriented edges and lines within specific positions and orientations, while complex cells showed less positional specificity but responded to oriented stimuli over larger areas. A pivotal observation was the hierarchical organization of neurons in the visual cortex. They found that simple cells in the primary visual cortex project to complex cells in higher visual areas, indicating a pathway for

progressively complex feature extraction and processing of visual information.

Their research not only elucidated fundamental principles of visual processing but also inspired advancements in the understanding of cognition and the development of posterior neural networks. This is the case of Fukushima's Neocognitron in 1980, a pioneering convolutional neural network (CNN) architecture aimed at emulating visual processing and feature extraction in the brain [42].

The progress continued by delving deeper into the brain's workings, developing algorithms like backpropagation in the 1970s [43, 44]. This algorithm allowed not a single perceptron but a multi-layer perceptron (MLP) to efficiently learn from training data, adjusting the strengths of connections between artificial neurons, mirroring the brain's synaptic plasticity.

It was the introduction of LeNet-5 in the 1990s [9], that made artificial neural networks (ANNs), especially CNNs, as we know them nowadays. This was one of the earliest successful applications of CNNs, demonstrating the effectiveness of convolutional layers, pooling layers, and fully connected layers in a systematic architecture (see Figure 1.3). This work laid the groundwork for deeper CNNs by showing how to hierarchically extract features from images. Furthermore, researchers found that increasing the depth of CNNs (i.e., stacking more convolutional and pooling layers) enabled better representation learning and hierarchical feature extraction. Thus, Alexnet [45] propelled CNNs into the spotlight, showcasing their capability in large-scale image classification tasks.

Recently, brain-inspired models have continued improving performance in different tasks. This is the exciting case of a vector-based navigation agent using grid cells-like representation that outperformed existing models [46], the

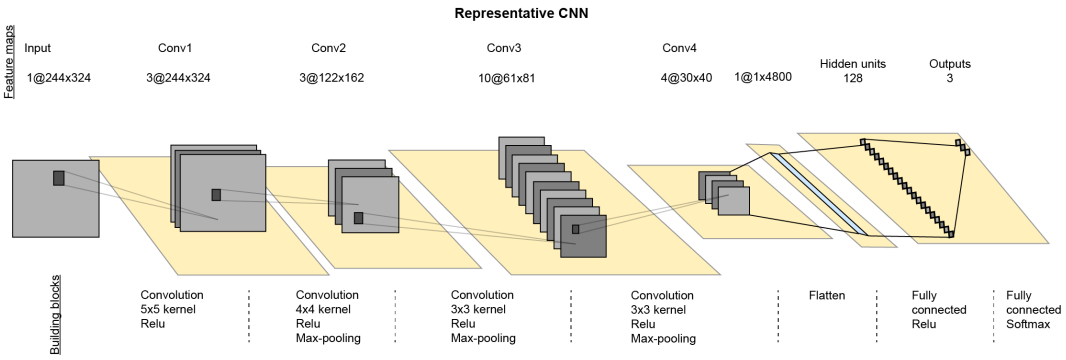


Figure 1.3: A diagram of a representative CNN architecture.

SlowFast model for video recognition based on the visual ventral/dorsal stream [47], and the c. elegans-based neural network policy for autonomous driving [48].

1.3.2 Drosophila's vision

Regarding the previous successful cases of brain-inspired AI, one model organism well established in neuroscience is the fruit fly *Drosophila melanogaster*. This insect offers an opportunity as a platform to develop new AI algorithms because its brain while being simpler than the mammalian brain still presents all the basic functionalities. Fly's brain has been dissected at a cell level via the connectomes released recently [1, 2] providing connectivity detail such as synaptic counts, synaptic neurotransmitter polarities, etc. Particularly, the optic lobe of about 80,000 neurons can perform multiple functions simultaneously through different neural circuits, such as motion detection, pattern recognition, looming detection, spatial navigation, etc. [49].

Fly's visual system has a columnar layer-based structure performing visual processing of different complexity upon deeper layers [5]. Electrical signals originating from photons are transduced through the rhabdomeres [50] within

the retina layer. This information then progresses through various layers of the visual system. Initially, it flows towards the early layers such as lamina and medulla, and subsequently to deeper layers including lobula and lobula plate [5] (see Figure 1.4 A). The optic glomeruli, a higher-order visual center, plays a crucial role in complex feature detection [6] (see Figure 1.4 A). The visual system follows a retinotopic structure where the visual surround is projected retinotopically onto the dendrites of subsequent neurons [3] (see Figure 1.4 B). The resolution of about $\sim$ 800 ommatidia at the retina layer is maintained through the layers up to the output of lobula/lobula plate cells where there is a biological pooling to about $\sim$ 40 resolution [1] (see Figure 1.4 B).

The visual system also encompasses distinct ON and OFF pathways, dedicated to detecting moving bright and dark edges respectively [51, 52, 53, 54]. These pathways ultimately converge onto motion-detecting neurons T4 and T5 [55, 56], enabling motion direction selectivity [4].

Deeper within the layers, specific neurons like LC11 and LC15, with dendrites in lobula and axons projecting to optic glomeruli, exhibit specialized responses. LC11 responds to small objects such as spots, while LC15 detects long objects like bars [7, 8]. The intricate connections upstream to these neurons and higher centers have also been meticulously mapped [57]. In the lobula, another crucial neuron, LPLC2, functions as a selective looming detector [58]. Complementing this function, an adjacent neuron, LC4, encodes the velocity of approaching objects [59].

Upon the recognition of the visual features of the scenery the fly will generate visual-reflexive behaviors. The visual centers process the visual input and the motor centers will activate the corresponding wing muscles to generate the proper reflexive behavior (see Figure 1.5 A). In the case of complex visual

environments, multiple visual features are processed in different visuomotor circuits that can trigger a shared actuator, such as the wing motor system. A theory of how to manage this coordination is an EC mechanism. This theory posits that during movement, the nervous system generates a copy of the motor command, which is then compared with sensory feedback to predict the consequences of the movement. Essentially, it helps animals distinguish self-generated sensory signals from externally generated ones [23]. In such cases the behavior of the fly is modulated not only by external visual input but also by an internal state [60] (see Figure 1.5 A).

One of these reflexive behaviors is the optomotor response which is the ability to track and respond to visual motion patterns, particularly the optic flow, which is the perceived motion of objects as the observer moves through the environment. When a fly is in motion, the visual world around it shifts accordingly. For instance, as it flies forward, objects in its field of view move backward. Once the optic flow is detected and analyzed, the fly's motor system is activated to generate appropriate behavioral responses. This typically involves adjusting the orientation and position of the fly's body and wings to counteract any deviations from its intended trajectory [15, 16, 17] in response to horizontal or vertical gratings. For example, if the optic flow suggests that the fly is veering to the left, it will adjust its wing movements to steer back to the right, effectively stabilizing its flight path (see Figure 1.5 B).

Another intriguing aspect of its behavioral repertoire is the bar fixation behavior. This behavior involves the fly's ability to fixate on and track a moving visual stimulus, such as a narrow bar or stripe, within its visual field. Bar fixation serves various purposes, including navigation and object tracking. When presented with a moving bar or stripe in their environment, *Drosophila*

exhibits a characteristic response wherein they actively orient themselves to keep the bar within their field of view [18, 61, 19] (see Figure 1.5 B). This behavior is observed both in tethered flies and in free-flying individuals in more naturalistic environments [20].

Contrary to the previous, spot aversion behavior in *Drosophila* refers to their instinctive response to avoid spots in their environment. When *Drosophila* encounters spots or short small objects, they typically exhibit a series of avoidance maneuvers [20, 19] (see Figure 1.5 B). These can include changes in locomotion, such as rapid turns or retreats, as well as alterations in wing posture. One proposed explanation for spot aversion behavior is that flies have evolved to associate dark spots with potential dangers, such as predators or obstacles. In nature, dark spots could indicate hiding places for predators or areas of reduced visibility where collisions with objects are more likely. By avoiding these areas, flies can increase their chances of survival.

Recent studies in *Drosophila* have proven the existence of EC signals in an array of visual neurons during both spontaneous and visually evoked rapid flight turns [24, 25, 26]. Thus, in cluttered visual environments when the fly may encounter gratings in the background and long stripes in the foreground EC can mediate to generate proper behavior (see Figure 1.5 B).

1.3.3 Fly vision models

Models of fly vision have been established to mimic its visual processing. In a classical approach, the Hassenstein-Reichardt motion detector model where a differential activation of photoreceptors with a temporal delay served as a foundational concept for understanding how flies detect and respond to visual motion [62]. In a quantitative analysis the torque response of the fly to

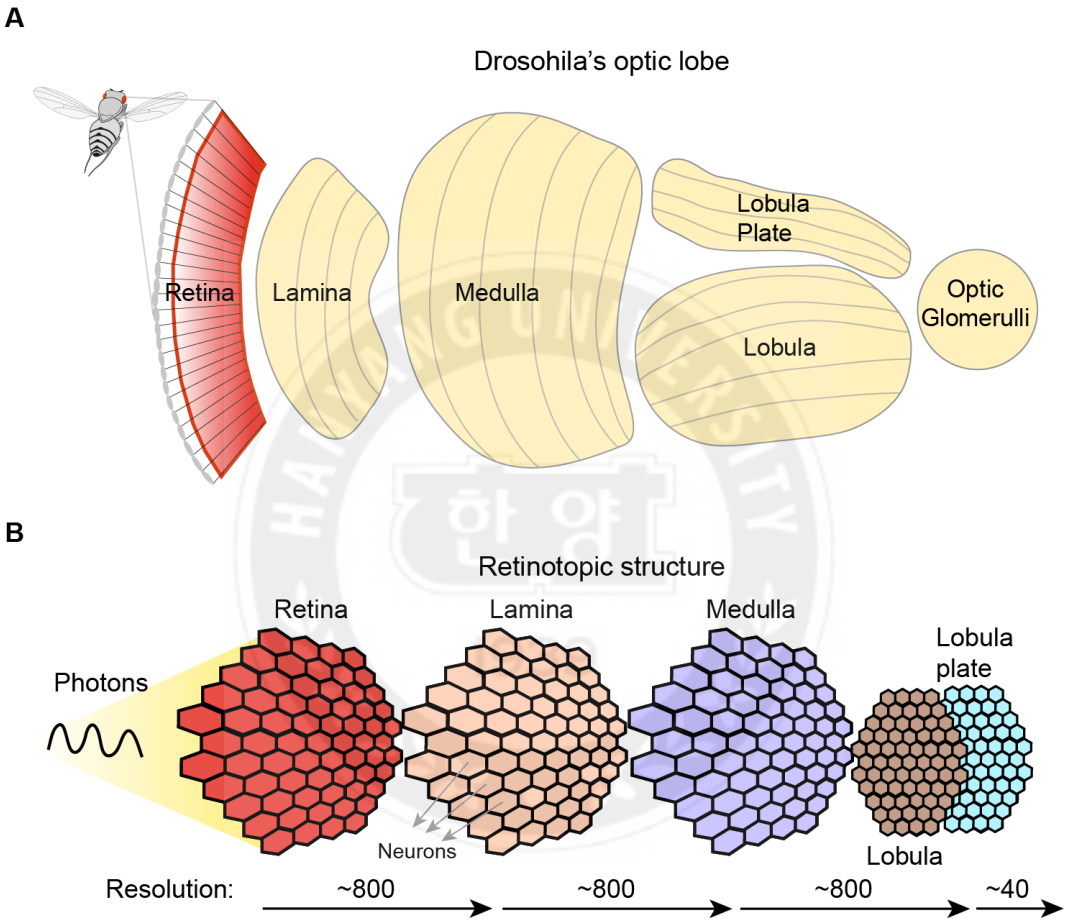


Figure 1.4: The fly's visual system structure. (A). The *Drosophila*'s optic lobe organized in subsequence layers. (B). The retinotopic projection structure of the fly visual system. It maintains $\sim$ 800 ommatidia resolution up to lobula/lobula plate cell outputs where it is reduced.

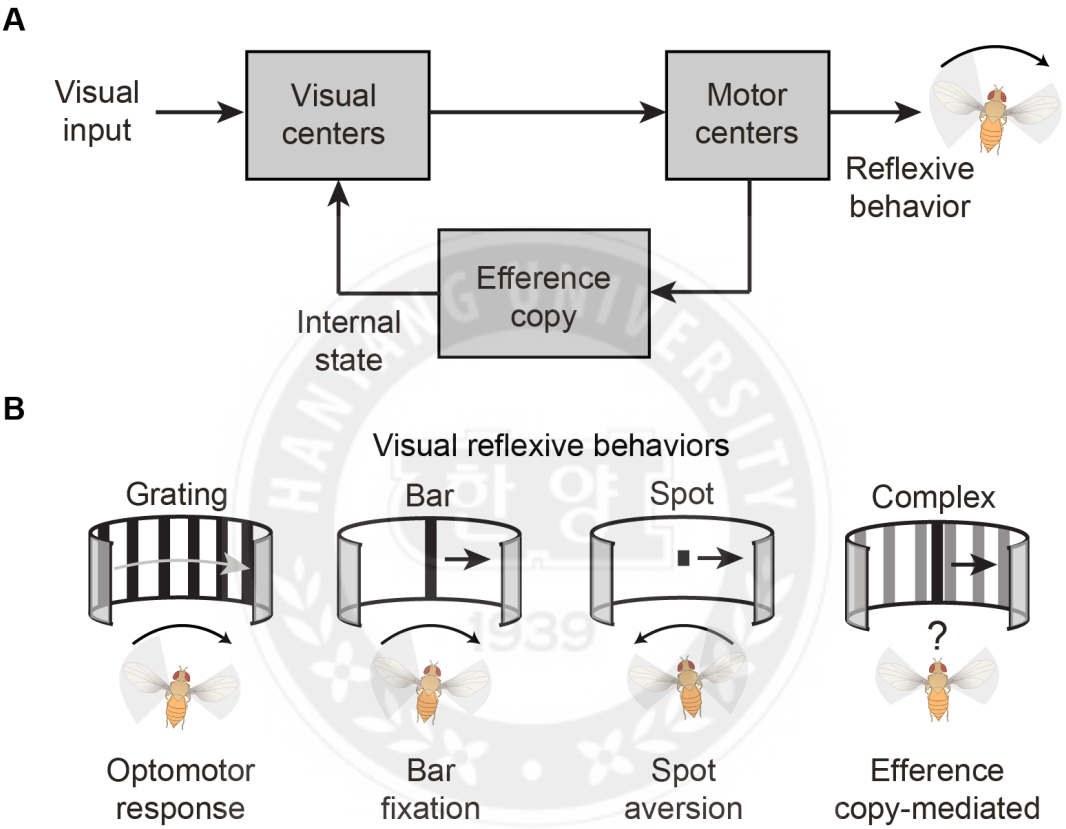


Figure 1.5: Depending on the visual inputs the fly generates different reflexive behaviors. (A). A concept diagram indicating the neural processing flow where the reflexive behavior depends on the visual input and the internal state. (B). Visual reflexive behaviors. In a complex visual environment, the behavior is mediated by EC.

a moving bar was mathematically defined by summation of direction-sensitive and direction-intensive components [63, 64]. Furthermore, dynamical system models using feedback control have shown the utility of fly vision for flight control [65, 17, 21]. Models using the Hassenstein-Reichardt motion detector allowed robots to navigate avoiding collisions [66, 67].

ANNs have been also used to model fly vision. Partial implementations have provided solutions in collision detection tasks [10, 11], and classification [12]. Also, connectome-based ANNs implementing early visual layers achieved direction selectivity-like responses [13, 14], providing a deeper understanding of neural processing that complements the connectome data and solving tasks such as optic flow detection.

1.3.4 Vision-based artificial agents

fly vision-inspired models are expected to be more power and memory efficient than those mimicking larger visual systems, so it might be useful in applications with memory and power limitations, such as drones. Particularly, fly vision models might be ideal for controlling small drones. In fact, there is a trend in shifting the technology and application of drones, transitioning from radio-controlled models to advanced systems driven by AI. This evolution has transformed drones from simple remote-controlled gadgets into sophisticated aerial platforms capable of autonomous decision-making and complex tasks [68, 69]. AI algorithms allow them to interpret and understand their surroundings in real-time, enabling autonomous flight and decision-making. AI-driven drones excel in object recognition and tracking. They can identify and track objects of interest such as vehicles, people, animals, or specific landmarks [68, 70]. Also, AI-driven drones employ advanced obstacle detection

and avoidance techniques based on vision. They can dynamically detect and navigate around obstacles in their path, ensuring safe and efficient flight even in complex environments [71].

Among drones, nano-sized drones present the most demanding memory and power constraints, becoming necessary to design optimized AI algorithms. In this direction, the introduction of AI microcontroller units (MCU) has permitted to run CNN models on an ultra-low power basis. These microcontrollers require networks of small size architectures and low memory. Well-established architectures for classification such as ResNet [72] need to be adapted for these types of controllers. Thus, the DroNet, based on ResNet, was embedded successfully in an AI processor optimized to execute CNNs, which was assembled on a low-power nano drone, and proved to be effective for steering control and collision avoidance [73]. Further optimization was possible up to a network of 8-bit parameter resolution, the Frontnet, that reduced significantly power consumption [74].

Chapter 2. FlyVisNet: an insect vision-inspired artificial neural network for the control of a nano drone

2.1 Summary

Flying insects exhibit remarkable visual computation capabilities despite having a limited number of neurons. Specifically, the fruit fly *Drosophila*'s visual system consists of $\sim 110,000$ neurons, and its neural circuits have been steadily dissected at the level of single cells. Among them, visual projection neurons have been found to sense different features such as a looming visual object, optic flow, and objects in specific shapes. In this work, we have developed a neuromorphic Convolutional Neural Network (CNN), FlyVisNet, in which we have implemented the visual projection neurons LC11, LC15, and LPLC2 as well as all their upstream connections including T4/T5 motion-detecting neurons from the ON/OFF pathways. We compared its accuracy on a classification task to other vanilla CNN architectures of a similar scale. We found that FlyVisNet achieved higher accuracy for two different datasets than the other CNNs, even with a reduced number of parameters, thereby requiring less memory. Furthermore, the artificial neurons of our network, after being trained for the classification, reproduced selectivies of real neurons, in terms of the shape and direction of visual pattern. Finally, we tested FlyVisNet embedded in a nano drone, in virtual as well as in real environments. With an onboard ultra-low-power parallel processor dedicated to FlyVisNet, the drone achieved high performance in terms of the response time and memory requirement. Our results highlight the importance of brain-inspired computations

for a deeper understanding of their biological counterparts, as well as for the application to other domains, particularly in embedded systems with memory and power limitations.

2.2 Introduction

The brain's neural processing has long inspired artificial intelligence, ranging from artificial neuron models to perceptrons and convolutional neural networks (CNNs) [38, 39, 42]. In particular, neocognitron, the first CNN, was developed based on the study of orientation-selective feature-detecting neurons in the cat's visual cortex [40, 41]. The convolution operation in this network was designed after the visual property of so-called simple cells in the visual cortex, which can be obtained by adding visual tuning of presynaptic neurons. Although the resulting CNN architecture has been foundational for recent progress in artificial intelligence, attempts to build an artificial neural network mimicking a more detailed organization of a visual system have been scarce.

With the advent of the full brain connectome in Drosophila, alongside premiere genetic toolkits, behavioral and physiological experimental assays, it has become possible to dissect the structure and function of single visual neurons in depth [1, 2]. Among $\sim 200,000$ neurons in the *Drosophila* brain, more than half were shown to be involved directly in visual processing [75, 76], making it one of the most visual organisms. The system consists of five early visual structures: retina, lamina, medulla, lobula, and lobula plate. Visual processing in these structures consists of columnar structures corresponding to each compound eye lens, detecting various local visual features while main-

taining the spatial dimensionality. When these structures pass local visual feature signals to central brain regions, the feature information converges from a large visual field to a singular structure, which exhibits responses to different visual patterns such as translating spots and bars or approaching objects [7, 8, 57, 58, 59]. These structures represent a concise visual system for visual feature detection, making it an ideal target for brain-inspired artificial neural networks.

The *Drosophila* visual system has been previously used for building visual feature detection systems [62, 66, 14, 13]. For example, the Hassenstein-Reichardt motion detector model based on dynamical systems where a differential activation of photoreceptors with a temporal delay served as a foundational concept for understanding how flies detect and respond to visual motion [62]. Further improvements of this model allowed robots to navigate without collisions [67]. Simulations of a part of the visual system have been reported through ANNs [14, 13, 12, 10], providing a deeper understanding that complements the connectome data and solving tasks such as image classification, optic flow detection, and collision avoidance. These implementations could be useful in applications with memory and power limitations such as drones.

Since the fly brain-inspired models are expected to be more energy efficient than those after a larger visual system, they might be useful for applications requiring less energy, such as a drone. Moreover, there has been a significant shift in the technology and application of drones, transitioning from radio-controlled models to advanced systems driven by ANNs [68, 69]. One key aspect of ANN-driven drones is their autonomy, allowing them to make decisions based on the data collected from their surroundings. They can navigate complex environments, avoid obstacles, and even collaborate with other drones

to achieve specific objectives [68, 70, 71]. Among such applications, nano-sized drones present one of the most challenging platforms to operate ANN-based algorithms because of their memory and power limitations. Alongside the development of ANN-compatible microcontrollers, *Drosophila* brain-inspired neural network may be adequate for nano-sized drones [73, 74].

In this study, we present the implementation of a neuromorphic CNN inspired by the structure of the Drosophila visual system. We named this network FlyVisNet and designed it to mimic the neural circuits crucial for detecting translating or approaching visual objects. FlyVisNet incorporates artificial neurons representing visual neurons transmitting visual feature information to the central brain, along with early visual neurons essential for computing these features. When trained to detect simple visual features, FlyVisNet demonstrated superior accuracy compared to the existing lightweight network, such as MobileNetV2, despite having a comparable network size. Furthermore, we conducted experiments to evaluate FlyVisNet's performance on a low-power nano-drone for fully onboard autonomous flight. Our results indicate that FlyVisNet enables successful visually guided course control behaviors.

2.3 Results

2.3.1 FlyVisNet – A fly brain-inspired CNN for visual feature detection

We designed a CNN model whose structure mirrors that of the *Drosophila* visual system. The early visual system of *Drosophila* consists of five neuropils

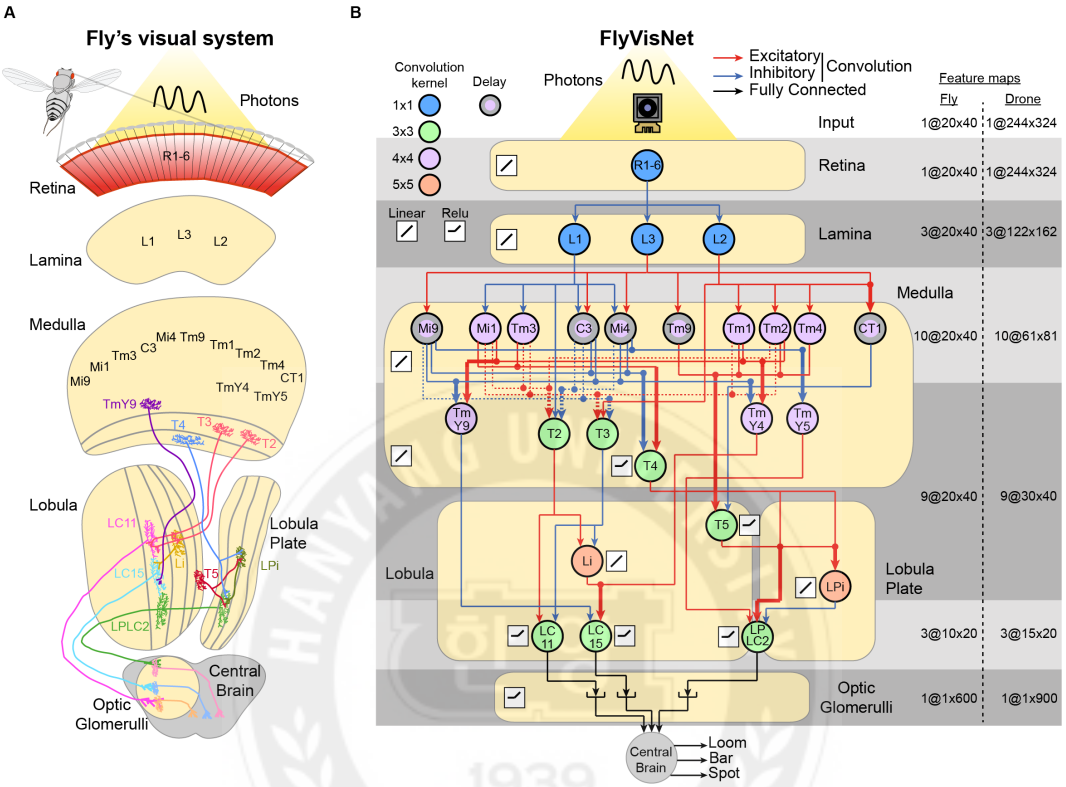


Figure 2.1: FlyVisNet mimics the Drosophila's visual system architecture through a CNN. (A). The visual system of Drosophila, showing the neural pathways from the retina to the central brain and highlighting the neurons that we have implemented, specially the feature-detecting neurons LC11, LC15, and LPLC2. (B). The FlyVisNet diagram in 2 versions, the fly version of 20x40 corresponding with the $\sim$ 800 ommatidia input resolution, and the drone version for a camera resolution of 244x324. Each neuron is implemented as a convolution layer. The kernel initialization values are uniform positive/negative for excitatory/inhibitory connection, respectively. The color of each neuron represents the kernel size. The gray backgrounded neurons process one frame delay. The activation functions used are linear and ReLU, which are indicated for each neuron. Convolution outputs from LC neurons are concatenated through fully connected layers. The network has 3 outputs for the classes loom, bar, and spot respectively.

in the optic lobe [5], and the output of these structures converge to small structures in the central brain, through different types of visual projection neurons (VPNs) [6, 77]. Among these VPNs, three cell types known for their high visual selectivity are LC11, LC15, and LPLC2 (see Figure 2.1 A). LC11 neurons get activated in response to the movement of a small object, regardless of the movement direction [7]. LC15 neurons depolarize selectively to the horizontal motion of a vertical stripe, again regardless of the direction [8]. Finally, LPLC2 cells respond to both an approaching object and a translating small object [58, 59]. We built a model around these three types of VPNs to sense translating or approaching visual objects.

Each neuron is implemented as a convolution layer with a main building block of convolution operation followed by activation and max-pooling (see Figure 2.1 B and see Figure 2.S1) (see Methods 2.5.1). We present 2 versions, the fly version following the retinotopic $\sim$ 800 ommatidia resolution [3] of input images and the drone version for its camera resolution of 244x324 (see Figure 2.1 B). In the fly version, the max-pooling is only applied in the last layer (see Figure 2.S1). The convolution kernels are initialized with positive or negative uniform values depending on whether their real counterpart is excitatory or inhibitory, respectively. These values are constrained to maintain the polarity during optimization. To implement a visual feature detection network centered around these three VPN types, we incorporated their presynaptic visual circuits into the model. LC11 cells receive their primary excitatory inputs from T2/T3 neurons, columnar neurons projecting from the medulla to the lobula [7]. LC15 cells receive the majority of their inputs mainly from TmY4 and TmY9 [8, 77]. To enhance its feature selectivity, we also added inhibitory interneurons that were hypothesized in a previous study [8], corresponding to

Li27 identified in a connectome [1] (see Figure 2.1 A, B). Finally, LPLC2 cells respond to a looming object pattern with the majority of their inputs from T4, T5, LPi, and Tm5f [58, 59, 77] (see Figure 2.1 A). T4/T5 neurons are the first to show direction selectivity to bright/dark moving edges respectively in a direction-dependent manner [4, 55, 56].

One recent study showed that LPLC2 neurons form their feature selectivity by integrating visual inputs from both the lobula and the lobula plate [78]. In the lobula, they receive the majority of inputs from trans-medullar neurons such as Tm5f. In the lobula plate, the majority of excitatory inputs come from T4/T5 neurons, which are well known for their direction-dependent motion response to local edges. LPLC2 neurons also receive inputs from inhibitory neurons, such as LPi11 neurons, to enhance their direction sensitivity [78]. To endow T4/T5 neurons with their direction-dependent motion sensitivity, we incorporated all the major medulla neurons presynaptic to these neurons, such as Mi1, Tm3, Mi4, Tm9, Tm1, Tm2, Tm4 cells. We implemented a processing delay to some of these medulla neurons (see Figure 2.1 B) by shifting the input batch one frame. Peripheral to medulla neurons, we incorporated all the R1-R6 neurons in the retina and L1-L3 neurons in the lamina.

LC neurons at the final stage of FlyVisNet are implemented as convolution layers with a relu activation function (see Figure 2.1 B and see Figure 2.S1). The LC neurons project their axons into the optic glomeruli and from there they connect with neurons in the central brain. We concatenated the outputs of LCs and through one hidden fully connected layer for 3 output classes (see Figure 2.1 B and see Figure 2.S1). Through this layer, we classified the input images into one of the three patter types: loom, bar, and spot.

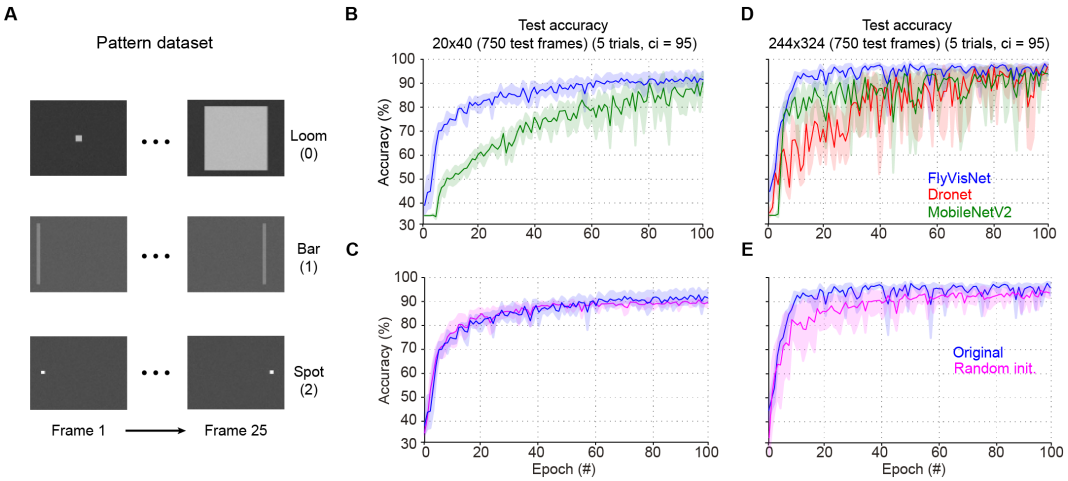


Figure 2.2: FlyVisNet trained for classification showed higher top accuracy on the pattern dataset than other networks of comparable size. (A). A pattern dataset was designed to match the representative shapes for Drosophila, whose images contain moving patterns labeled as loom, bar, and spot, respectively. Each sequence contains 25 frames. (B). FlyVisNet, and MobileNetV2 test accuracies on 750 frames of the pattern dataset for frames of size 20x40 for 5 trials with a confidence interval (ci) of 95. (C). FlyVisNet with original and with random kernel initialization testing accuracies on the pattern dataset for frames of size 20x40 with ci of 95. (D). Same as in B for size 244x324, including Dronet accuracy. (E). Same as in C for size 244x324.

2.3.2 Classification performance of FlyVisNet

To train our network to classify images through feedforward visual processing, We made a moving pattern dataset that presents visual features for *Drosophila*. We generated images in sequences of 25 frames featuring moving bars and spots of random width. We also included sequences of looming squares expanding from random positions (see Figure 2.2 A). The brightness of the background and the objects is generated randomly. We also added a uniform random noise component of 5% to each pixel for a better training generalization of natural conditions. For the loom squares, we applied loom label. For moving bars and spots, we used bar and spot labels respectively (see Figure 2.2 A). We generated a set of 7500/750 training/testing frames of

244x324 size (drone camera resolution), and another set of 20x40 size, aiming to emulate the fly's 800 ommatidia resolution [3]. For an additional comparison, we chose the COIL-100 dataset [79], which we resized to the previous resolutions. This dataset contains images of 100 different objects filmed on a rotating table and captured from different angles (see Figure 2.S2 A). The objects have different complexity, being some of them natural-based like fruits and vegetables, which could be familiar to the actual fly, whereas others are more artificial like cars and cans that could be challenging to classify. Since our network was to be embedded in the drone, we checked how well our network could do on the classification task of these datasets in comparison to the other networks of similar size designed for embedding applications. We divided the datasets into batches for the training loop and optimized on accuracy (see Methods 2.5.2).

For the accuracy performance comparison, we chose Dronet [73] and MobileNetV2 [80], architectures with a similar range of parameters. We varied the hyperparameter α to maintain MobileNetV2 size similar to FlyVisNet's. We found that after training on the pattern dataset of resolution 20x40, FlyVisNet achieved a 92.80% test top accuracy, 2.57% higher than MobileNetV2 (see Figure 2.2 B and Table 2.1). To achieve this accuracy FlyVisNet used 78,226 parameters and 1200KB, which is slightly less than MobileNetV2. Additionally, when we initialized the convolution filter weights as random uniform instead of maintaining the synaptic polarity initialization, the accuracy dropped by 1% (see Figure 2.2 C and Table 2.1). These results suggest that our architecture is light and powerful enough to classify simple patterns in a similar way as the fly can do. For a resolution of 244x324, FlyVisNet achieved a 98.53% test top accuracy, slightly higher than Dronet and 2.07% higher than MobileNetV2 (see

Figure 2.2 D and Table 2.1). To achieve this accuracy FlyVisNet used 116,626 parameters and 1660KB, which is significantly less memory than Dronet and slightly less than MobileNetV2. Additionally, when we initialized the convolution filter weights as random uniform instead of maintaining the synaptic polarity initialization, the accuracy dropped by 3% (see Figure 2.2 E and Table 2.1).

When training on COIL-100 dataset of resolution 20x40, FlyVisNet obtained 99.14%, which was slightly higher accuracy than in the pattern dataset (Table 2.1). This is most likely due to the absence of uniform noise in this dataset as well as the empty background of the frames. Our architecture could classify successfully the objects of different complexities with 6.65% higher accuracy than MobileNetV2 (see Figure 2.S2 B and Table 2.1), and also higher than the randomly initialized version (see Figure 2.S2 C and Table 2.1). For a resolution of 244x324, FlyVisNet obtained 99.23% with slightly higher accuracy than Dronet, and MobileNetV2 (see Figure 2.S2 D and Table 2.1), and also higher than the randomly initialized version (see Figure 2.S2 E and Table 2.1). The training curves for each dataset and resolution case followed the same trend as the test curves (see Figure 2.S3 A-D). By quantizing FlyVisNet from 32-bit to fixed 8-bit, the resolution of the parameters was reduced to an ultra-light memory of 143KB for a resolution of 244x324 still with a 91.00% accuracy being higher than that of the quantized MobileNetV2 (Table 2.1). This quantized version is what we needed for embedding and using the drone camera resolution. But we also compared a quantized version for the 20x40 resolution case, which showed an impressive memory reduction to just 96KB with 80.33% accuracy (Table 2.1). Even if we only quantized this version for comparison purposes, the result suggests that this could be useful in an

Table 2.1: Classification accuracy on pattern and COIL-100 datasets. Best mean ± std is shown for 5 trials.

Architecture	**Top accuracy (mean ± std %)**		**Parameters**	**Memory (kB)**	**Resolution**
	Pattern	**COIL-100**			
Dronet	98.33 ± 0.004	98.91 ± 0.006	343,267	4,183	244x324 (drone)
MobileNetV2 ($\alpha = 0.180$)	96.46 ± 0.021	99.05 ± 0.003	117.813	1,807	
FlyVisNet	**98.53** ± 0.002	**99.23** ± 0.003	116.626	**1,660**	
FlyVisNet (random init.)	95.53 ± 0.010	98.73 ± 0.007	116.626	1,660	
MobileNetV2 ($\alpha = 0.145$)	90.23 ± 0.027	92.49 ± 0.050	78,761	1,354	20x40 (fly)
FlyVisNet	**92.80** ± 0.019	**99.14** ± 0.005	78,226	**1,200**	
FlyVisNet (random init.)	91.86 ± 0.006	98.86 ± 0.014	78,226	1,200	
MobileNetV2 (fixed 8)	89.63	-	117.813	156	244x324 (drone)
FlyVisNet (fixed 8)	**91.00**	-	116.626	**143**	
FlyVisNet (fixed 8)	80.33	-	78,226	96	20x40 (fly)

extremely memory-demanding application.

In a further test, we checked how variations in the architecture would affect the network performance on the pattern classification. We performed a selective pruning of different convolution layers and a general pruning of 80% sparsity. For the resolution of frames 20x40, by removing the R16 convolution layer the top accuracy dropped by 0.80% (Table 2.S1). The network needed similar training epochs to converge to the top accuracy (see Figure 2.S4 A). When we removed T4 and T5 layers the accuracy dropped by about 3% which proved the importance of these layers (see Figure 2.S4 A and Table 2.S1). By applying a pruning of 80% sparsity, the accuracy dropped significantly by about 10% still reducing a significant amount of non-relevant parameters and memory (Table 2.S1). In the case of resolution 244x324, the accuracy drop rates were similar to the lower resolution case (Table 2.S1). However, the pruned versions needed more training epochs to converge to the top accuracy (see Figure 2.S4 B).

2.3.3 Emergence of feature selectivity in FlyVisNet

After training the network on the pattern classification task, we explored the neural responses of the artificial neurons to check whether they were correlated with their biological counterparts. Since FlyVisNet follows the fly's visual system architecture, we were specially interested in targeting the motion direction and feature-detecting neurons T4, T5, LC11, LC15, and LPLC2 (see Figure 2.3 A). To obtain the 1D activation, that is the response over time, we passed to the network a visual stimulus through a sequence of frames. Then, we captured the 2D activations, that is the output of each activation function after the convolution, and computed the mean over all the neurons that made up the layer to obtain the averaged 1D response (see Figure 2.3 B). The polarity constraints on the kernels ensured their polarity during training (see Figure 2.S5), so the feature maps from the main inputs to T4/T5 maintained the polarity predominance for a moving bar (see Figure 2.S6). When targeting T4/T5 neurons, we presented to the network a bright moving bar in a preferred direction (PD), that is from left to right, and in a non-preferred direction (ND) (see Figure 2.3 C). We found that the 1D activation response of T4 showed greater amplitude to the PD bright edge of the moving bar, as previously reported for the biological counterpart [55] (see Figure 2.3 C). In the case of T5 response, which detects dark moving edges, it also showed greater amplitude for PD dark edge of the moving bar as has been reported [56] (see Figure 2.3 C). We found that the oscillations in these responses were due to the max pooling operation. We targeted the same neurons for the 20x40 fly version without the pooling and indeed did not exhibit any oscillations (see Figure 2.S7 A). Nonetheless, we have used as main architecture the drone ver-

sion with pooling given the benefits in accuracy performance. We also checked the response of the original FlyVisNet before being trained. The motion direction selectivity still worked but not as well as with the training (see Figure 2.3 D). When the convolution kernels were randomly initialized we found a loss in selectivity in T5 or even inversion in T4 (see Figure 2.3 E). We also measured the 2D activations averaged over time that showed the preference for the PD (see Figure 2.S8 A). So the artificial neurons T4, and T5 trained exclusively for a classification task, exhibited direction selectivity in line with those of the fly. When targeting LC11 and LC15, we showed to the network a moving bar and a moving spot. For the moving bar case, LC15 showed a greater amplitude response than LC11 (see Figure 2.3 F) as it was found in the fly [8]. The fly version of the network did not produce oscillations (see Figure 2.S7 B). The 2D activations also showed this preference (see Figure 2.S8 B). For the moving spot, LC11 response was greater than LC15's (see Figure 2.3 G) as reported previously for the fly [7]. The non pooling version neither showed oscillations (see Figure 2.S7 C). The 2D activations additionally demonstrated this selectivity (see Figure 2.S8 C). Therefore LC11 and LC15 showed feature selectivity. For the LPLC2 case, we presented to the network a looming spot and compared LPLC2 to LC15 responses. LPLC2 showed greater amplitude response showing looming selectivity (see Figure 2.3 H) as it was found previously [58, 59]. The 2D activations complemented these results (see Figure 2.S8 D). To quantify the selectivity of each neuron, we computed a selectivity index (see Equation 2.1) and found that at least all of them are beyond 0.21 value (see Figure 2.3 I). LPLC2 neuron showed the greatest selectivity with an index value of 0.785.

$$SI_a = \frac{max(N_a) - max(N_b)}{max(N_a) + max(N_b)} \tag{2.1}$$

where N_a is the array containing the response over time of neuron a, and N_b is the array containing the response over time of neuron b.

In the untrained case, all the neurons showed proper selectivity (see Figure 2.3 J, K, and L), but with lower index values (see Figure 2.3 M). In the case of a random initialization of the network, LC15 exhibited preference for the moving bar but also for the spot over LC11 (see Figure 2.3 N, O). LPLC2 response was weaker than LC15 to the looming spot (see Figure 2.3 P). The selectivity indexes for the random initialization case showed worsened or even loss in direction and feature selectivity (see Figure 2.3 Q). These results along with the accuracy drop highlight the importance of the initialization following the actual synapse signs to improve the network performance.

We further checked the responses of the different pruned versions of Fly-VisNet (see Figure 2.S9 A). When we removed the R16 layer, the direction selectivity of T4 and T5 was inverted with respect to the original (see Figure 2.S9 B, and C). This is somehow expected since the R16 layer is initialized with negative values and works as an inverter. LC15 responded greater to both bar and spot than LC11 showing noisier traces than in the original version (see Figure 2.S9 D, E, F, G, H, and I). LPLC2 neither responded greater to the looming spot than LC15 (see Figure 2.S9 J). So as a result the selectivity index was inverted for T4/T5 and was also lower for the LC neurons than in the original version (see Figure 2.S9 K). In the case of removing the T4 layer, LC11 responded greater to bar and spot than LC15 (see Figure 2.S9 L, M). Interestingly, LPLC2 responded greater to the loom but did not decay after

the onset expansion of the spot (see Figure 2.S9 N). This is most likely due to the absence of T4, that is the ON contribution to the LPi neuron which is an inhibitory input to LPLC2. The index value showed the loss of selectivity of LC15 in this case (see Figure 2.S9 O). When removing T5, LC11 responded slightly greater to both moving bar and spot than LC15 (see Figure 2.S9 P, Q). LPLC2 response was greater than LC15 to the loom and showed decay (see Figure 2.S9 R). The selectivity in this case was lost for LC15 and the index value was lower for LC11 and LPLC2 than in the original version (see Figure 2.S9 S).

Regarding the emerged selectivity in FlyVisNet we should highlight the importance of the visual processing delay of specific neurons in the medulla layer that allowed a proper direction selectivity. We compared the original version with a non-delayed version of the mentioned neurons, and found that the direction selectivity of T4/T5 neurons was worsened in the absence of delay (see Figure 2.S10 A). The looming detection of LPLC2 was also worsened when distinguishing expanding from shrinking square (see Figure 2.S10 A). Another important element is the presence of LPi neuron for the looming detection. By removing this neuron, the LPLC2 response did not decay after onset (see Figure 2.S10 B) highlighting the role of LPi in contributing to the detection of a looming motion.

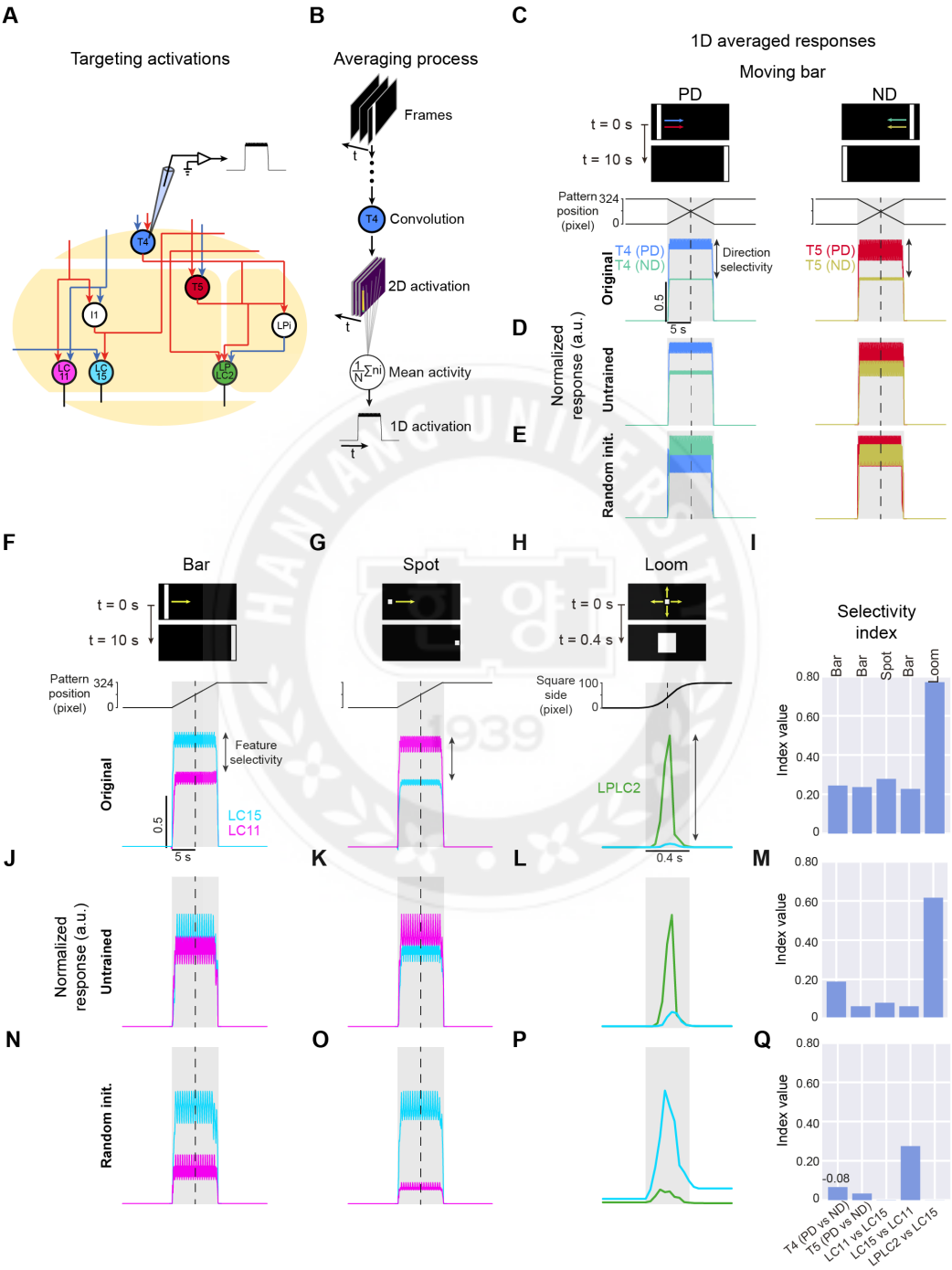

A
Targeting activations
T4
T5
l1
LPi
LC 11
LC 15
LPLC2
B
Averaging process
Frames
Convolution
2D activation
Mean activity
1D activation
C
1D averaged responses
Moving bar
PD
ND
t = 0 s
t = 10 s
Pattern position (pixel)
324
0
T4 (PD)
T4 (ND)
T5 (PD)
T5 (ND)
Direction selectivity
Original
0.5
5 s
D
Normalized response (a.u.)
Untrained
E
Random init.
F
Bar
t = 0 s
t = 10 s
Pattern position (pixel)
Feature selectivity
Original
LC15
LC11
0.5
5 s
G
Spot
H
Loom
t = 0 s
t = 0.4 s
Square side (pixel)
100
LPLC2
0.4 s
I
Selectivity index
Bar
Bar
Spot
Bar
Loom
Index value
0.80
0.60
0.40
0.20
0
J
Normalized response (a.u.)
Untrained
K
L
M
Index value
N
Random init.
O
P
Q
Index value
-0.08
T4 (PD vs ND)
T5 (PD vs ND)
LC11 vs LC15
LC15 vs LC11
LPLC2 vs LC15

Figure 2.3: FlyVisNet's artificial neurons exhibited feature selectivity. (A). Targeted neurons T4, T5, LC11, LC15, and LPLC2. (B). Averaging process to obtain 1D layer activations over time. (C). Neural response comparison of T4, and T5 to a bright moving bar in the PD and ND. Responses are normalized for better comparison. (D). Same as in C but before training the network. (E). Same as in C but when the network is trained being initialized with random filter values. (F). Neural response comparison of LC11, LC15 to a bright moving bar. (G). Same as in F for a bright moving spot. (H). LPLC2 to LC15 comparison to a looming stimulus. (I). Selectivity index comparison. The index is computed for each compared case in C, F, G, and H. (J), (K), (L), (M). Same as in F, G, H, and I but before the network is trained. (N), (O), (P), (Q). Same as in F, G, H, and I but when the network is initialized with random filter values.

2.3.4 FlyVisNet for artificial agents

We aimed to unveil the potential of using this feature selectivity emergence for autonomous flight control. Once the network had been trained, we designed a virtual 3D world environment to test an artificial agent that runs FlyVisNet on board (see Figure 2.4 A). We prepared an arena layout for a visually guided flight test that had three attached patterns on its walls, a square, a rectangle, and a circle (see Figure 2.4 A) (see Methods 2.5.3). To perform the flight test we retrained FlyVisNet by including as label the X centroid coordinate of the objects to be trained through regression. We designed a flight algorithm that provides commands to the drone depending on the detected class by the network. In this algorithm, the expected sequence is to fly straight while using the predicted X coordinate to keep the object in the center of the field of view (see Figure 2.4 B). When the object is too close will be detected as loom and the drone will turn to the left if the previously detected class is spot. It will turn to the right if the previously detected class is bar. If a loom is detected and the number of detections is reached it will land (see Figure 2.4 B). We

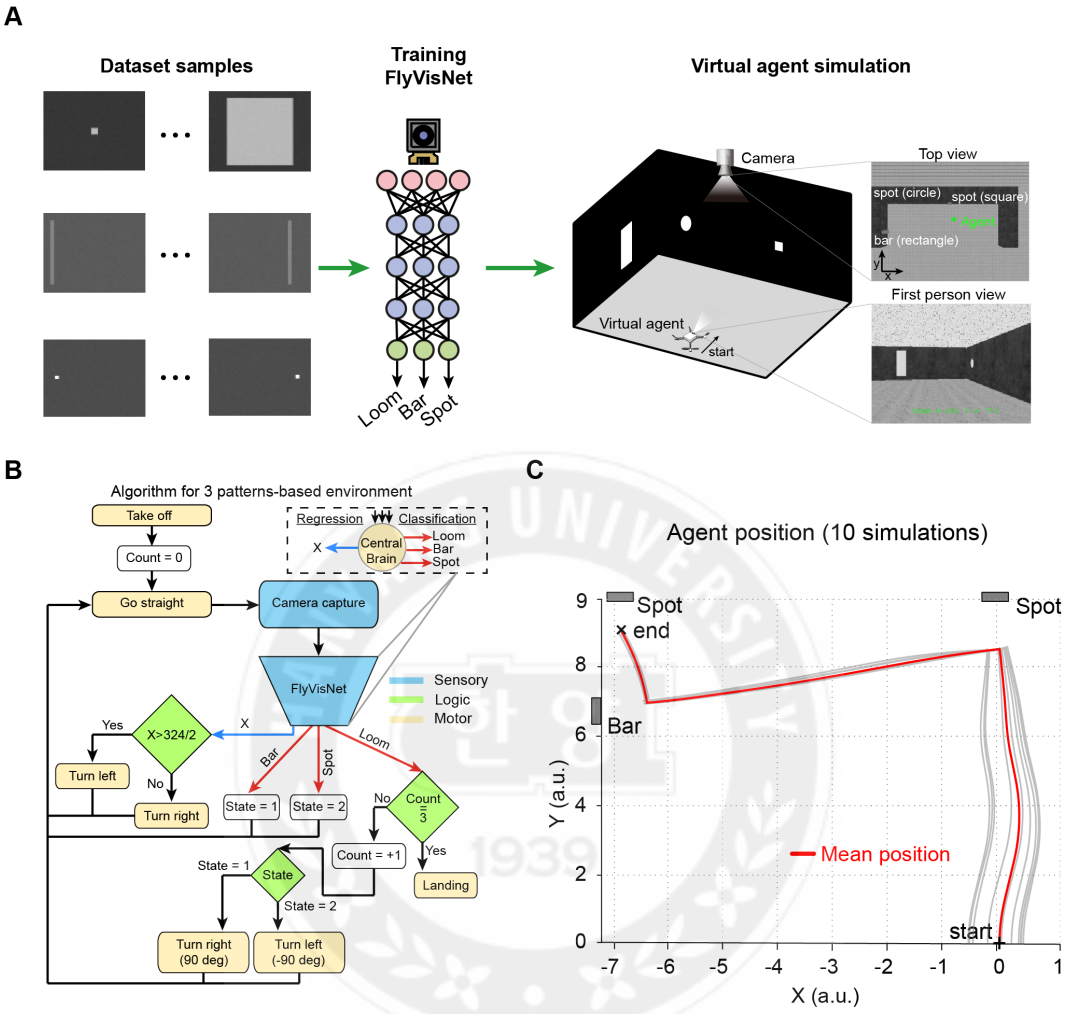


Figure 2.4: A virtual agent achieved an autonomous navigation by using FlyVisNet's inference in a 3D world environment. (A). After trained on the pattern dataset, the network is tested by simulating an artificial agent in a virtual environment. The virtual environment designed for visually guided flight simulation with a camera capturing the agent's position from the top. (B). The algorithm for the 3 patterns-based environment flight where the agent captures images from its subjective perspective and makes decisions depending on the FlyVisNet's inference. The X coordinate of the object's centroid is added as a regression output of the network. Sensory, logic, and motor-related operations are represented by blue, green, and yellow colors respectively. (C). The trajectories of the agent during 10 simulations at random starting points, and the mean trajectory.

placed the agent in the environment and by using FlyVisNet, obtained the centroid coordinate of the object X and the class from the first-person view of the scene (see Figure 2.4 A, first person view). The agent navigated through the environment based on the predictions of the network (see Figure 2.4 A, top view). We simulated 10 times, where the agent started at a random position in front of the spot (square) and moved forward by keeping the object in front of the visual field. When it was close enough to the spot to detect loom, it turned to the left and moved forward keeping the bar (rectangle) in the center, until the next loom was detected. Then, it turned right and moved forward until the loom was detected when close to the spot (circle) and stopped (see Figure 2.4 C and Movie 2.S1).

We prepared a second layout that had 2 attached rectangles on opposite walls (see Figure 2.S11 A) and was thought to perform a flight based on the Buridan's paradigm [81]. In this paradigm, a fruit fly walks persistently back and forth between two inaccessible landmarks. The fly is attracted by the landmark and when it is close enough senses the landmark as a looming object, so it turns away getting attracted by the opposite landmark. We designed a flight algorithm accordingly, where the drone will fly straight controlling the X coordinate as before. It will fly until loom is detected, then turn away 180 deg to go towards the opposite rectangle and repeat (see Figure 2.S11 B). We placed the agent in this arena environment (see Figure 2.S11 A, first-person view and top view). We simulated 10 times, where the agent started at a random position in front of the right bar and moved forward maintaining the bar in front of the visual field until loom detection. Then it turned away 180 degrees and moved towards the opposite bar repeating this loop several times (see Figure 2.S11 C and Movie 2.S2). Additionally, we found how the artificial

neuron activations showed greater activity to the preferred visual objects in real-time during the simulation (Movie 2.S3).

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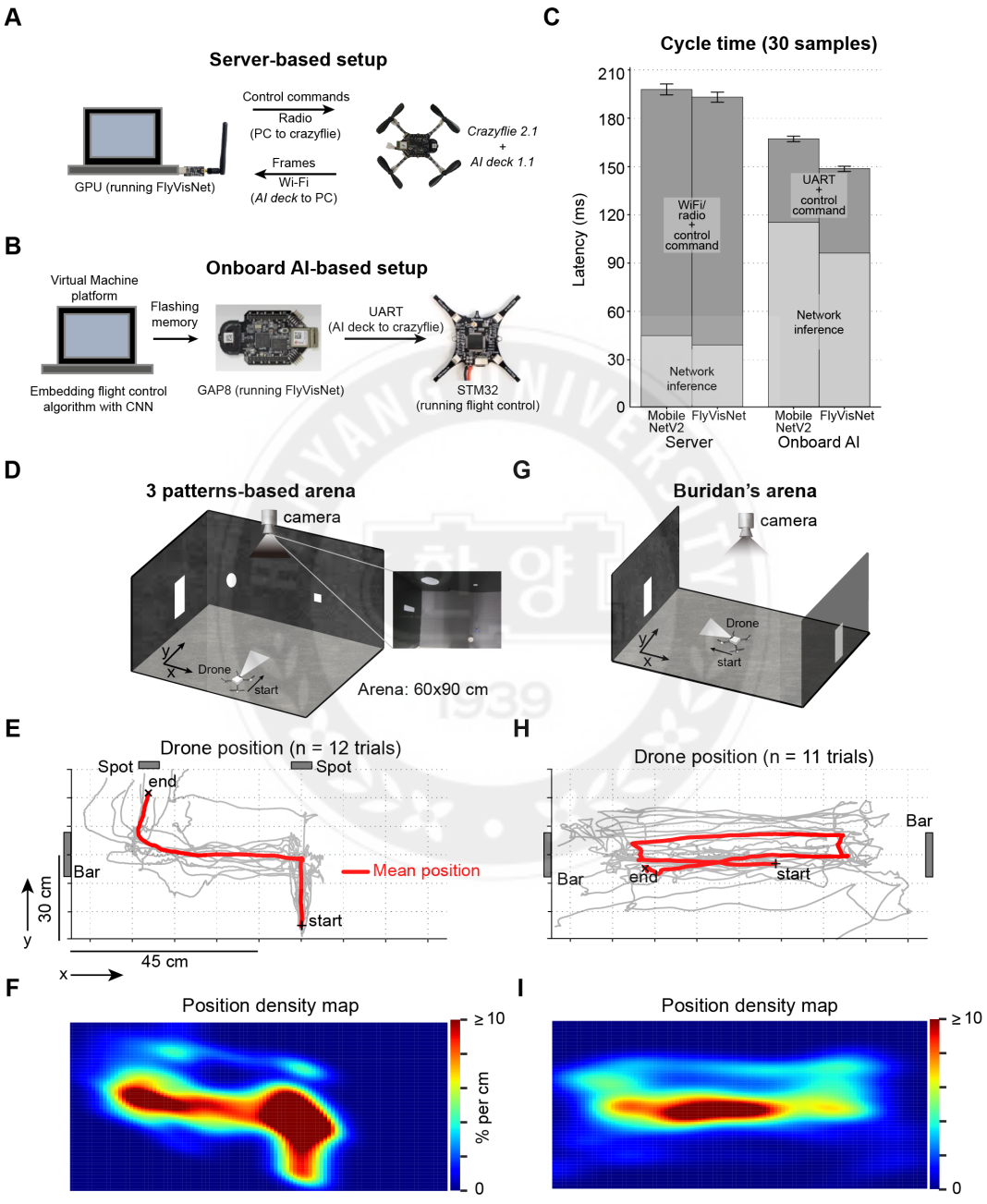

A
Server-based setup
Control commands
Radio
(PC to crazyflie)
Frames
Wi-Fi
(AI deck to PC)
GPU (running FlyVisNet)
Crazyflie 2.1
+
AI deck 1.1
B
Onboard AI-based setup
Virtual Machine platform
Flashing memory
UART
(AI deck to crazyflie)
Embedding flight control algorithm with CNN
GAP8 (running FlyVisNet)
STM32
(running flight control)
C
Cycle time (30 samples)
Latency (ms)
210
180
150
120
90
60
30
0
WiFi/ radio + control command
Network inference
UART + control command
Network inference
Mobile NetV2
FlyVisNet
Server
Onboard AI
D
3 patterns-based arena
camera
Drone
start
Arena: 60x90 cm
G
Buridan's arena
camera
Drone
start
E
Drone position (n = 12 trials)
Spot
Spot
end
Bar
Mean position
start
30 cm
45 cm
H
Drone position (n = 11 trials)
Bar
Bar
end
start
F
Position density map
≥ 10
% per cm
0
I
Position density map
≥ 10
0

Figure 2.5: FlyVisNet onboard allowed navigation of a drone in two different arena layouts. (A). A server-based setup refers to sending frames from the AI deck in the drone by Wi-Fi to the PC for network inference in a GPU and sending back the corresponding commands by radio. (B). An onboard AI-based setup refers to quantizing the network in a virtual machine platform that allows flashing both the network in the GAP8 (AI deck) processor and the flight control application code in the STM32 (crazyflie). The result of the inference is sent via UART internally from the GAP8 to the STM32, where the flight commands are executed. (C). Bar plot comparison between FlyVisNet and MobileNetV2 showing the mean cycle time for 30 samples for each setup, which is the time period between capturing an image and executing a control command. Errors bars represent the standard deviation of the samples. (D). The 3 patterns-based arena where the camera on top was used to record each trial and to track the drone's position. (E). The position of the drone during the test for 12 trials and the mean position in red color. (F). Heat map of most repeated positions during trials. (G). Same as in D for a Buridan's paradigm flight test layout of the arena. (H). Same as in E for 11 trials for the Buridan's paradigm flight. (I). Same as in F for the Buridan's paradigm flight.

Once we had tested FlyVisNet with the virtual agent, we repeated the process with the nano drone. We followed an end-to-end process to deploy the network on the drone and perform an autonomous flight. We chose Bitcraze's crazyflie 2.1[1] drone (equipped with an STM32 processor), which includes as an accessory an AI-based electronic module, the AI deck 1.1[2]. This module is equipped with a camera and a GAP8[3] processor, which allows to execute embedded CNNs. To control the drone during flight we tested two strategies, a server-based and an onboard AI-based strategy. In the server-based strategy, the AI deck module sends the captured frames to a computer via WiFi where the network runs the inference. Depending on the inference results different motor commands are sent via radio from the computer (by using the

1. https://www.bitcraze.io/products/crazyflie-2-1/

2. https://www.bitcraze.io/products/ai-deck/

3. https://greenwaves-technologies.com/gap8_mcu_ai/

crazyradio PA 2.4 GHz USB dongle) to the crazyflie drone (see Figure 2.5 A). For the onboard AI-based strategy (see Methods 2.5.4), we quantized the trained network on a virtual machine platform to a fixed 8-bit resolution and developed a control flight algorithm application code. We flashed the network application code in the memory of the GAP8 processor and the flight control algorithm in the STM32 (see Figure 2.5 B). In this strategy, the AI deck module captures the image, the GAP8 runs the inference using the quantized version of FlyVisNet, and the result is sent to the crazyflie via UART. Depending on the inference result specific motor commands are directly computed in the drone's STM32 (see Figure 2.5 B). In the GAP8, we set an internal clock frequency of 250MHz for the Fabric Control (FC) and 175MHz for the Cluster (CL). We compared both strategies in terms of cycle time, that is the time needed between capturing a frame and executing a motor command. As a reference, we also compared the cycle time with respect to MovileNetV2. The onboard strategy showed faster cycle time than the server-based, being FlyVisNet faster than MovileNetV2 in both strategies (see Figure 2.5 C). In the server-based approach, the cycle time was about 195ms for FlyVisNet, around 40ms for the inference, and 155ms for communications and executing control commands. The difference with respect to the MobileNetV2's inference was just about 5ms (see Figure 2.5 C). Given that the inference was performed in the computer's GPU, this difference was significantly small. The cycle time of FlyVisNet in the onboard setting was near 150ms in total. The network inference took about 99ms and the UART transmission along with the flight control algorithm execution about 51ms. The difference with respect to the MobileNetV2 inference was approximately 16ms (see Figure 2.5 C). The significant cycle time difference between both strategies highlighted the value of

a fully onboard system rather than relying on external communications with a computer that would otherwise delay the response. Also in the onboard setting the size and memory burden of the networks had a higher impact, becoming greater the inference time difference between FlyVisNet and MobileNetV2.

To perform a flight test with the drone, we built two high-contrast arenas by mimicking the virtual environments that we created previously (see Methods 2.5.5). We performed the drone flight via the on-board strategy. In the layout for the 3 patterns-based flight (see Figure 2.5 D), for 12 trials we placed the drone in the same initial position. Similarly to the simulated agent, the drone took off and approached the spot (square), when loom was detected turned left, flew towards the bar, and upon detection turned right. It landed when loom was detected while approaching to the spot (circle) (see Figure 2.5 E, and Movie 2.S4). The position density map showed that the trajectory was consistently within the same area (see Figure 2.5 F), validating the FlyVisNet utility for a visually guided flight. In the second layout (see Figure 2.5 G) for 11 trials we placed the drone again in the same initial position. The drone took off, flew towards the bar until loom was detected, then turned 180 degrees, flew towards the opposite bar until loom was detected, and turned away again repeating this loop several times (see Figure 2.5 H, and Movie 2.S5). The mean trace showed a displacement in the y-axis, which was due to the fact that when turning 180 degrees away from the bar, the wind generated by the rotors pushed the drone laterally for being close enough to the walls (see Figure 2.5 H). The position density map showed that the drone was consistently in the center of the arena (see Figure 2.5 I). This test proved that FlyVisNet allowed the drone to reproduce the Buridan's paradigm [81] as in the case of the virtual agent.

2.4 Discussion

The utility of mimicking the *Drosophila*'s visual system for artificial neural networks has been highlighted, as reported in the case of a classification task of conspecifics through a reduced part of the visual system [12]. Through the release of the recent connectomes [1, 2], the implementations can be put into cell level. This was shown in a study where a hexagonal connectome-based neural network also captured the direction selectivity of T4/T5 neurons and the spatiotemporal receptive field of many of the neurons of the visual system [14]. However, there has not been a suitable applicability in the use of artificial agents or autonomous systems which have power and memory limitations. In fact, insect vision-inspired algorithms have been applied in robotics for flight control or navigation tasks based on optic flow detection of the fly via elementary motion detectors [66, 67, 82]. Also, an implementation of a looming detecting neuron algorithm from the locust allowed collision avoidance of a mobile robot [83, 84]. Specifically, *Drosophila*'s LPLC2 neuron mechanism for looming detection has been implemented in silico showing its potential utility [10, 11, 85]. Still, these implementations were focused on a specific neural pathway for a specific function that allowed collision avoidance tasks. In this work, we have implemented different neural visual pathways that perform different functions simultaneously. We have not only implemented the ON/OFF pathways that end up in T4/T5 neurons but also the deeper LC neurons in the Lobula layer that can identify objects of different shapes. We have discovered that the neurons of our artificial neural network while being trained exclusively for classification, naturally exhibited emergence of direction selectivity to moving objects in T4/T5. It also emerged selectivity to small objects

in LC11 and long objects in LC15 as well as LPLC2 exhibited selectivity to expanding objects. This resulted in a capital characteristic that permitted to perform a visually-guided flight of a virtual agent as well as a nano drone.

Our network also showed higher accuracy in classification tasks of simple and complex objects while being memory efficient. It also allowed to control the nano drone within a faster cycle time than the representative embedded-oriented architecture MobileNetV2. The onboard AI-based strategy demonstrated faster flight control of the drone without additional external communications. Furthermore, our embedding level achieved here provides an alternative approach to other extensively used and power-demanding hardware such as Raspberry Pi. In fact, the GAP8 processor at a maximum internal clock frequency of 250MHz for the FC consumes 10mW[4] along with 175MHz for the CL can consume a total power of about 100mW [74], whereas a basic Pi model such as Raspberry Pi 2 B consumes 2.3W[5]. Therefore, we provided a fully autonomous flight-controlled drone that can operate remotely at very low power and memory.

However, our system has shown to be highly sensitive to changes in the luminance level, making sometimes wrong pattern recognitions in brighter spaces. In addition, fast maneuvers should be also avoided, because the captured image should be stable to allow a proper recognition. Crazyflie equipped with the AI deck showed to consume the drone battery quite fast, which led to short autonomous flight time. The drone is small enough to be also sensitive to weak gusts of wind, which led to a loss of stability.

4. https://greenwaves-technologies.com/wp-content/uploads/2021/04/Product-Brief-GAP8-V1_9.pdf

5. https://www.raspberrypi.com/documentation/computers/raspberry-pi.html#typical-power-requirements

Given the pros and cons of our flight system, it could be useful in applications with power and size constraints, such as monitoring and management of areas of difficult access, or those areas dangerous for people due to pollution, toxicity, etc. Additionally, it could be also applied in surveillance tasks, and as a remote security system.

Still, FlyVisNet has a partial portion of neural circuits implemented, which opens new future lines of work to expand the design by including additional functionalities associated with the visual system of *Drosophila*. Among these functionalities, we find interesting spatial navigation, color brightness learning, and tracking targets. This could solve one of the problems found regarding the luminance sensitivity.

2.5 Materials and methods

2.5.1 FlyVisNet model

The network processes the images in batches of 25 frames. The equations describing the visual processing flow through the FlyVisNet model, $y = FlyVisNet(x, w)$, are as follows:

$$R16 = x \otimes w_{R16} + b_{R16}$$

$$L1 = MaxPooling2D(R16 \otimes w_{L1} + b_{L1})$$

$$L2 = MaxPooling2D(R16 \otimes w_{L2} + b_{L2})$$

$$L3 = MaxPooling2D(R16 \otimes w_{L3} + b_{L3})$$

$$Mi1 = MaxPooling2D(L1 \otimes w_{Mi1} + b_{Mi1})$$

$$Tm3 = MaxPooling2D(L1 \otimes w_{Tm3} + b_{Tm3})$$

$$C3 = Roll(MaxPooling2D(Concat(L1, L3) \otimes w_{C3} + b_{C3}), shift = 1, axis = 0)$$

$$Mi4 = Roll(MaxPooling2D(Concat(L1, L3) \otimes w_{Mi4} + b_{Mi4}), shift = 1, axis = 0)$$

$$Tm1 = MaxPooling2D(L2 \otimes w_{Tm1} + b_{Tm1})$$

$$Tm2 = MaxPooling2D(L2 \otimes w_{Tm2} + b_{Tm2})$$

$$Tm4 = MaxPooling2D(L2 \otimes w_{Tm4} + b_{Tm4})$$

$$Mi9 = Roll(MaxPooling2D(L3 \otimes w_{Mi9} + b_{Mi9}), shift = 1, axis = 0)$$

$$Tm9 = Roll(MaxPooling2D(L3 \otimes w_{Tm9} + b_{Tm9}), shift = 1, axis = 0)$$

$$CT1 = Roll(MaxPooling2D(Concat(L3, L2) \otimes w_{CT1} + b_{CT1}), shift = 1, axis = 0)$$

$$TmY9 = MaxPooling2D(Concat(Tm2, Mi1, C3, Mi4, Tm1, Mi9) \otimes w_{TmY9} + b_{TmY9})$$

$$TmY4 = MaxPooling2D(Concat(Tm2, Mi1, C3, Mi4, Tm1, Mi9) \otimes w_{TmY4} + b_{TmY4})$$

$$TmY5 = MaxPooling2D(Concat(Mi4, Mi9) \otimes w_{TmY5} + b_{TmY5})$$

$$T2 = MaxPooling2D(Concat(L1, Tm2, Mi1, Tm3, C3, Mi4, Tm9, CT1) \otimes w_{T2} + b_{T2})$$

$$T3 = MaxPooling2D(Concat(L2, Tm1, Tm2, Tm3, Mi1, Mi9, C3, Tm9, CT1) \otimes w_{T3} + b_{T3})$$

$$T4 = MaxPooling2D(Relu(Concat(Mi1, Tm3, C3, Mi4, Mi9) \otimes w_{T4} + b_{T4}))$$

$$T5 = MaxPooling2D(Relu(Concat(Tm1, Tm2, Tm4, Tm9, CT1) \otimes w_{T5} + b_{T5}))$$

$$Li = MaxPooling2D(Concat(T2, T3) \otimes w_{Li} + b_{Li})$$

$$LPi = MaxPooling2D(Concat(T4, T5) \otimes w_{LPi} + b_{LPi})$$

$$LC11 = MaxPooling2D(Relu(Concat(T2, T3) \otimes w_{LC11} + b_{LC11}))$$

$$LC15 = MaxPooling2D(Relu(Concat(Li, TmY9, TmY4) \otimes w_{LC15} + b_{LC15}))$$

$$LPLC2 = MaxPooling2D(Relu(Concat(T4, T5, LPi, TmY5) \otimes w_{LPLC2} + b_{LPLC2}))$$

$$CB = Flatten(Concat(LC11, LC15, LPLC2))$$

$$CB = Relu(CB \cdot w_{128} + b_{CB}))$$

$$y = Softmax(CB \cdot w_3 + b_{out}))$$

where $\otimes$ represents convolution operation, x is the batch of input images of shape $(batchsize, height, width, channel)$, w is the kernel of each convolution layer, b is the bias, and y is the output vector containing the probability for each class.

2.5.2 FlyVisNet training

We used the Python library TensorFlow to train the network, which allows to set up different configurations of loss functions, optimizers, layer activations, etc. (see Table 2.2). Once trained, we converted FlyVisNet model graph to a TensorFlow Lite model, which allows it to be quantized.

The training algorithm we used is as follows:

Table 2.2: Training settings

Settings	**Selected option**
Loss function	Categorical cross-entropy
Optimizer	Adam[86]
Batch size	25
Learning rate	$1 \cdot 10^{-3}$
Epochs	100
Conv. filter weights initialization exc./inhib.	U[pos. low, pos. high]/U[neg. low, neg. high]
FC weights initialization	glorot uniform
Activations (Conv./FC/Output)	linear, relu/relu/softmax

Algorithm 1 Training algorithm

Being: Training set (X, Y), validation set $(\hat{X}, \hat{Y})$, neural network $FlyVisNet(x, w) \mapsto y$, loss $L(y, \hat{(y)}) \mapsto \mathbb{R}$, accuracy $A(y, \hat{y}) \mapsto \mathbb{R}$, mini-batch of size k
Initialize weights w
$a_{best} \coloneqq 0$
for $iteration = 1, M$ **do**
  for $i = 1, \lfloor |X|/k \rfloor$ **do**
    (x, y) is a random batch of size k from (X, Y)
    $w \coloneqq w - \alpha \frac{\partial L(y, FlyVisNet(x,w))}{\partial w} \nabla$ stochastic gradient descent
  end for
  $a_e \coloneqq \frac{1}{|\hat{X}|} \sum_{(x,y)\in(\hat{X},\hat{Y})} A(y, FlyVisNet(x, w))$ validation accuracy
  if $a_e < a_{best}$ **then**
    $a_{best} \coloneqq a_e$
    $w_{best} \coloneqq w$
  end if
end for
return w_{best}

where a is the accuracy, L is the categorical cross-entropy loss, $\alpha = 1 \cdot 10^{-3}$ is the learning rate, and $M = 100$ is the number of training iterations. The accuracy function is given by:

$$A(y, FlyVisNet(x, w)) = \begin{cases} 1, & \text{if } y = FlyVisNet(x, w) \\ 0, & \text{otherwise} \end{cases}$$

2.5.3 Building an environment for the virtual agent

To design the 3D virtual world environment we used OpenGL, a cross-platform graphics API that allows the rendering of 2D and 3D vector graphics in real-time. Its corresponding Python library provides bindings to the API, which allowed us to generate a scene, where we built the layouts for the 3 patterns and Buridan's paradigm flight, placed the objects, and applied real-

istic textures to the walls, the ceiling, and the floor. We simulated the virtual agent in the two layouts and captured the scene from 2 different view cameras, one on the top, and the other for the first-person view. The position x and y coordinates of the agent were also recorded in real-time during the simulation.

2.5.4 Embedding process

Once we had our trained FlyVisNet TensorFlow Lite model, we placed it in the virtual machine platform equipped with the necessary deployment tools (see Figure 2.5 B) provided by Bitcraze. In the first place, we quantized the FlyVisNet TensorFlow Lite model to a fixed 8-bit, which consists of reducing the resolution of the network parameters from 32 to 8 bits. This allows to reduce the used memory, and it is a necessary step for embedding in micro-controller units (MCU), such as the GAP 8 processor. To do so, we used the GAPflow tools provided by GreenWaves Technologies, which are included in the virtual machine. Through this process, we obtained the quantized CNN model into header C files, which can be included in the application code that was flashed and run in the AI deck's GAP8 unit. In the second place, we included the algorithm for autonomous flight in the application code that was flashed and run in the crazyflie's STM32 unit. Both units are connected via UART and run each respective application code simultaneously.

2.5.5 Building an environment for the drone

In the case of the environment for the drone, we mimicked the virtual one and built two high-contrast arenas one for the 3 patterns-based and the other for a Buridan's paradigm flight of 60x90 cm. We used black boards of

cork-like material as background, attached white paper patterns to their walls, and covered the floor with a shock-absorbing elastic material to avoid damage to the drone in case of crashing. The 3 patterns-based arena had attached on its walls, a square, a rectangle, and a circle (see Figure 2.5 D). The arena for the Buridan's paradigm flight test had 2 attached rectangles on opposite walls (see Figure 2.5 G). To obtain the position traces of the drone for each flight trial, we placed a camera on top of the arena (see Figure 2.5 D, G). We recorded each trial and preprocessed the output videos to obtain the real-time position of the drone. Specifically, we applied the AdaBoost tracking algorithm [87] from the visual processing Python library OpenCV to obtain the x and y coordinates.

Code availability. The essential code used to generate the primary results and conduct the simulations for this study as well as to perform the deployment is available in our GitHub repository (https://github.com/A-Canelo/Insect-inspired-image-recognition-CNN).

2.6 Supplementary information

Movie S1. A movie showing the simulated 3 patterns-based flight in the 3D world. The left part of the frame shows the first-person view of the agent and the right part the top view.

Movie S2. A movie showing the simulated Buridan's paradigm flight in the 3D world. The left part of the frame shows the first-person view of the agent and the right part the top view.

Movie S3. A movie of the real-time layer activations of the FlyVisNet during the simulation in the 3D world. It also shows the labels applied to the first-person view.

Movie S4. A movie that shows the flight test of the drone from a top view in the 3 patterns-based arena.

Movie S5. A movie that shows the flight test of the drone from a top view in the layout of the arena for the Buridan's paradigm flight.

Table 2.S1: FlyVisNet versions comparison on pattern dataset

Architecture	**Top accuracy (%)**	**Parameters**	**Memory (kB)**	**Resolution**
Original	**98.53**	116,626	1,660	244x324 (drone)
R1-R6 pruned	97.66	116,599	1,652	
T4 pruned	92.33	116,546	1,649	
T5 pruned	92.66	116,529	1,642	
80% sparsity	88.33	23,325	623	
Original	**92.80**	78,226	1,200	20x40 (fly)
R1-R6 pruned	91.99	78,199	1,192	
T4 pruned	89.23	78,146	1,188	
T5 pruned	89.66	78,130	1,181	
80% sparsity	82.66	15,645	418	

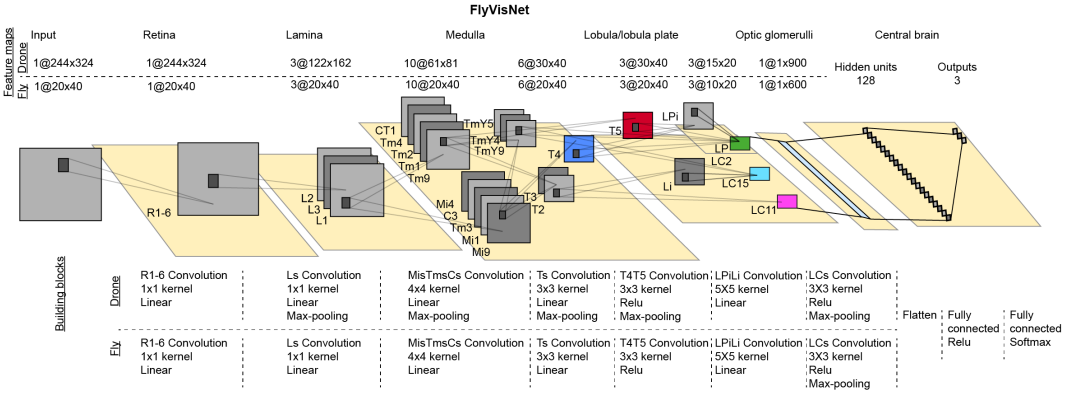


Figure 2.S1: FlyVisNet extended architecture diagram.

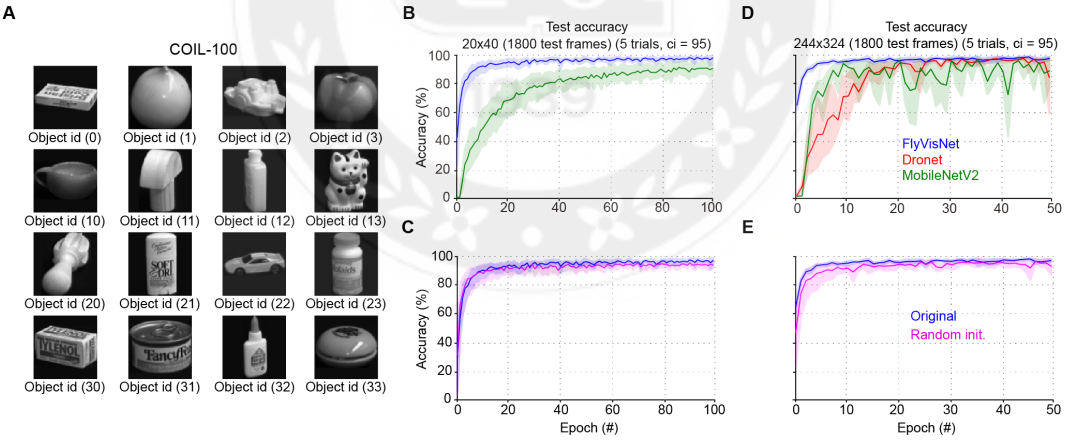


Figure 2.S2: FlyVisNet trained for classification showed higher top accuracy on COIL-100 dataset than other networks of similar size. (A). The COIL-100 dataset is made up of 100 different objects filmed on a rotating table and captured from different angles. (B). Same as in Figure 2.2 B for COIL-100. (C). Same as in Figure 2.2 C for COIL-100. (D). Same as in Figure 2.2 D for COIL-100. (E). Same as in Figure 2.2 E for COIL-100.

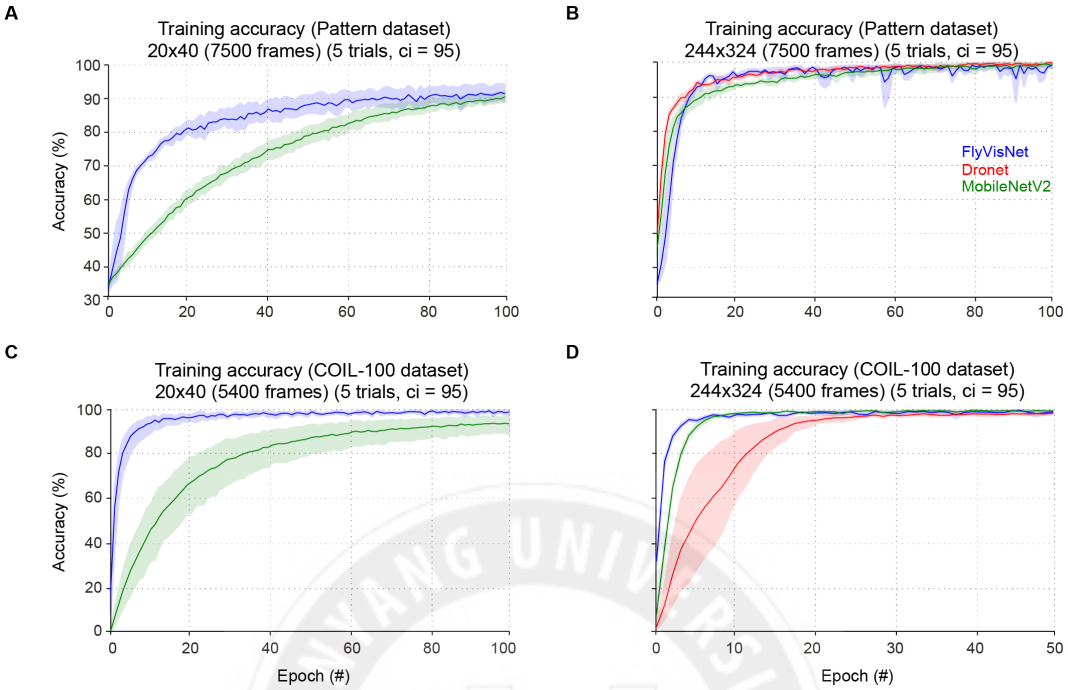


Figure 2.S3: Training performance comparison on pattern dataset and COIL-100. (A). FlyVisNet and MobileNetV2 training accuracies on the pattern dataset for frames of size 20x40. (B). Same as in A for size 244x324, including Dronet. (C). Same as in A for COIL-100. (D). Same as in C for size 244x324, including Dronet.

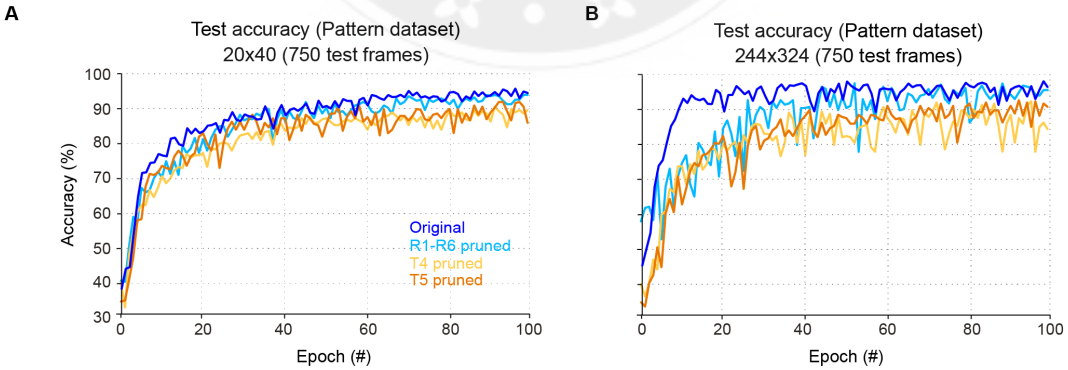


Figure 2.S4: Testing performance comparison between pruned versions of FlyVisNet on pattern dataset. (A). Original, R1-R6 pruned, T4 pruned, and T5 pruned testing accuracies on the pattern dataset for frames of size 20x40. (B). Same as in A for size 244x324.

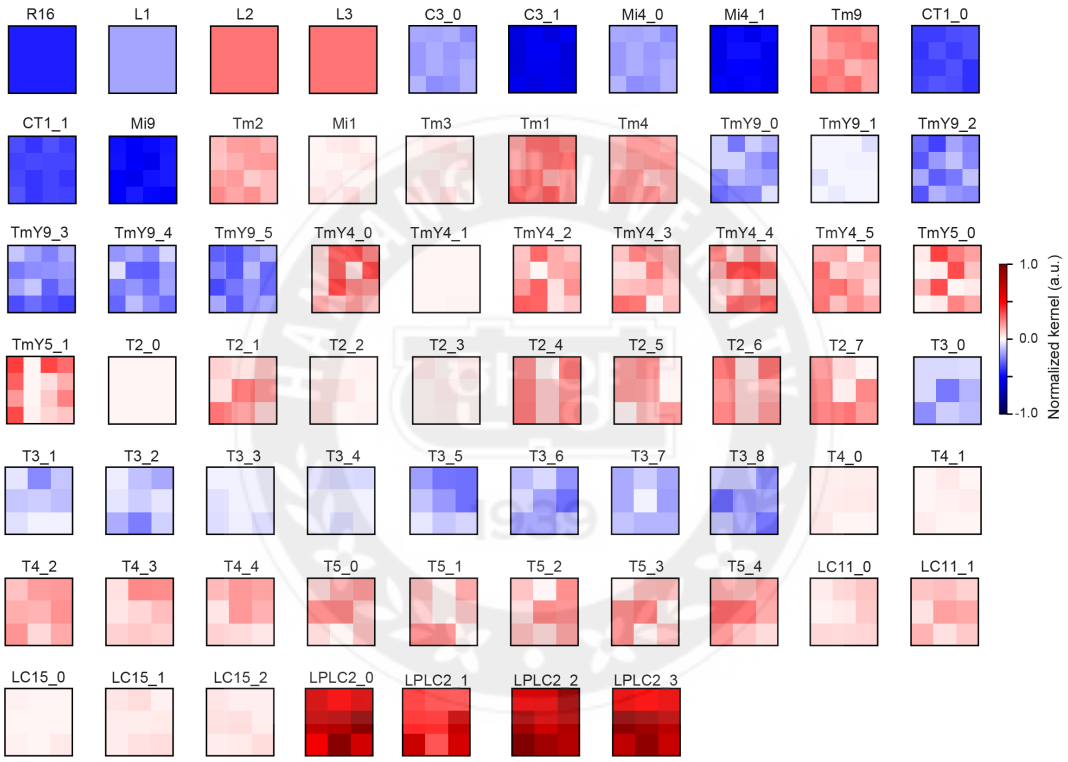


Figure 2.S5: Convolution kernels learned via the pattern dataset.

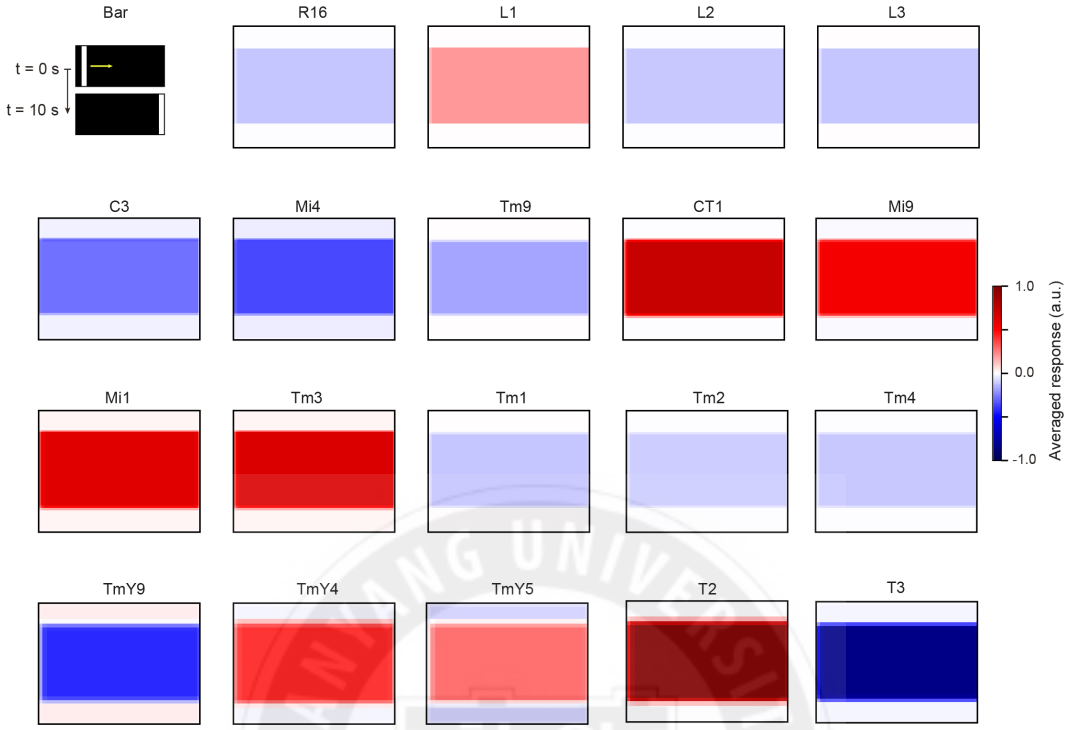


Figure 2.S6: 2D averaged responses to a moving bar showed the polarity predominant values in the feature maps in the retina, lamina, and medulla layers.

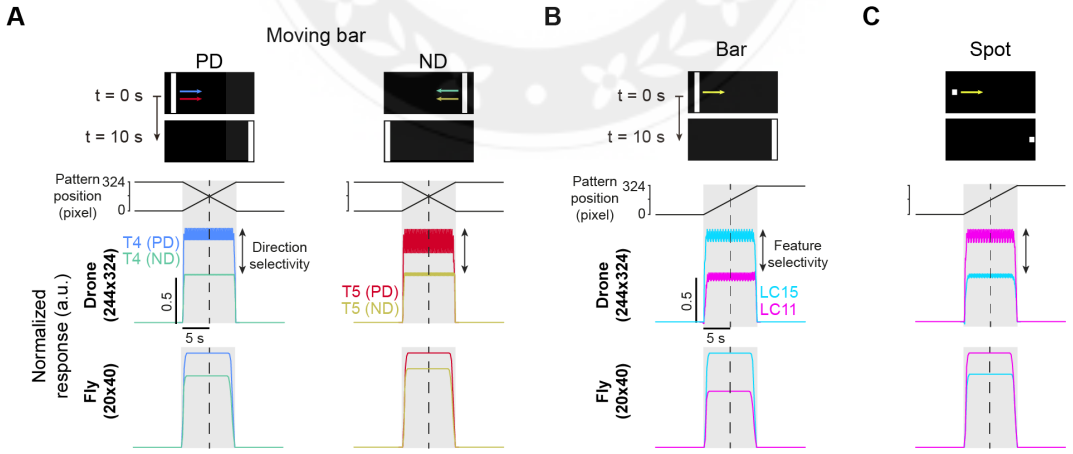


Figure 2.S7: Max pooling operation produces oscillations in the neural responses of the artificial neurons. (A). Same as in Figure 2.3 C but for the comparison between drone and fly versions. (B), (C). Same as in Figure 2.3 F and G but for the comparison between drone and fly versions.

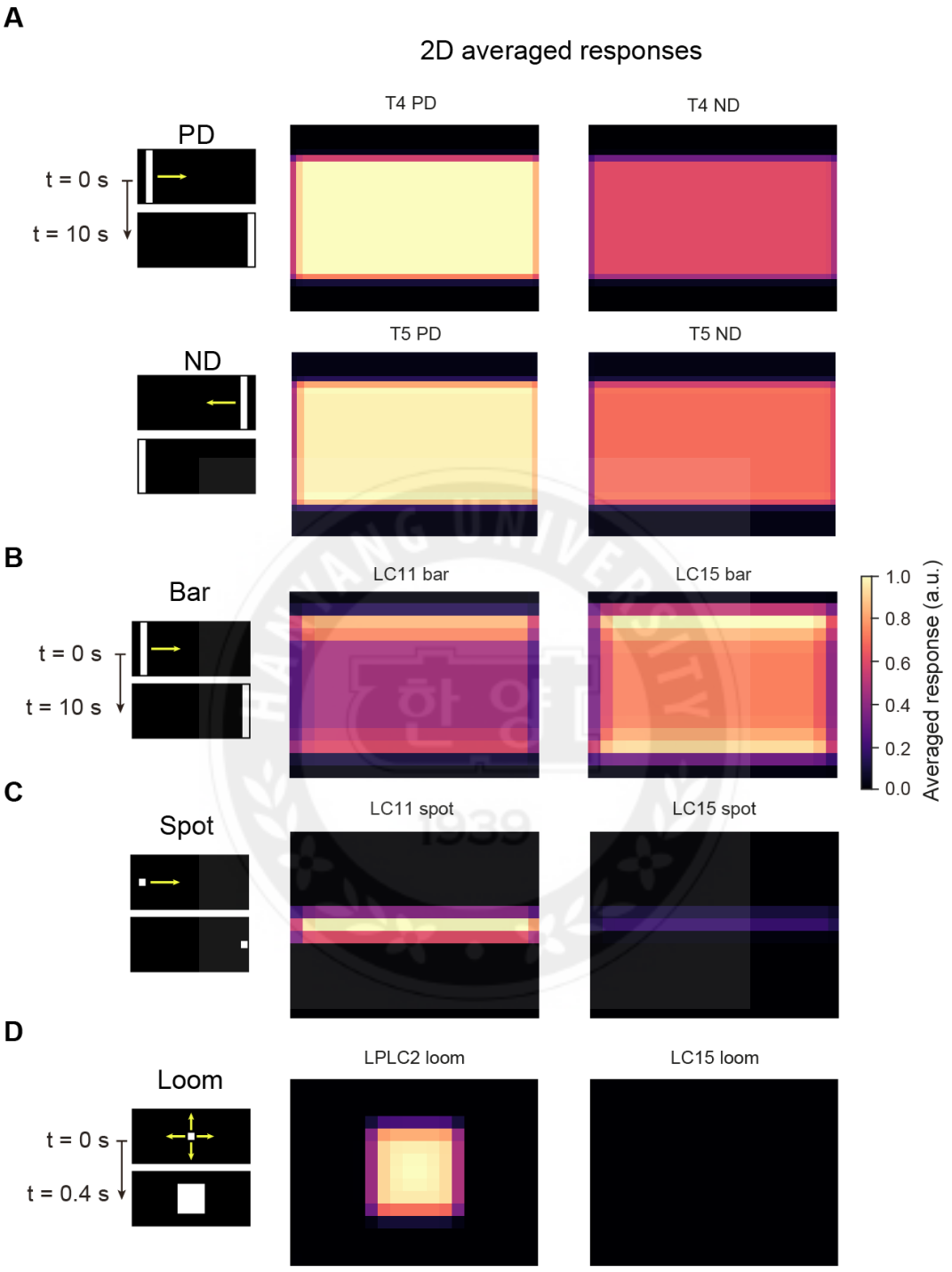


Figure 2.S8: 2D averaged feature map responses showed the preferred visual feature. (A). 2D time averaged activations of T4/T5 neurons to a moving bar from left to right (PD), and from right to left (ND). (B). 2D time averaged activations of LC11, and LC15 neurons to a moving bar from left to right. (C). Same as in B to a moving spot. (D). 2D time averaged activations of LPLC2, and LC15 neurons to a loom.

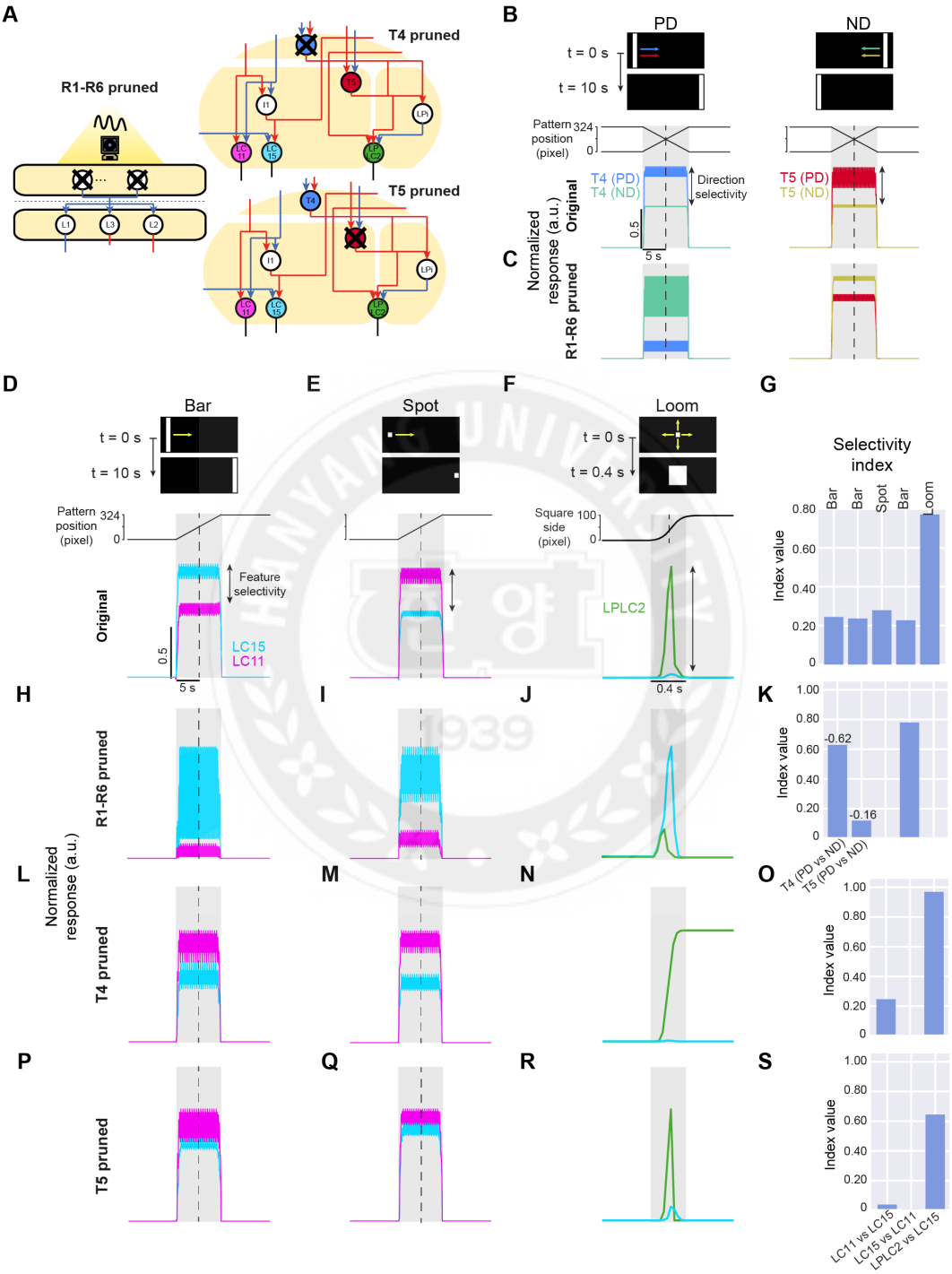

A
R1-R6 pruned
T4 pruned
T5 pruned
B
PD
ND
t = 0 s
t = 10 s
Pattern position (pixel)
324
0
T4 (PD)
T4 (ND)
T5 (PD)
T5 (ND)
Direction selectivity
Original
0.5
5 s
C
Normalized response (a.u.)
R1-R6 pruned
D
Bar
E
Spot
F
Loom
t = 0.4 s
Square side (pixel)
100
Feature selectivity
LC15
LC11
LPLC2
0.4 s
G
Selectivity index
Index value
Bar
Spot
Loom
H
I
J
K
-0.62
-0.16
T4 (PD vs ND)
T5 (PD vs ND)
L
M
N
O
T4 pruned
P
Q
R
S
T5 pruned
LC11 vs LC15
LC15 vs LC11
LPLC2 vs LC15

Figure 2.S9: FlyVisNet's artificial neurons lose feature selectivity when pruning different neurons. (A). Targeted neurons to be pruned R1-R6, T4, and T5. (B). Neural response comparison of T4, and T5 to a bright moving bar in the PD and ND. Responses are normalized for better comparison. (C). Same as in B but when are R1-R6 pruned. (D). Neural response comparison of LC11, LC15 to a bright moving bar. (E). Same as in D for a bright moving spot (F). LPLC2 to LC15 comparison to a looming stimulus. (G). Selectivity index comparison. The index is computed for each compared case in B, D, E, and F. (H), (I), (J), (K). Same as in D, E, F, and G but when R1-R6 are pruned. (L), (M), (N), (O). Same as in D, E, F, and G but when T4 is pruned. (P), (Q), (R), (S). Same as in D, E, F, and G but when T5 is pruned.

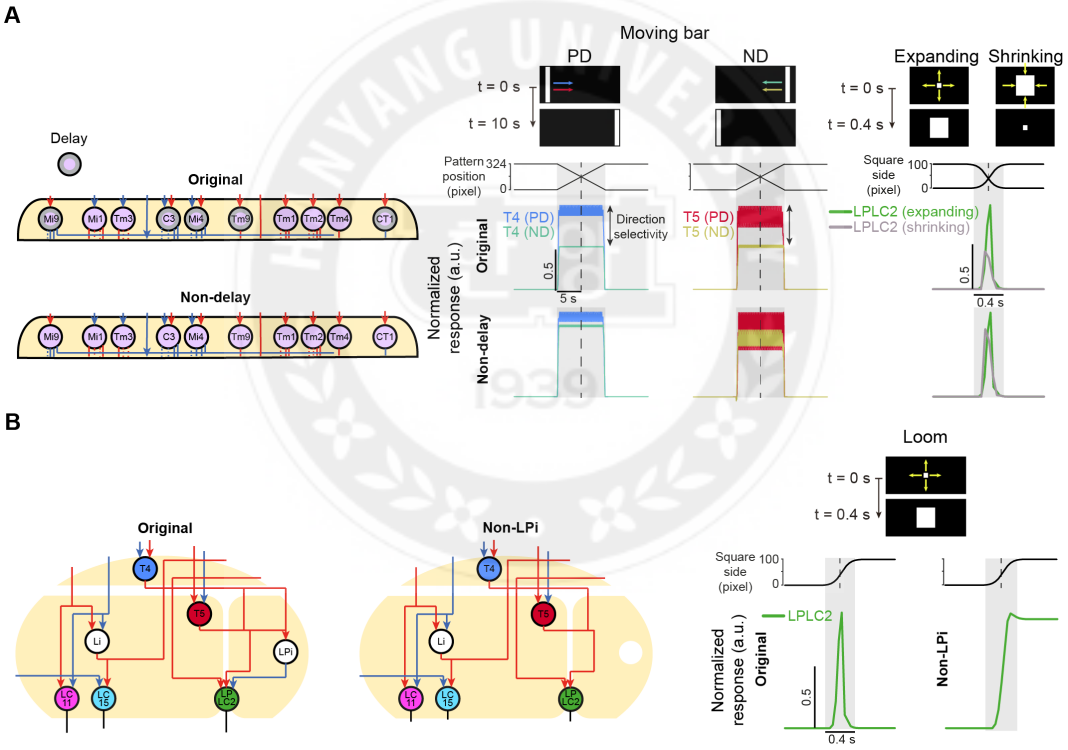


Figure 2.S10: Implementing delay processing improved direction selectivity in T4/T5 neurons and inhibition from LPi allowed looming detection in LPLC2. (A). The neurons with one frame delay of the original architecture and the implementation by removing the delay. Same as in Figure 2.3 C but for the Non-delay implementation. Also, LPLC2 selectivity is improved with the delay for expanding and shrinking square. (B). The original architecture and the implementation by removing LPi neuron. Same as in Figure 2.3 H but for the LPi elimination.

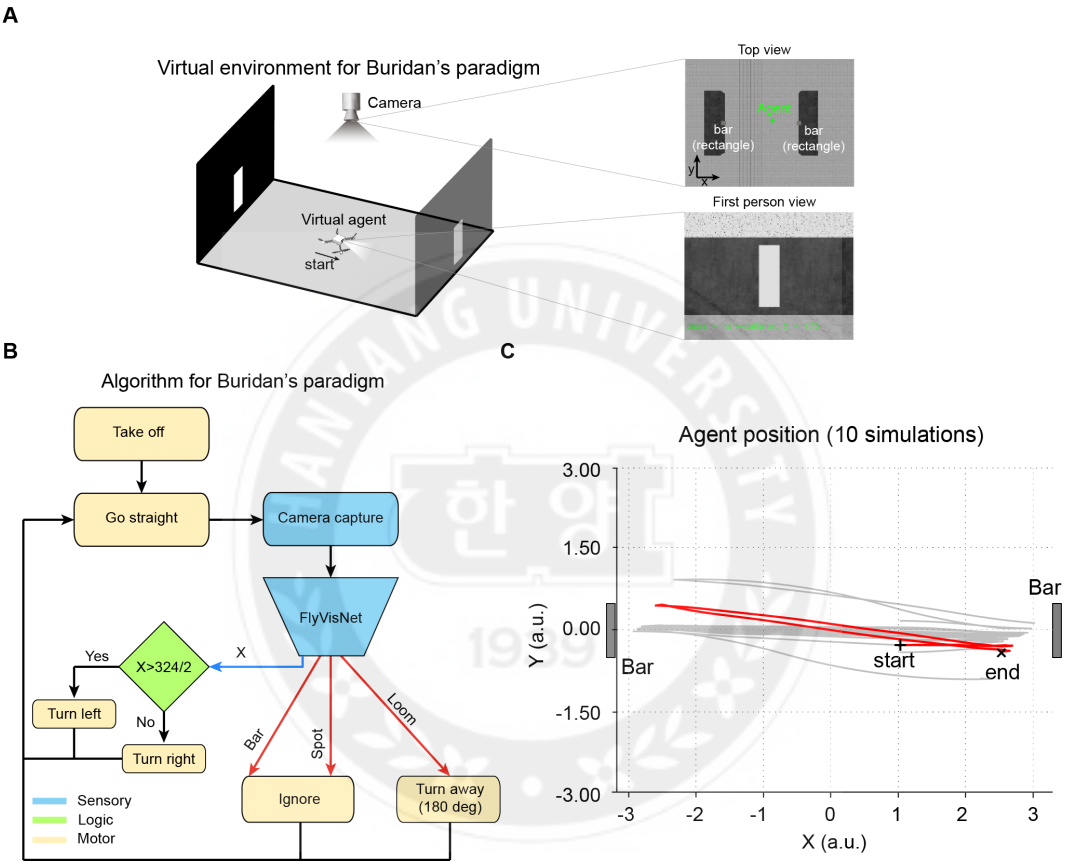


Figure 2.S11: A virtual agent reproduced Buridan's paradigm by using FlyVisNet's inference in a 3D world environment. (A). Same as in Figure 2.4 A for the Buridan's paradigm layout of the environment. (B). Same as in Figure 2.4 B for the Buridan's paradigm flight algorithm. (C). Same as in Figure 2.4 C for the Buridan's paradigm flight environment.

Chapter 3. A visual EC-based navigation algorithm in *Drosophila* for complex visual environments

3.1 Summary

Drosophila visuomotor processing has been intensively studied in recent years, leading to a detailed understanding of individual neural circuits. However, the collective operation of these circuits during naturalistic behaviors, in which flies encounter a mixture of complex visual stimuli—including those caused by their own actions—remains poorly understood. In this study, we developed an integrative model of *Drosophila* visuomotor processing, wherein multiple visuomotor circuits interconnect through an efference copy (EC) mechanism. We first derived simple visuomotor models by analyzing wing responses to individual, rotating visual patterns. We then combined these models to predict wing responses to the superposition of the visual patterns, using three different strategies. We compared orientation behaviors of these models with those of flying *Drosophila* that rotates their body freely in complex visual environments. The results of these experiments supported one of the EC-based models, in which the amplitude of the flight turn is unaffected by the background scene, irrespective of the visual environment. Together, our "virtual fly" model provides a formal description of vision-based navigation strategies of *Drosophila* in complex visual environments and offers a novel framework for assessing the role of constituent visuomotor neural circuits in real-world contexts.

3.2 Introduction

Fruit flies demonstrate impressive agility in flight [88, 89, 90, 91], and vision plays a pivotal role in flight control, often translating directly into precise flight maneuvers. Neural circuit mechanisms underlying many of these visuomotor behaviors have been studied intensively in recent years, both structurally and functionally [92, 93, 94, 51, 95, 96, 49]. While we have gained a qualitative grasp of the neural substrates underlying various visual behaviors, a quantitative understanding remains less explored [59, 4, 97, 8].

In the pioneering era of insect vision in the 1950s to the 1970s, research heavily leaned towards computational modeling. For instance, the Reichardt-Hassenstein elementary motion detector was formulated to explain visual locomotion behaviors observed in beetles [62]. Another example is the model of EC (EC) that was put forward to describe the visual behavior of hoverflies [98, 23]. While some recent studies have offered quantitative models for Drosophila visual circuits [99, 100, 55, 101], most focused on circuit mechanisms for visual feature detection, without considering its behavioral context. To contextualize the function of these circuits within behavior, a phenomenological model integrating these visual functions within a moving animal would be invaluable.

One important consideration that needs to be made in an integrative model is the interaction among different visuomotor circuits tuned to different visual features. Since these circuits in complex visual environments may activate multiple motor programs simultaneously, circuit mechanisms by which distinct sensorimotor pathways are integrated adaptively are necessary. A foundational theory regarding the interaction of multiple sensorimotor pathways is the theory of EC, which underscores the necessity for a feedback signal that

filters out undesired sensory input [23]. Previous behavioral studies argued that such a mechanism may indeed exist in some dipteran species [98, 102]. Recent studies in Drosophila have uncovered the neural correlates of the EC in an array of visual neurons during both spontaneous and visually evoked rapid flight turns [26, 24, 25].

In this study, we present an integrative model of Drosophila visuomotor behaviors that is derived from and validated by behavioral experiments. Results of the experiments provided clear behavioral evidence of an EC in flying Drosophila, which was in line with one type of integration strategy that we used in the model. The model was able to predict the steering behavior of flying Drosophila in response to various visual patterns when they are presented individually as well as in superposition. Our model provides a framework in which detailed neural circuit models for Drosophila visuomotor processing can be implemented and tested in a real-world context.

3.3 Results

3.3.1 Flight response of Drosophila for singly presented visual patterns

To develop a quantitative model of the visuomotor control in flying Drosophila, we set out by measuring wing responses to singly presented visual patterns. In particular, we placed a tethered, flying fly at the center of a cylindrical LED display and presented a rotating visual pattern on the display (see Figure 3.1 A). To measure the angular torque generated by the fly, we subtracted the right wingbeat amplitude (R WBA) from the left wingbeat amplitude (L

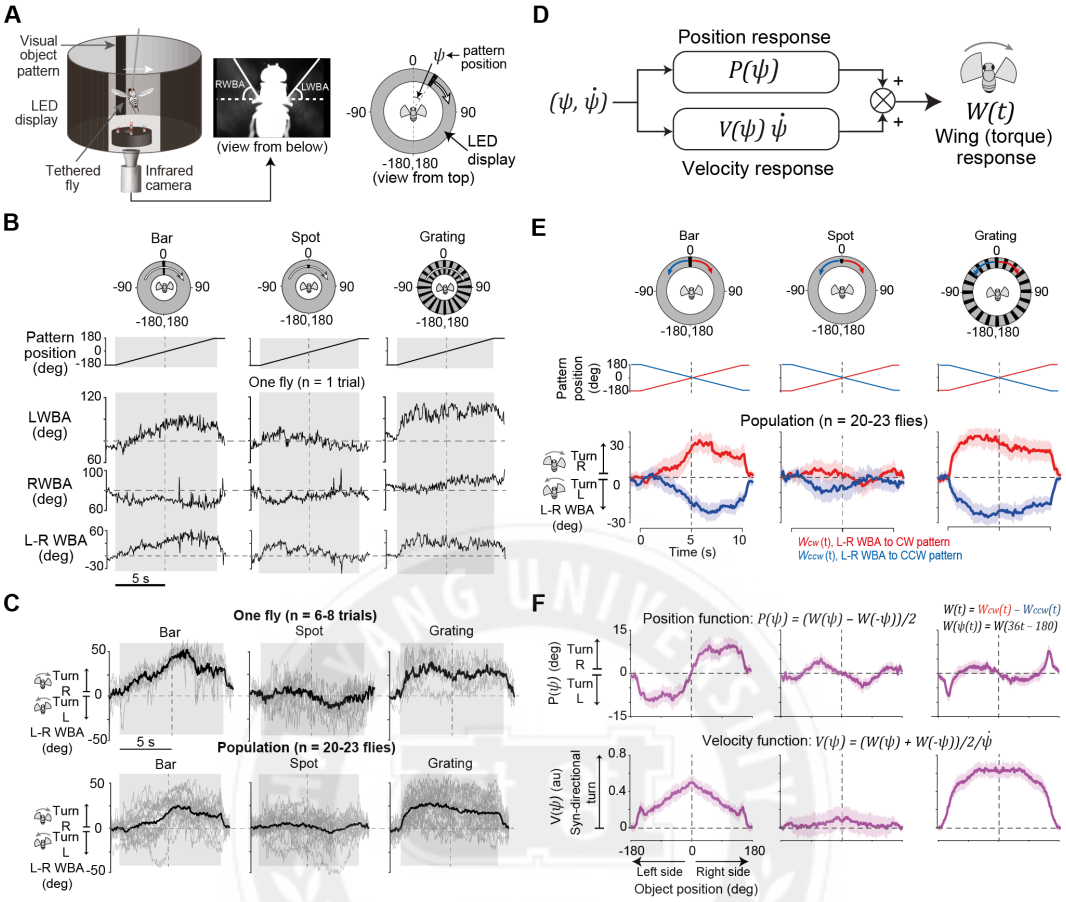


Figure 1.

Figure 3.1: Construction of flight control models for singly presented visual patterns. (A) Schematic of the experimental setup (left), a frame captured by the infrared camera (middle), and a simplified schematic of the experimental setup (right). The annulus surrounding the fly schematic represents the visual display viewed from above. (B) Wing responses of a sample fly to three rotating visual patterns: bar, spot, and grating. L-R WBA at the bottom row represents the angular torque of the fly, calculated by subtracting the right wingbeat amplitude (RWBA) from the left wingbeat amplitude (LWBA). (C) L-R WBA traces of a sample fly in response to the three visual patterns. The thick black lines indicate the average of all trials (top) or the average of all flies (bottom). Thin gray lines indicate individual trials (top) or fly averages (bottom). (D) Schematic of the position-velocity-based flight control model. (E) Average wing responses of a population of flies in response to the three rotating visual patterns, either in a clockwise (red) or counterclockwise (blue) direction. Top traces show the position of each pattern. The red and blue shadings at the bottom indicate the 95% confidence interval. (F) Position and velocity functions estimated from wing responses in E. Light purple shadings indicate the 95% confidence interval.

WBA), resulting in a time signal termed L-R WBA (see Figure 3.1 B). Previous studies showed that L-R WBA is correlated with the angular torque that a fly exerts on its body [103, 16].

We presented three visual patterns rotating a full circle at $36^{\circ}/s$: a dark vertical bar, a dark spot, and a vertical grating (see Figure 3.1 B). These simple patterns, when tested in different velocities and sizes, were previously shown to elicit robust visuomotor reflexes in flying Drosophila [15, 78, 20, 16]. In response to a clockwise-rotating bar starting from the back, flies generated wing responses corresponding to a fictive rightward turn throughout the duration of the stimulus (see Figure 3.1 B-C). The response increased gradually and peaked after crossing the midline, then declined. In response to the moving spot, flies exhibited L-R WBA indicating turning away from the position of the spot (see Figure 3.1 C). In response to the moving grating, flies showed strong wing responses that would exert angular torque in the direction of the grating movement, also consistent with previous studies [21, 17, 16].

3.3.2 Flight control models of Drosophila for singly presented visual patterns

We built a simple model for these visual behaviors by adopting a classical approach that was originally applied to flying *Musca Domestica* [63, 64]. In this approach, the torque response of an animal, $W()$, is modeled as a sum of two components: one defined purely by a position response $P(\psi)$ and the other by a position-dependent velocity response $V(\psi)\dot{\psi}$ (see Figure 3.1 D).

$$W(\psi) = P(\psi) + V(\psi)\dot{\psi} \tag{3.1}$$

In this model, the position response can be experimentally obtained if we add torque responses to a visual pattern rotating in the clockwise and counter-clockwise directions, but with respect to the same object position.

$$P(\psi) = (W_{cw}(\psi) + W_{ccw}(\psi))/2 \tag{3.2}$$

$$V(\psi)\dot{\psi} = (W_{cw}(\psi) - W_{ccw}(\psi))/2 \tag{3.3}$$

where $W_{cw}(\psi)$ denotes the wing response for the clockwise-rotating pattern, and $W_{ccw}(\psi)$ for the counterclockwise-rotating pattern. The velocity response, $V(\psi)\dot{\psi}$, can be obtained similarly, but by a subtraction.

To apply the above approach, we tested flies with both clockwise- and counterclockwise-rotating visual patterns (see Figure 3.1 E). Wingbeat signals to the patterns were largely symmetric for the two directions, but with subtle asymmetries, which are likely due to experimental variabilities such as misalignment of the fly's body orientation with respect to the display arena. To minimize the influence of such experimental anomalies, we combined the two L-R WBA signals for a pair of symmetrical patterns after flipping one for its sign as well as the time and used them for the further analyses (see Data Analysis in Methods).

When we computed the position functions, $P(\psi)$, for the spot and bar patterns with the above method, we found that they were symmetric with respect to the origin (see Figure 3.1 F). That is, for the rotating bar, the position response was positive when the bar was in the left hemifield, but negative in the right hemifield. This result is consistent with fly's fixation behavior to a vertical bar as this position response would have made flies turn toward the bar in both sides. For the spot pattern, the amplitude of

the position function was much smaller than that for the spot, and its overall direction was opposite to that for the bar, which would have made flies turn away from the spot.

The velocity function, $V(\psi)$, was nearly zero for the spot, except around the frontal visual field (see Figure 3.1 F), and this is consistent with weak direction dependency in the wing response (see Figure 3.1 E). By contrast, the velocity function for the bar is consistently positive (see Figure 3.1 F), poised to contribute to the flight turn in the direction of the bar movement [64]. The amplitude of the velocity response, however, varied depending on the object position, unlike the nearly constant velocity function estimated in blow flies in response to a moving bar [64].

In response to the moving grating, the velocity response was consistently positive, but its amplitude varied depending on the position (see Figure 3.1 E). The grating pattern consists of 12 periods of bright and dark vertical stripes, with each stripe spanning 15 degrees, which is larger than the fruit fly's inter-ommatidial angle ($\sim 5^\circ$). However, because we do not see any periodicity in the wing response to the grating (see Figure 3.1 E), we reasoned that the change in position function is not because of the positional variation of the pattern. Instead, the time-dependency in the position and velocity response was likely due to non-visual factors such as adaptation or fatigue in the motor control caused by exceptionally large L-R WBA. Furthermore, the relatively slow L-R WBA change at the beginning and the end of the grating stimulus suggests a dynamical system transformation between the visual stimulus and the wing response. Thus, we expanded our models with a set of dynamical system equations involving a fatigue variable (see Figure 3.1–figure supplement 1), which successfully reproduced L-R WBAs to the three visual patterns as

well as the position and velocity responses associated with each pattern. We also performed this experiment and analysis in a different strain of wild type flies (Canton S) and found that the profile of the position and velocity functions remain largely unchanged (see Figure 3.1–figure supplement 2).

3.3.3 Simulating the orientation behavior of the virtual fly model to individual visual patterns

To test the orientation behavior of the model, we developed an expanded model, termed "virtual fly model" hereafter. In this model, we added a biomechanics block that transforms the torque response of the fly to the actual heading change according to kinematic parameters estimated previously [104, 22] (see Figure 3.2 A, see Equation 3.4 in Methods and Movie S1). The virtual fly model, featuring position and velocity blocks that are conditioned on the type of the visual pattern, can now simulate the visual orientation behavior of flies in the free flight condition. This simulation experiment is reminiscent of the magnetically tethered flight assay, where a flying fly remains fixed at a position but is free to rotate around its yaw axis [105, 106, 107]. Additionally, we simplified the position and velocity functions for the bar and spot pattern with mathematically simple representations (see Figure 3.2 B, see Equation 3.5-7 in Methods). For the grating pattern, we approximated the position response as zero and the velocity response as a constant value (see Figure 3.2 B), where no fatigue effect was considered for the sake of simplicity (see Figure 3.1–figure supplement 1).

We tested the model using visual patterns that moved horizontally with three distinct dynamics: a rapid sigmoidal shift, sinusoidal oscillations, and

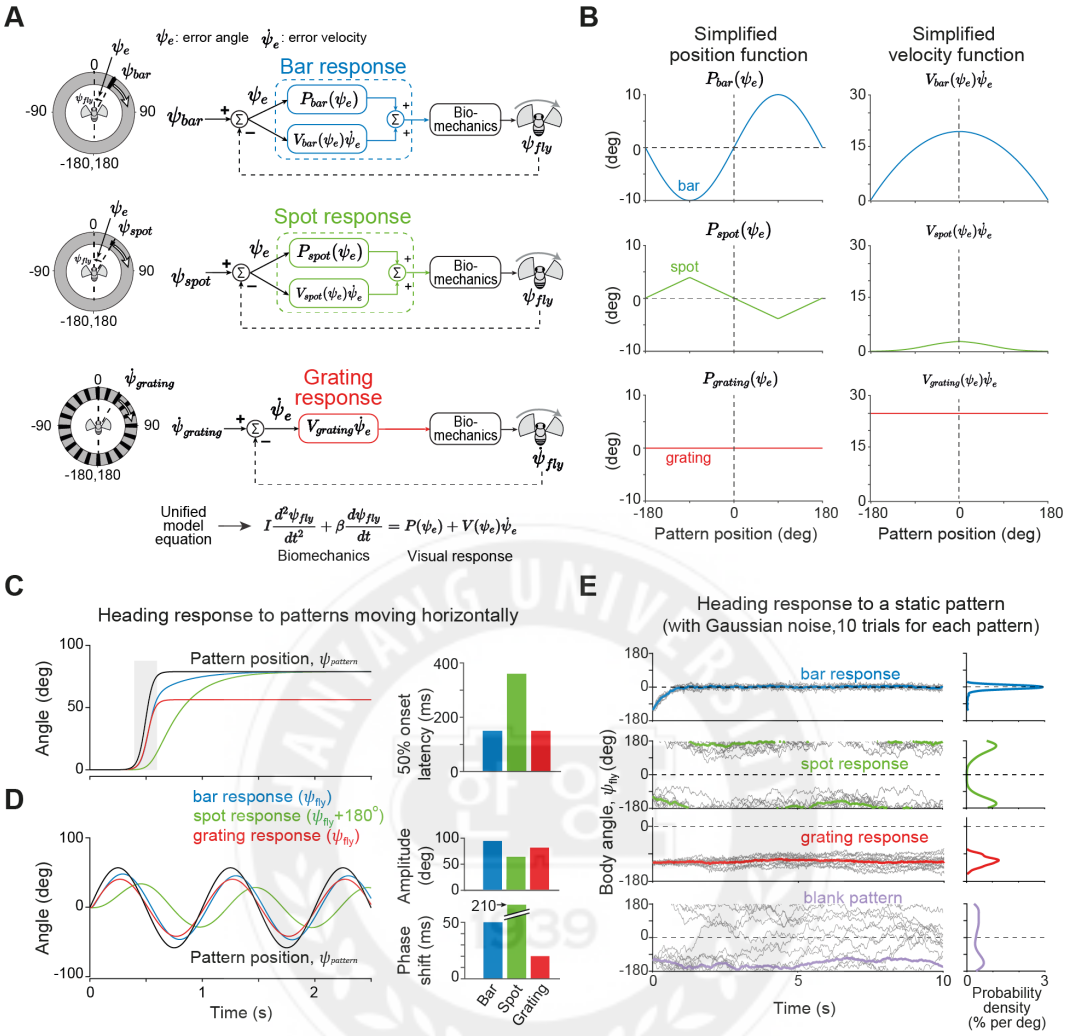


Figure 3.2: The flight control models with a biomechanics block predicted the dynamics of the orientation behavior to individual visual patterns. (A) Schematics of three visuomotor response models with a biomechanics block. (B) Simplified version of the position and velocity responses for each pattern. (C) Simulation results for the three patterns (bar, spot, and grating) moving in a sigmoidal dynamics. The spot response was plotted with an 180° offset to facilitate comparison. Bar plots on the right show the latency of body angle with respect to the stimulus onset, measured at the 50% point of the pattern movement. (D) Simulation results for the three patterns moving in a sinusoidal dynamics. In the bar plots on the right, the amplitude was measured as the peak-to-peak amplitude, and the phase shift was calculated by measuring the peak time of the cross-correlation between the pattern and the fly heading. (E) Same as in C, but for visual patterns remaining static at 0 degree position. The simulation was performed 10 times with a Gaussian noise component (gray lines). The mean response was plotted in thick colored lines. The probability density function of the body angle is shown on the right for each pattern.

Gaussian noise. The virtual fly model's response to the sigmoidally shifting visual patterns varied depending on the pattern (see Figure 3.2 C). For the bar, the fly responded quickly toward the bar and fixated on it (see Equation 3.8, 9 in Methods). The spot elicited a heading change in the opposite direction, albeit at a slower pace and with the pattern positioned behind the fly at the end (see Equation 3.10 in Methods). In the case of the grating, the response was the most rapid, but the heading change ceased early once the visual pattern stopped (see Equation 3.11 in Methods). This is due to the absence of positional cue in the grating and consistent with experimental results in a previous study [98]. We measured the visual response latencies at 50% of the bar movement for each pattern: 150 ms for the bar and grating, and 360 ms for the spot. These heading latencies are considerably longer than the wingbeat response latency reported previously to similar visual patterns [78]. This is in part due to the additional delay caused by the biomechanics block in our model. We also tested our models by incorporating a delay unit between the vision and the biomechanics blocks to take into account different neuronal delays reported previously for each visual pattern (see Figure 3.2–figure supplement 1 A-B). For the sake of simplicity, however, we used the models without the delay block in the following.

For sinusoidally oscillating patterns, we measured the peak-to-peak amplitude and the phase shift of the wing responses (see Figure 3.2 D and see Figure 3.2–figure supplement 1 C). The amplitude of heading change was comparable to the amplitude of the bar and grating stimuli, but much smaller for the spot pattern (see Figure 3.2 D). The phase shift was the longest for the spot, the shortest for the grating, and intermediate for the bar, consistent with the delay observed in response to the rapidly shifting patterns [78] (see Figure

3.2 C). We also simulated our models under the static visual patterns, but with Gaussian noise added to its body torque (see Figure 3.2 E, see Equation 3.12 in Methods). We observed fixation to the bar, anti-fixation to the spot, and stabilization to the grating, whereas the virtual fly model drifted randomly when no visual cue was present (see Figure 3.2 E, bottom).

Finally, one important locomotion pattern that a flying *Drosophila* exhibits while tracking an object is rapid orientation changes, called "saccades". To be specific, while tracking a slowly moving bar, for example, flies perform relatively straight flights interspersed with saccadic flight turns [108, 109]. During this behavior, it was proposed that visual circuits compute an integrated error of the horizontal bar position with respect to the frontal midline and triggers a saccadic turn toward the bar when the integrated value reaches a threshold [110, 109]. We expanded our bar fixation model to incorporate this behavioral strategy (see Figure 3.2–figure supplement 2). The overall structure of the modified model is akin to one proposed in a previous study [109], and the amplitude of a saccadic turn was determined by the sum of the position and velocity functions (see Figure 3.2–figure supplement 2A, see Equation 3.13 in Methods). When simulated, our model successfully reproduced previous experimental observations in terms of the saccade dynamics for different object velocities (see Figure 3.2–figure supplement 2B-D) [109]. Together, visual behaviors of our models in response to singly presented visual patterns recapitulated the results of previous behavioral observations [98, 111, 78, 20, 109].

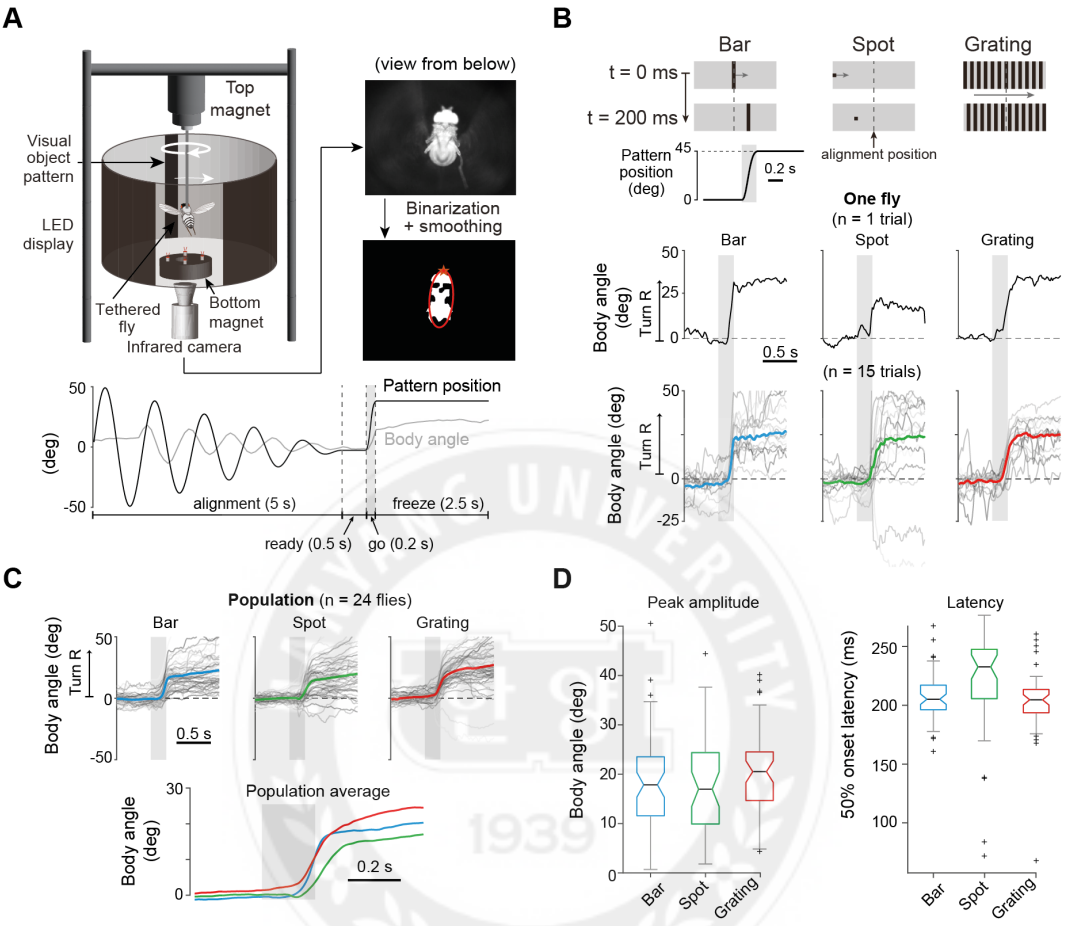


Figure 3.3: Magnetically tethered flight experiments confirmed orientation changes predicted by the virtual fly model. (A) Schematic of the magnetically tethered flight assay with an LED display (left). The image acquired from below was analyzed to estimate the body angle (right). The stimulus protocol (bottom) consisted of four phases: alignment, ready, go, and freeze. (B) Body orientation responses of a single fly for the bar, spot and grating patterns moving horizontally. (C) Same as in B for a population of flies. The population averages were replotted at the bottom to facilitate the comparison of their dynamics. (D) Amplitude and latency of the body orientation responses. The box represents the interquartile range (IQR), with the median indicated by the horizontal black line. The whiskers extend to the minimum and maximum values within 1.5 times the IQR. Outliers are denoted by "+" marks beyond the whiskers.

3.3.4 Orientation behaviors of magnetically tethered flying Drosophila to rapidly moving visual patterns

We validated the prediction of our models with flies flying in an environment similar to the one used in the simulation (see Figure 3.3 A). A fly was tethered to a short steel pin fixed at a position by a vertically oriented magnetic field that also allowed it to rotate around its yaw axis with little friction (see Figure 3.3 A). We captured the image of the fly from below in response to visual stimuli and calculated the body angle post hoc (see Figure 3.3 A, Methods). To present the visual patterns at a defined position relative to the fly's heading, we coaxed the fly to be aligned to a reference angle by oscillating a stripe pattern widely across the visual display ("alignment" phase, see Figure 3.3 A bottom). We then presented for 0.5 s the initial frame of the visual pattern ("ready" phase), which is then rotated rapidly for 0.2 seconds in a sigmoidal dynamics ("go" phase) (see Figure 3.3 A). After the movement, the pattern remained static for 2.5 seconds ("freeze" phase).

For all visual patterns, the fly turned in the direction that was predicted by the models, which we have developed according to wing responses to slowly moving visual patterns (see Figure 3.2). Namely, flies turned toward the moving bar, away from the moving spot, and in the direction of the grating (see Figure 3.3 B, C). The mean amplitude was comparable for the three patterns at around 20 degrees, which was about half that of the angle of the pattern displacement (see Figure 3.3 C, D). While the small response angle was expected for the grating pattern (see Figure 3.2 C), the bar and spot response amplitudes were unexpectedly small compared to the model predictions. This is in contrast to the match between the model prediction and behavioral re-

sults for a slow moving object (see Figure 3.1–figure supplement 1 C). One hypothesis is that strong upward magnetic force or a blunt top end of the steel pin significantly dampens the flies' flight turns. To determine if our assay applies additional friction compared to other assays used in previous studies, we analyzed the dynamics of spontaneous saccades during the "freeze" phase (see Figure 3.3–figure supplement 1 A). We found their duration and amplitude to be within the range reported previously [105, 109] (see Figure 3.3–figure supplement 1 B-D). Another potential explanation arises from recent studies demonstrating that proprioceptive feedback provided during flight turns in a magnetically tethered assay strongly dampens the amplitude of wing and head responses [106, 112]. According to these studies, our models – derived from the behavioral assay with no proprioceptive feedback (see Figure 3.1 A) – are expected to predict larger visual responses than actual flies in a magnetically tethered assay, thus explaining the relatively small response amplitude. Additionally, the response latency was the longest for the spot, and similar between the bar and grating (see Figure 3.3 D), in line with the virtual fly model (see Figure 3.2 C).

With these experimental results in a magnetically tethered assay, we confirmed that our virtual fly models captured essential dynamics of the vision-based flight control in *Drosophila* when the patterns are presented individually. In natural conditions, however, multiple patterns may be presented simultaneously in a visual scene. For example when a fly is trying to land on a fruit that is on the foreground and there are objects in the background such as trees, bushes, etc. This led us to further develop the model to add a capacity of controlling their orientation under more complex visual environments.

3.3.5 Integrative visuomotor control models in a complex visual environment

In natural environments, various visual patterns might simultaneously activate distinct visuomotor circuits. These circuits, if activated, may normally trigger either synergistic or opposing locomotive actions. For instance, when a fly turns to track a foreground object moving against a static background (see Figure 3.4 A), it experiences rotational optic flow from the background in a direction opposite to the flight turn. This optic flow would immediately activate the visual stability reflex that opposes the object-evoked turn. Therefore, this raises the question as to how outputs from different visuomotor circuits integrate to control a shared actuator, such as the wing motor system.

This problem has been studied in sensory systems of many different animals, typically by analyzing physiological recordings made in behaving conditions [113, 98, 114]. According to the results of these studies, we developed three integrative models. The first model is termed "addition-only model", in which output of the bar and the grating response circuits are added to control the flight orientation (see Figure 3.4 B, see Equation 3.14 in Methods). In the second and third models, an EC is used to set priorities between different visuomotor circuits (see Figure 3.4 C, D). In particular, the EC is derived from the bar-induced motor command and is sent to the grating response system to nullify visual input associated with the bar-evoked turn [113, 98, 114]. In some sensory systems, such motor-related inputs are strong enough to completely silence sensory processing [114], whereas in others the motor-related inputs counteract only undesirable part of the sensory feedback [113, 115].

Recent studies in *Drosophila* characterized cell-type-specific EC-like in-

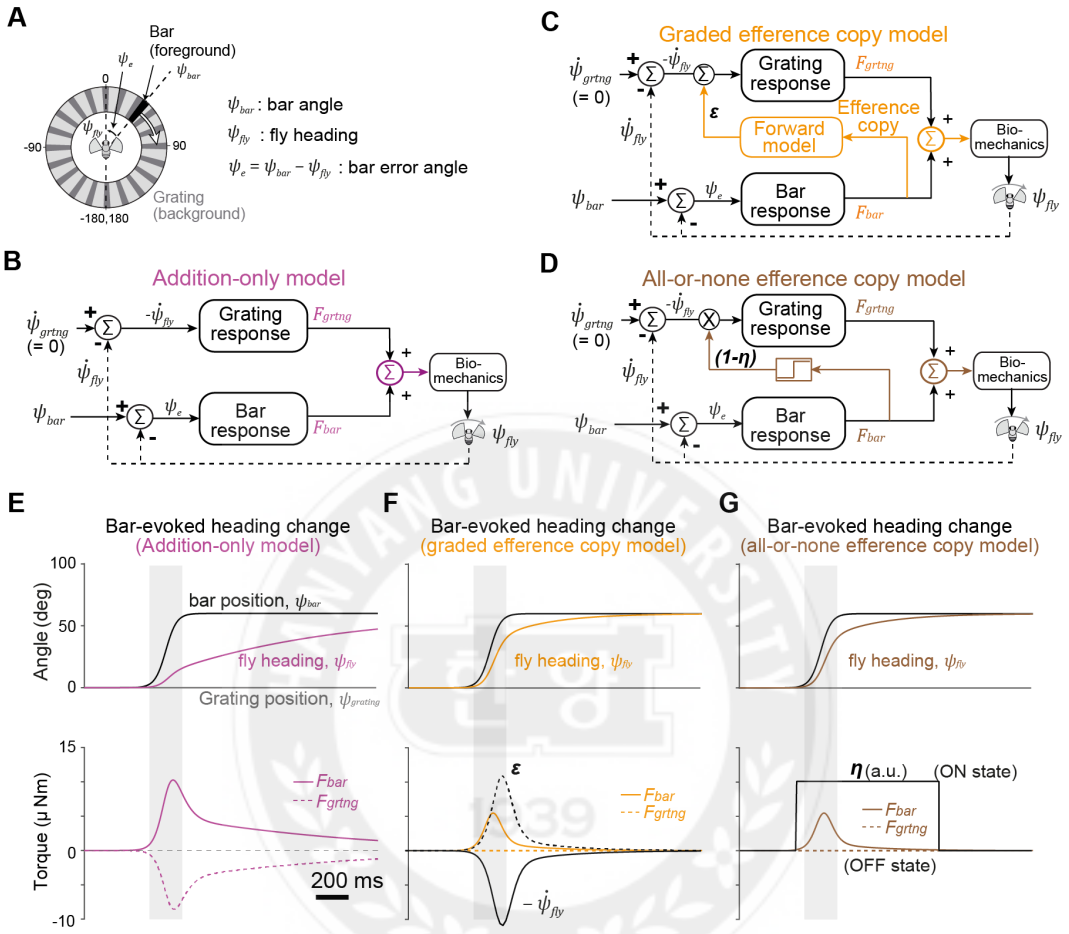


Figure 3.4: Three integrative models of the visuomotor control and their predictions in a complex visual environment. (A) Schematic of the visual environments used in the simulation. A moving bar is presented as a foreground object over the static grating background. (B) A diagram of the addition-only model. Bar and grating response circuits are joined at their output by addition. (C) A diagram of the graded EC model. An EC block translates the bar-evoked motor command to the negative image of the predicted grating input to counteract visual feedback. (D) A diagram of the all-or-none EC model. An EC switches off the grating response circuit during the bar-evoked turn. (E,F,G) Simulation results of the three models. The bar position and the heading of the virtual fly model (top). The associated torques as well as the EC signal (bottom).

puts in *Drosophila* visual neurons [26, 24, 25]. However, it has not been addressed clearly whether the amplitude of these inputs is fixed or modifiable depending on the amplitude of the turn, within a cell type. Thus, we proposed two EC-based models. In the "graded" EC model (see Figure 3.4 C, see Equation 3.15 in Methods), the amplitude of the EC, ϵ, is adjusted according to the predicted optic flow feedback. An example of such an EC has been characterized in detail in the cerebellum-like structure in the electric fish [113]. On the other hand, in the all-or-none EC model (see Figure 3.4 D, see Equation 3.16 in Methods), the grating response is blocked entirely by the binarized (i.e., all or none) EC, η, during the object-evoked turn regardless of its amplitude. An example of such a type of EC has been reported in the auditory neurons of crickets during stridulation [114].

When we simulated the addition-only model, we observed that the virtual fly model reduced the error angle rather slowly, as expected (see Figure 3.4 E, magenta trace). The slowdown is due to the torque generated by the grating, Fgrtng, which has the opposite sign to that of the bar-evoked torque, Fbar (see Figure 3.4 E, bottom). In the graded EC model, we observed that the bar response was not impeded (see Figure 3.4 F, orange trace in the top), and the heading dynamics matched that of the simulation with the empty background (see Figure 3.2 C). The torque due to the grating, F_{grtng}, remained at zero because the graded EC (see Figure 3.4 F, dotted black trace in the bottom) counteracted the input to the grating response (see Figure 3.4 F, solid black trace in the bottom). In the all-or-none EC model, the response to the bar was not dampened either (see Figure 3.4 G, brown trace in the top) because the grating response was fully suppressed by the EC signal during the turn (see Figure 3.4 G, dotted brown trace in the bottom).

Our results suggest that the EC can be used to combine multiple visual reflexes and allow for a high-priority sensorimotor circuit (e.g., object tracking/avoidance) to silence low-priority pathways (e.g., visual stability response). We also tested our models in other visual settings and found that both the EC-based models exhibit shorter latencies than the addition-only model (see Figure 3.4–figure supplement 1). Our EC-based models provide potential solutions for visuomotor integration in complex visual environments.

3.3.6 For changing visual environments, the graded EC model requires auto-tuning, but not the all-or-none EC model

Both the EC-based models were shown to disengage the visual stability reflex during object-evoked flight turns. The two models, however, differ in that the graded EC model is capable of responding to subtle mismatches between the predicted and actual background optic flow (see Figure 3.4–figure supplement 1 C), whereas the all-or-none EC model should always exhibit the same dynamic regardless of the background. For example, when the visual environment changes suddenly, such as when moving from a forest to a desert (see Figure 3.5 A), the dynamics of object-evoked flight turns would be unaffected in the all-or-none EC model, whereas significant heading response change would occur in the case of graded EC model.

This also suggests that the graded EC model is required to update its EC according to changing visual environments, adjusting the prediction of reafferent inputs adaptively. Such modifiable ECs were reported in the electric fish's cerebellum-like structure [115]. We thus implemented an algorithm by which the amplitude of EC is auto-tuned by a multi-layer perceptron (MLP).

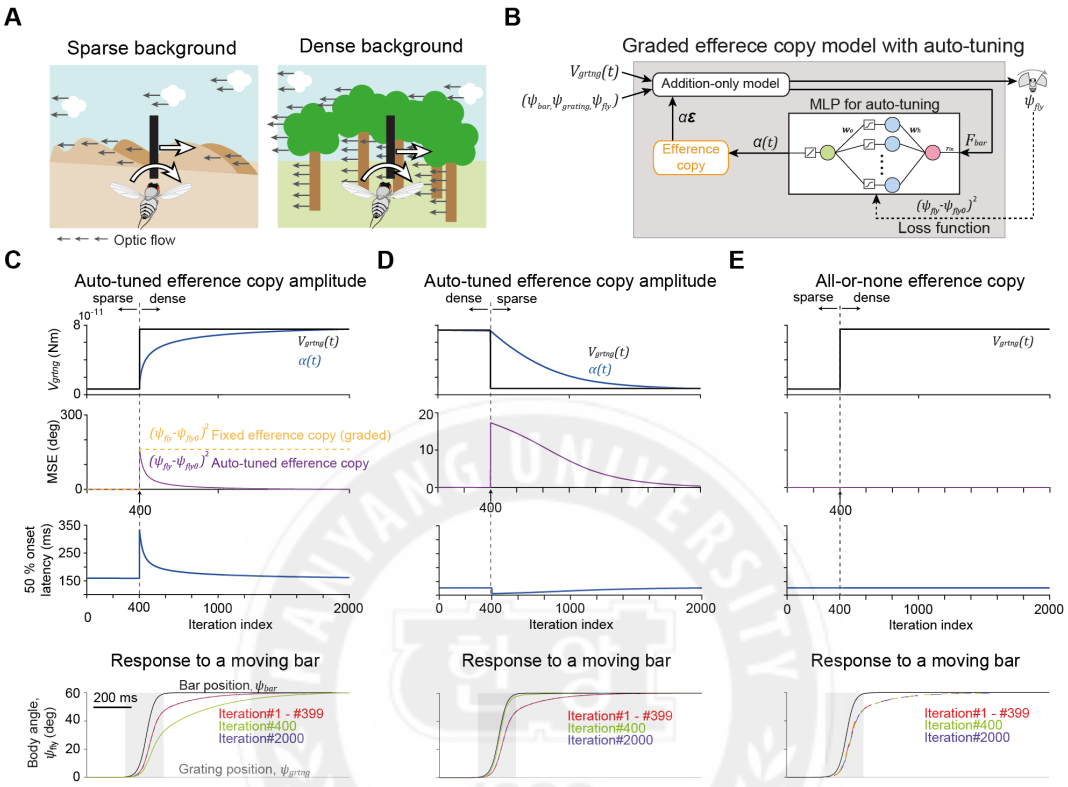


Figure 3.5: Behavior of EC-based models in changing visual backgrounds. (A) Two different visual environments, one with few objects in the background representing a weak visual feedback and the other with many objects such as trees representing a strong visual feedback. (B) A diagram of the auto-tuned, graded EC model, where a multi-layered perceptron (MLP) is used to adapt the EC amplitude to the changing visual feedback. The loss function is computed based on the mean squared error (MSE) between the angular position of the fly when the EC amplitude matches that of the visual feedback. (C) Simulation results for a moving bar with a static background over iterations when the visual feedback increases at the 400th iteration. The top plot depicts the change of the feedback amplitude and the EC amplitude, the second panel the MSE, and the third panel the 50% latency of the heading response. The bottom plot show the sample traces before and after the feedback change. (D) Same as in C, but when the feedback decreases at the 400th iteration. (E) Same as in C, but for the all-or-none EC model.

The MLP generates the EC amplitude, α, as a prediction for the current feedback amplitude, $V_{grating}$. In this model, the tracking error during each iteration of the visuomotor response was calculated as a loss function, and the weight parameters in the MLP were updated in a direction minimizing this error (see Figure 3.5 B, see Algorithm 2 in Methods).

Using this model, we simulated a moving bar over a static grating for 2000 iterations, where the background changes suddenly from a sparse optic flow to a dense one at the 400th iteration (see Figure 3.5 C). We observed that the response of the virtual fly model to the bar pattern slowed down significantly when the background changed, and as a result the error and the latency increased significantly (see Figure 3.5 C). Over iterations, the EC amplitude increased toward the new feedback amplitude, and the latency decreased toward the initial value (see Figure 3.5 C, bottom two rows). When the background changed from the dense to sparse optic flow, we observed the opposite behavior (see Figure 3.5 D). For the all-or-none EC model, however, the background change did not result in any change in the flight turn dynamics, as expected (see Figure 3.5 E).

3.3.7 Behavioral evidence of a visual EC in flying Drosophila

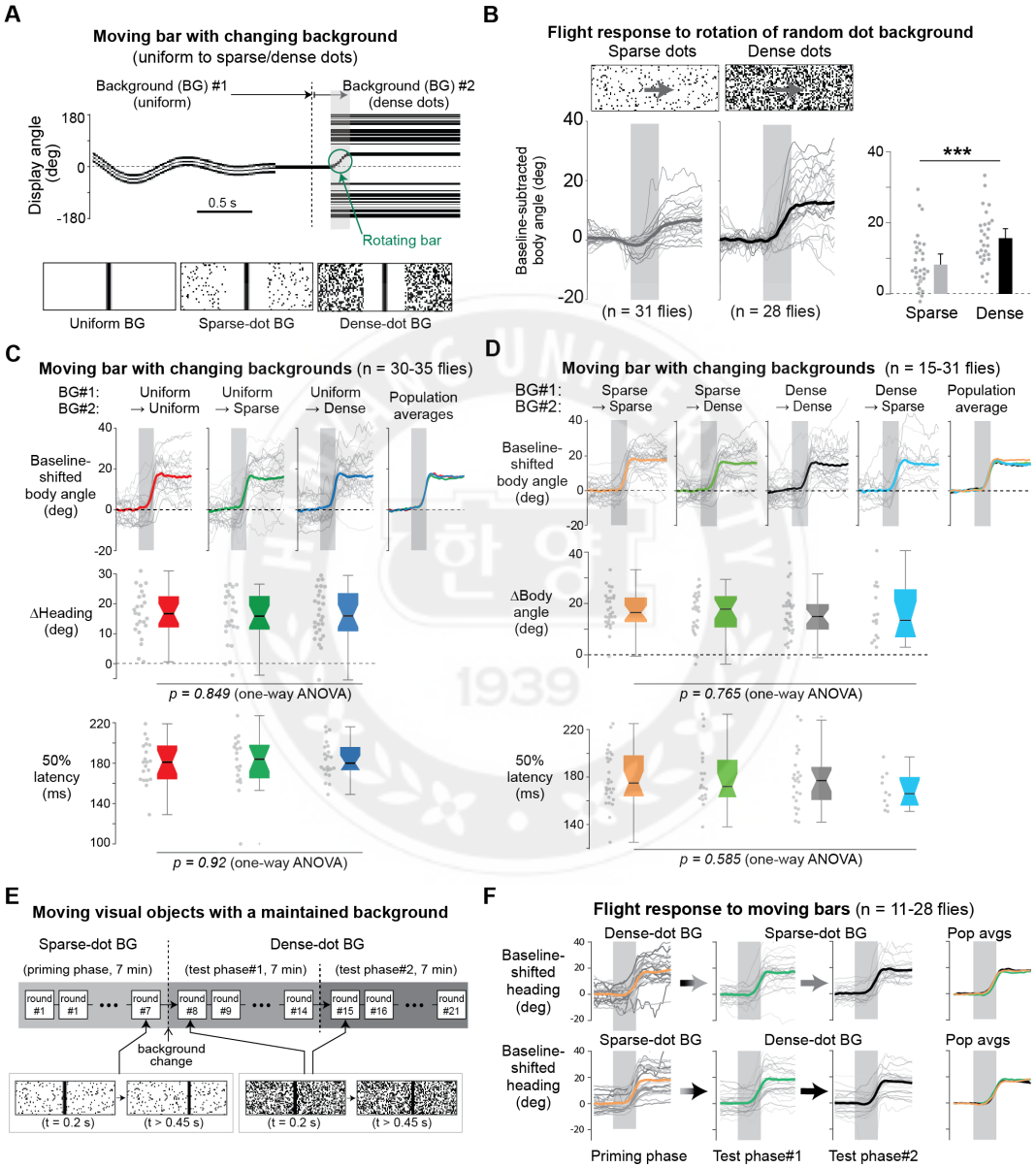

A
Moving bar with changing background
(uniform to sparse/dense dots)
Background (BG) #1
(uniform)
Background (BG) #2
(dense dots)
Display angle (deg)
180
0
-180
0.5 s
Rotating bar
Uniform BG
Sparse-dot BG
Dense-dot BG
B
Flight response to rotation of random dot background
Sparse dots
Dense dots
Baseline-subtracted body angle (deg)
40
20
0
-20
(n = 31 flies)
(n = 28 flies)
***
Sparse
Dense
C
Moving bar with changing backgrounds (n = 30-35 flies)
BG#1:
BG#2:
Uniform → Uniform
Uniform → Sparse
Uniform → Dense
Population averages
Baseline-shifted body angle (deg)
ΔHeading (deg)
p = 0.849 (one-way ANOVA)
50% latency (ms)
p = 0.92 (one-way ANOVA)
D
Moving bar with changing backgrounds (n = 15-31 flies)
BG#1:
BG#2:
Sparse → Sparse
Sparse → Dense
Dense → Dense
Dense → Sparse
Population average
Baseline-shifted body angle (deg)
ΔBody angle (deg)
p = 0.765 (one-way ANOVA)
50% latency (ms)
p = 0.585 (one-way ANOVA)
E
Moving visual objects with a maintained background
Sparse-dot BG
Dense-dot BG
(priming phase, 7 min)
(test phase#1, 7 min)
(test phase#2, 7 min)
round #1
round #7
round #8
round #9
round #14
round #15
round #16
round #21
background change
(t = 0.2 s)
(t > 0.45 s)
(t = 0.2 s)
(t > 0.45 s)
F
Flight response to moving bars (n = 11-28 flies)
Dense-dot BG
Sparse-dot BG
Pop avgs
Baseline-shifted heading (deg)
Sparse-dot BG
Dense-dot BG
Pop avgs
Baseline-shifted heading (deg)
Priming phase
Test phase#1
Test phase#2

Figure 3.6: Dynamics of object-evoked flight turns were not affected by the background pattern. (A) Temporal profile of the visual stimulus used to test the bar response when the background changes from the uniform to the dense-dot background. In each frame, the horizontal midline of the display is sampled and plotted over time (top). Sample pattern images at the moment of the bar movement onset (bottom middle). (B) Body angle changes to the rotation of the random dot backgrounds in two different densities (left). The amplitude is significantly larger for the dense background than the sparse one (right, Wilcoxon rank sum test). (C) Average body orientation traces in response to horizontally moving bars when the initial uniform, bright background is kept the same or changed to one of the two random-dot backgrounds. The thick colored lines in the top plots represent the population average, whereas the gray lines represent an average for individual flies. Box-and-whisker plots depict the amplitude (middle) and latency (bottom) of the body angle change. Error bars (bottom) indicate 95% confidence interval. (D) Same as in (C), but for different combinations of background patterns. (E) Schematic of the stimulus protocol for testing adaptation in the bar response with a maintained background. The background was maintained in either sparse or dense dot patterns for 7 minutes ("priming phase") and then changed to a different density. After this change, the same background pattern was maintained for 14 minutes ("test phase" 1 and 2). (F) Population-averaged body angle traces in response to a moving bar in three different phases.

To determine whether or not flying Drosophila actually uses an EC-based visual integration mechanism and also to dissociate between the two EC-based models, we replicated our simulation in the behavioral experiment using the magnetically tethered flight assay. In particular, we measured the amplitude of the orientation response to the moving vertical bar against two different random dot backgrounds that differ in their densities (see Figure 3.6 A and Movie S2). We first tested how different the visual stability responses are between the two random dot patterns. As expected, the body orientation change was considerably larger with the dense background compared to the sparse background (see Figure 3.6 B). Note also that the response amplitude to the dense-dot background was comparable to that of the grating motion

that we observed previously (see Figure 3.3 D). This indicates that, unless the stability reflex is suppressed during the flies' object-evoked turns, the turn should slow down more strongly with the dense background than the sparse one.

We then assessed the fly's bar-evoked flight turns with changing visual background (see Figure 3.6 A, C), as in the simulation (see Figure 3.5 C-E). That is, we ran the "alignment" and "ready" phase with a uniform, bright background and then either kept it the same or changed to one of the two random dot backgrounds at the moment of the bar movement onset (see Figure 3.6 A). To match the saliency level of the bar pattern the same in two different backgrounds, we kept the foreground visual field ($\pm 55^\circ$ from the center) the same as the initial, uniform background. Surprisingly, we found that the bar response was almost identical regardless of the background dot density (see Figure 3.6 C).

To further corroborate this result, we carried out the same experiment, but with the dots filling the entire panoramic visual field and the background changed 0.25 s prior to the bar movement onset (see Figure 3.6–figure supplement 1A for the depiction of the protocol). We tested flies with four different combinations of background patterns: sparse-to-sparse, sparse-to-dense, dense-to-dense, and dense-to-sparse. We discovered that the amplitude of the flight turn, in terms of both the body orientation and the head yaw angle, was not significantly influenced by the background pattern (see Figure 3.6 D, Figure 3.6–figure supplement 1 B). Together, these results challenged the addition-only model, which suggested the strong dependency of heading responses on the visual feedback from the background pattern. Furthermore, our results also counter the graded EC model because in this model, the background

change at the time of or immediately prior to the bar movement would have caused a mismatch between predicted and actual optic flow feedback, leading to response change, as shown in the simulation (see Figure 3.5 C, D). We did not observe such changes when compared between the maintained vs changing background.

To further test the possibility of the auto-tuned, graded EC, we tested the fly's bar response against a maintained background for an extended period of flight ($\sim$ 14 minutes). Prior to this extended exposure to a single background pattern, we primed the fly with a random-dot background in a different density for $\sim$ 7 minutes (see Figure 3.6 E). The resulting data revealed no significant changes in the flight turn amplitude across the three phases: a priming phase with one background ($\sim$ 7 min), an early test phase with the other background ($\sim$ 7 min), and a late test phase with the maintained background ($\sim$ 7 min) (see Figure 3.6 F). Altogether, we observed that flies' object-evoked flight turns were not significantly affected by the density of random dots in the background, at least in the time scale that we tested. Finally, we repeated the above experiment (see Figure 3.6 C, D) in response to a looming disc pattern instead of the moving bar and observed no significant change in the heading response for different background patterns (see Figure 3.6–figure supplement 1C-E), suggesting that the all-or-none suppression of visual stabilization reflex may be a general mechanism during visually evoked rapid flight turns. It is also noteworthy that unexpected visual feedback during spontaneously generated flight turns does not alter the dynamics of the turn significantly [116]. Altogether, our results suggested that flying *Drosophila* uses the all-or-none EC strategy in complex visual environments (see Figure 3.4 D), wherein the visual stability response is entirely suppressed during object-evoked turns (see

Figure 3.5 E).

3.4 Discussion

Drosophila visuomotor circuits have been successfully dissected in recent years, but in a fragmented manner for limited sensory and behavioral contexts. To build an integrative framework for quantitatively testing these circuits and their interactions in complex visual environments, we developed a computational model of visuomotor flight control. In this model, multiple visual features collectively determine flight heading through an EC mechanism. The model accurately predicted the flight orientation changes in response to singly presented patterns as well as to their superpositions. Our results also provide clear behavioral evidence for a visual EC in *Drosophila* and offer a framework for the future implementation and testing of detailed models for feedforward and feedback neural circuits in the *Drosophila* visuomotor processing.

3.4.1 Models of Drosophila visuomotor processing

Visual systems extract features from the environment by calculating spatiotemporal relationships of neural activities within an array of photoreceptors. In *Drosophila*, these calculations occur initially on a local scale in the peripheral layers of the optic lobe [55, 117]. When these features enter the central brain, they are integrated across a large visual field by converging either to a small brain region or to a single cell [97, 95]. These features are subsequently relayed to other central brain regions or descending neural circuits, which ultimately control action [118, 119].

The mechanisms for key visual computations such as direction-dependent

motion computation, collision avoidance, stability reflex, object tracking, and navigation are under active investigation, and in silico models for some of these sensory computations are already available [99, 55]. In the motor end, the neuromechanical interaction between the animal's actuator and external world was modeled in a recent study for walking [120]. What remains to be explored is the mapping of the features in the visual space onto a coordinated activation pattern of motor neurons by higher-order brain circuits, in a way to optimize the organism's objective functions (i.e., survival and reproduction). Our study provides an integrative model of the visuomotor mapping in *Drosophila*.

3.4.2 Visual EC in Drosophila

Under natural conditions, various visual features in the environment may concurrently activate different motor programs, which may interfere with each other through reafferent inputs. It is thus crucial for the central brain to arbitrate the motor signals originating from different sensorimotor circuits. The EC is defined as a copy of a motor command that plays a central role in this process by modifying or inhibiting specific sensory signaling based on the motor context [98, 23]. Recent studies reported such EC-like signals in *Drosophila* visual neurons [26, 24, 25]. One type of EC-like signals were identified in a group of wide-field visual motion-sensing neurons that were shown to control the neck movement for gaze stability [25]. The EC-like signals in these cells were bidirectional depending on the direction of flight turns, and their amplitudes were quantitatively tuned to that of the expected visual input in each cell type. Although their amplitude varied across cell types, it remained inconclusive whether it also changed within a cell type to match the expected visual feedback, which differs depending on the turn dynamics. The

detailed analysis of individual turn amplitude and the associated membrane potential change showed no significant correlation (data not shown). In a more recent study, the amplitude of flight turns in response to a rapidly looming visual object was analyzed with respect to the amplitude of EC-like signals in the same stability cells [26]. The result showed a strong correlation between wing response and the EC-like inputs, but the authors argued that it could be due to binary behaviors of flies in response to a looming visual object: no flight turn vs. saturated flight turn. Thus, these studies does not completely disambiguate between graded vs. all-or-none EC signaling. Another type of EC-like signals observed in the *Drosophila* visual system possesses contrasting features. It entirely suppresses visual signaling, irrespective of the direction of the self-generated turn [24].

Despite the cellular evidence of the EC found in multiple visual pathways, the behavioral role of these EC-like signals has not been clear until now. Our behavioral experiments confirmed that the *Drosophila* brain, during visually evoked flight turns, suppresses optic flow-induced stability reflexes in an all-or-none manner. Why would *Drosophila* use the all-or-none EC instead of the graded one? Recent studies on electric fish suggested that a large array of neurons in a multi-layer network are crucial for generating a modifiable EC signal matched to the current environment [121]. Given their small-sized brain, flies might opt for a more economical design for suppressing unwanted visual inputs regardless of the visual environment. Circuits mediating such a type of EC were identified in the cricket auditory system during stridulation [114], for example. Our study strongly suggests the existence of a similar circuit mechanism in the *Drosophila* visual system. To definitively prove this mechanism, one may need to record from neurons that are responsible for the visual stabil-

ity response for the body orientation, but the identity of such neurons has been elusive. Future studies, along with the identification of a novel visual pathway for the visual stability response, will be required to demonstrate whether such a visual pathway is modulated by all-or-none EC signals.

3.5 Materials and methods

Fly stocks and rearing. We used female Oregon-R *Drosophila melanogaster* 2-7 days post-eclosion for the experiments, unless mentioned otherwise. Flies were reared on standard cornmeal agar in $25^{\circ}C$ incubators with a 12h-12h light/dark cycle and in 70% humidity. For all experiments, flies were cold-anesthetized at $4^{\circ}C$ and tethered either to a tungsten pin for the rigidly tethered experiments (see Figure 3.1) or to a steel pin for the magnetically tethered experiments (see Figure 3.3, 3.6).

Visual display. We used a modular LED display system composed of 8x8 dot matrix LED arrays [122]. The visual stimuli were displayed on the interior of a cylindrical display arena featuring green LED pixels with a 570-nm peak wavelength, facing inward. The display covered 360 degrees (96 pixels) in the azimuth and 94 degrees (32 pixels) in elevation. From the animal's perspective, each LED pixel subtended less than or equal to 3.75 degrees, which is narrower than the inter-ommatidial angle of the fly ($\sim$ 5 degrees) [123].

Magnetic tethering setup. We constructed a magnetic tethering setup following the designs used in previous studies [124, 25, 109]. The support frame for the two neodymium magnets was designed in Fusion 360 (Autodesk,

Inc.) and manufactured using a 3D printer (Ender 3 S1, Creality) (see Figure 3.3 A). The bottom magnet had a ring shape with an outer diameter of 32 mm, an inner diameter of 13 mm, and a height of 10mm. The top magnet was cylinder-shaped with a diameter of 12 mm and a height of 10 mm. The distance between the top of the bottom magnet and the bottom of the top magnet was 25 mm. A V-shaped jewel bearing (Swiss Jewel Company) was attached to the bottom surface of the top magnet using epoxy and was used to anchor a steel pin-tethered fly. We positioned a Prosilica GE-680C camera (Allied Vision Inc.) with a macro lens (Infinistix, 1.5x zoom, 94-mm working distance) on the ground to capture the fly's behavior from below. To illuminate the fly without interfering with its vision, we affixed four infrared LEDs (850-nm peak wavelength) concentrically to the top surface of the bottom magnet support. Furthermore, we applied matte black acrylic paint (Chroma Inc.) to the bottom surface of the top magnet to minimize background glare in the acquired images and to the top surface of the bottom magnet to minimize visual interference to flies.

Visual stimuli. To estimate the position and velocity functions, we measured flies' wing responses in response to three visual patterns moving at the speed of $36°/s$ for 10 seconds (see Figure 3.1). The bar pattern was a dark vertical stripe with a width of 15° (4 pixels) on a bright background that moved laterally from the back to back for a full rotation in either clockwise or counterclockwise directions. The spot pattern was a 2x2-pixel dark square on a bright background and moved in the same dynamics as the bar. The grating pattern consisted of 12 cycles of dark and bright vertical stripes, each with a width of 15° (4 pixels). For the magnetically tethered flight experiments

(Figs. 3,6), the patterns moved rapidly 45 degrees in a sigmoid dynamics. Each pattern consisted of 4 distinct phases: alignment, ready, go, and freeze. The vertical pattern used for the alignment consisted of 3 dark, 2 bright, 1 dark, 2 bright, and 3 dark pixel vertical stripes, spanning a total of 41.25° (11 pixels) in width. To align the orientation of the fly to the reference angle, this pattern moved horizontally in a sinusoidal dynamics for 4-5 seconds at the frequency 1 period/s, with its envelope amplitude decreasing linearly (Movie S2). The bar pattern turned to a dark uniform bar over a bright background with a width of 11.25° (3 pixels) and was used for the rest of the stimulus period. The random dot pattern was created by adding dark pixels at a random position in the display arena. The sparse dot pattern had these dots in 7% of the total pixels, while the dense dot pattern had 40%.

Behavioral experiment. For the rigidly tethered flight experiments (see Figure 3.1), we glued the anterior-dorsal part of the fly's thorax to a tungsten pin using a UV-cured glue (Bondic). The other end of the pin was inserted into a pin receptacle located at the center of the visual display. The fly body orientation was tilted down from the horizontal by 30° to match the normal flight attitude [125]. The fly was illuminated by four 850-nm LEDs on the ring-shaped platform positioned on the top of the camera (see Figure 3.1 A). The fly was imaged from the bottom by a GE-680 camera at 60 frame/s with a macro zoom lens (MLM3X-MP, Computar, NC, USA) and an infrared long-pass filter. Wingbeat signals were analyzed in real-time by FView and motmot package [126]. For the magnetically tethered flight experiments (see Figure 3.3, 3.6), we instead used a short steel pin to levitate the fly by a magnetic field.

Data analysis. For quantifying wing responses from the rigidly tethered flight experiment, we first subtracted the left wingbeat amplitude from the right wingbeat amplitude to calculate the L-R WBA for each recording (see Figure 3.1 B). For each fly, we determined the stimulus-triggered L-R WBA for every pattern, provided that the number of trials in which the animal continuously flew was at least 6. We then computed the mean L-R WBA across the trials for each fly, and from these individual means, we determined the population-averaged L-R WBA (see Figure 3.1). In the magnetically tethered flight experiment, we captured images of the flies at 60 frames per second (see Figure 3.3, 3.6). The camera was externally triggered by an Ubuntu computer via a USB-connected microcontroller board (Teensy, PJRC), and the trigger signal was stored on the Windows computer using WinEDR (University of Strathclyde, Glasgow), along with the stimulus type and position signals.The acquired images, along with their timestamps, were saved on the Ubuntu computer using FView [126]. To analyze body kinematics (see Figure 3.3, 3.6), we wrote a custom machine vision code in Matlab. Namely, we calculated the body angle by binarizing each frame, performing an eroding operation to remove small non-fly objects, and then obtaining the orientation of the largest region using *regionprops()* function in Matlab (Movie S3). To measure the amplitude of the visually evoked responses (see Figure 3.3 D, 3.6 B-D), we subtracted the mean body angle during the 300-ms interval immediately prior to the stimulus onset from the mean in the 300-ms interval starting 200 ms after the onset. The 50% latency was measured as the time at which the body angle crosses the 50% of the total pattern displacement from the baseline, with respect to the pattern movement onset (see Figure 3.3 E). Additionally, we measured the head angle (see Figure 3.6–figure supplement 1 B) by track-

ing the position of both antennae with respect to the neck (Movie S3) through the deep learning-based software DeepLabCut [127]. The body kinematic variables were stored separately from the stimulus parameters and combined post hoc for further analyses. To synchronize these signals, we inserted a 2-s pause in the camera trigger at the beginning and end of each experiment. These no-trigger intervals appeared in both sets of data files and were used to align kinematic variables acquired from the image data to the stimulus parameters. To compensate for any asymmetry that may exist in the rigidly or magnetically tethered flight experiments, each pattern was presented multiple times in two opposing directions after inverting all the frames with respect to the front midline. We then inverted all the kinematic signals in the counterclockwise trials and merged them with the clockwise trials. To analyze spontaneous saccades (see Figure 3.3–figure supplement 1) we filtered out the body angle traces through a bandpass filter of 15Hz cut-off frequency. We then computed the derivative of the filtered signal to obtain the angular velocities and set a threshold of 250 deg/s. If the velocities were greater than the threshold it was considered as a spontaneous saccade. We identified all the spontaneous saccades in the freeze period (see Figure 3.3–figure supplement 1A) through the smoothed signal. With these indexes we took the corresponding indexes on the original body angle traces with 120 elements backward and onwards. We computed the amplitude as the 80% of the maximum value of the body angle. The duration was computed as the time difference between the amplitude's corresponding time mark and the 80% of the minimum value's time mark.

Statistical analysis. All statistical analysis was performed in Matlab. Heading responses to random dot patterns in the two different densities were

analyzed by Wilcoxon rank-sum test (see Figure 3.6 B). Responses to moving bars (see Figure 3.6 C-D) and to looming discs (see Figure 3.6–figure supplement 1 D-E) for different background profiles were analyzed by one-way ANOVA test.

Modeling visually guided flight control in *Drosophila* (see Figure 3.2) The orientation of the fly is obtained through a second order dynamical system that represents the biomechanics of the fly's body on the left side and the visually-evoked torque response on the right (Figure 2A).

$$I\ddot{\psi}_{fly}(t) + \beta\dot{\psi}_{fly}(t) = P(\psi_e) + V(\psi_e)\dot{\psi}_e \tag{3.4}$$

where ψ_{fly} is the angular position of the fly, $\dot{\psi}_{fly}$ the angular velocity, $\ddot{\psi}_{fly}$ the angular acceleration, $I = 6 \cdot 10^{-14} kg \cdot m^2$ the moment of inertia of the fly's body, $\beta = 1 \cdot 10^{-11} kg \cdot m^2 \cdot s^{-1}$ the drag coefficient of its wings [104, 22], ψ_e the error angle between the fly and the pattern, and $\dot{\psi}_e$ the error velocity.

We approximated the experimentally obtained position and velocity functions with the following mathematical representations for each pattern (see Figure 3.2 B).

$$P_{bar}(\psi) = 10sin(\psi), V_{bar}(\psi)\dot{\psi} = 20 - 2\psi^2 \tag{3.5}$$

$$P_{spot}(\psi) = -5\frac{4}{2\pi}\left|(\psi - \frac{2\pi}{4})mod2\pi - \frac{2\pi}{2}\right| + 5, V_{spot}(\psi)\dot{\psi} = \frac{5}{1 + e^{-2(\frac{3}{8}\pi - |\psi|)}} \tag{3.6}$$

$$P_{grating}(\psi) = 0, V_{grating}(\psi)\dot{\psi} = 25 \tag{3.7}$$

To solve Equation 3.4 using an ODE solver (ode45 in Matlab), we refor-

mulated it into an array of first-order ODEs as in the following.

$$\begin{aligned} \dot{\psi}_{fly} &= z \\ \dot{z} &= \frac{1}{I}(-\beta z + F_{obj}) \end{aligned} \tag{3.8}$$

where the visually evoked torque response, F_{obj}, is defined as follows for different visual patterns.

$$F_{bar} = P_{bar}(\psi_{bar} - \psi_{fly}) + V_{bar}(\psi_{bar} - \psi_{fly})(\dot{\psi}_{bar} - \dot{\psi}_{fly}) \tag{3.9}$$

$$F_{spot} = P_{spot}(\psi_{spot} - \psi_{fly}) + V_{spot}(\psi_{spot} - \psi_{fly})(\dot{\psi}_{spot} - \dot{\psi}_{fly}) \tag{3.10}$$

$$F_{grating} = V_{grating}(\dot{\psi}_{grating} - \dot{\psi}_{fly}) \tag{3.11}$$

To simulate the noisy orientation behavior of flies in a static visual environment (see Figure 3.2 E), we added a Gaussian noise, n, to the torque, as follows.

$$\begin{aligned} \dot{\psi}_{fly} &= z \\ \dot{z} &= \frac{1}{I}(-\beta z + F_{obj} + n) \end{aligned} \tag{3.12}$$

To consider a saccadic-based steering in the virtual fly we expanded the model for bar with a saccade mechanism (see Figure 3.2–figure supplement 2 A). Through a leaky integrator, the integrated error angle (Q) triggers a specific threshold (C) that provides a trigger signal to a switch. When the switch is triggered a saccade is generated with an amplitude provided by the summation of the position and velocity functions. In Equation 8 F_{obj} becomes a conditional value as follows.

$$F_{obj} = \begin{cases} F_{bar}, & \text{if } Q \geq C \\ 0, & \text{otherwise} \end{cases} \text{ where } Q = \int \psi_e e^{\frac{-t}{\tau}} dt \tag{3.13}$$

We picked the values of C and τ (see Figure 3.2–figure supplement 2 C) that minimize the objective function given by $\sqrt{(\psi_{ampl} - \psi_{truth})^2 + (t_{interval} - t_{truth})^2}$. Where ψ_{ampl} is the saccade amplitude and $t_{interval}$ is the inter-saccade interval. Their ground-truth values are $\psi_{truth} = 9\ degrees$ and $t_{truth} = 0.3\ s$, which are approximated from [109].

Integrative models of the visuomotor control (see Figure 3.4) The addition-only model (see Figure 3.4 B) was developed by additively joining the torque response from the bar system, F_{bar}, and that from the grating system, $F_{grating}$, as in the following.

$$\ddot{\psi}_{fly} = \frac{1}{I}(-\beta\dot{\psi}_{fly} + F_{bar} + F_{grating}) \tag{3.14}$$

For the graded EC-based model (see Figure 3.4 C), we defined the integration of the torque responses through an additional ODE that created an EC, ϵ, according to the fly velocity, fly. The EC signal, ϵ, was then subtracted from the grating torque.

$$\begin{aligned} F_{grating} &= V_{grating}(\dot{\psi}_{grating} - \dot{\psi}_{fly} + \epsilon) \\ \dot{\epsilon} &= \frac{1}{I}(-\beta\epsilon + F_{bar}) \end{aligned} \tag{3.15}$$

For the all-or-none EC-based model (see Figure 3.4 D), the grating torque, $F_{grating}$, was switched off to 0 when the bar-evoked torque became non-zero.

$$F_{grating} = V_{grating}(\dot{\psi}_{grating} - \dot{\psi}_{fly})(1 - \eta) \tag{3.16}$$

where η is an indicator function that takes the value of 1 when $|Fbar| > 1.5 \cdot 10^{-11}$ and 0 otherwise.

Auto-tuning mechanism for the graded EC-based model (see Figure 3.5 B) We added a multi-layer perceptron (MLP) to the graded EC-based model (Equation 3.15). The output of the MLP is used to calculate the EC (see Figure 3.5 B). The equations and pseudocode for these computations are as follows:

Algorithm 2 Auto-tuning EC-based model

Initialize MLP with random parameters w_{out}, w_h
for $iteration = 1, M$ **do**
  $V_{grating} = V_{grating} + \Delta V_{grating}$
  for $t = 1, T$ **do**
    $\dot{\psi}_{fly} = z$
    $\dot{z} = \frac{1}{I}(-\beta z + F_{bar} + F_{grating})$
    $F_{bar} = P_{bar}(\psi_{bar} - \psi_{fly}) + V_{bar}(\psi_{bar} - \psi_{fly})(\dot{\psi}_{bar} - \dot{\psi}_{fly})$
    $F_{grating} = V_{grating}(\dot{\psi}_{grating} - \dot{\psi}_{fly} + \epsilon)$
    $\dot{\epsilon} = \frac{1}{I}(-\beta\epsilon + \alpha F_{bar})$
    $r_{in} = F_{bar}$
    $r_h = \frac{1}{1+e^{(-w_h r_{in})}}$
    $r_{out} = w_{out} r_h$
    $\alpha = r_{out}$
    Update MLP parameters:
    $d_{out} = (\psi_{fly0} - \psi_{fly})^2$
    $d_h = r_h(1 - r_h)(w_{out} d_{out})$
    $w_{out} \leftarrow w_{out} + \gamma(d_{out} r_h)$
    $w_h \leftarrow w_h + \gamma(d_h r_{in})$
  end for
end for

where $M = 2000$ is the number of iterations, $T = 2.5s$ is the time range,

$\Delta V_{grating}$ is the change in the visual feedback, α is the output of the MLP, r_h is the output of the hidden layer, r_{in} is the input, ψ_{fly0} is the heading of the fly in a fixed visual feedback condition, and $\gamma = 1 \cdot 10^{-4}$ is the learning rate.

Fly simulator. We have developed a graphic user interface (GUI) application in Matlab where users can simulate the magno-tethered flying fly for three visual patterns and their superpositions (e.g., bar and grating). The GUI also allows users to select from three integrative models (i.e., the addition-only and EC-based models) and choose both the displayed pattern and the maximum angular position amplitude of the stimulus. During the simulation, the GUI plots real-time heading and torque traces, as well as a simplified fruit fly that rotates in response to visual patterns (Movie S1). Finally, simulation results can be saved as a video file.

Code availability. The essential code used to generate the primary results and conduct the simulations for this study is available in our GitHub repository (https://github.com/nisl-hyu/flightsim).

3.6 Supplementary information

Movie S1 (Related to Figure 3.2). A movie showing a simulation using the GUI fly simulator. The simulation corresponds to a moving bar stimulus. On the left the virtual fly model animation, and the simulated LED display, on the right the plots showing the bar position, body angle, and the torque.

Movie S2 (Related to Figure 3.6). A movie showing visual patterns used for testing object-evoked flight turns in changing backgrounds.

Movie S3 (Related to Figure 3.3, 3.6). A fly movie corresponding to an experimental trial in the magno-tethering setup showing the original frame, the processed frame to obtain the body angle, and the frame where the body parts are tracked via DeepLabCut. In the plots below the frames, the stimulus position, body orientation, and head angle are shown respectively. The motion of the body, and head are synchronized with the video. The video was captured at 60 frames per second.

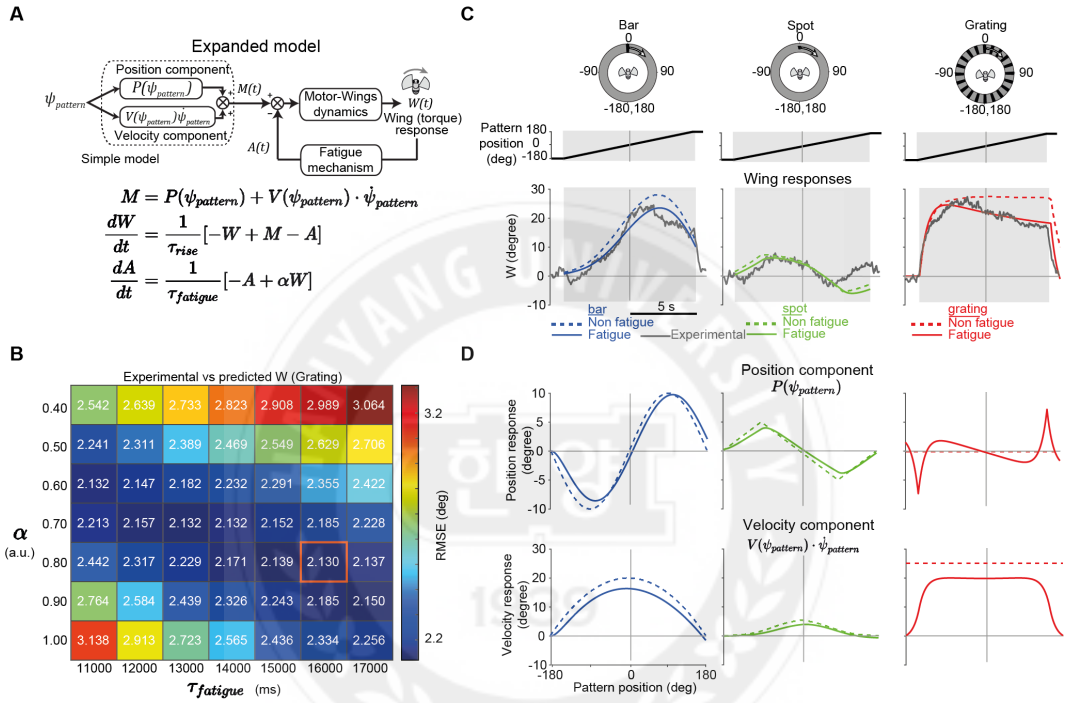


Figure 3.1–figure supplement 1. An expanded model predicts the history-dependent dynamics in the wing responses. (A) Schematic of an expanded flight control model with a motor-wing dynamics and a fatigue block. (B) A heat map indicating the errors in the grating responses for different combinations of the fatigue block parameters. (C) Visually evoked wing responses to the three rotating patterns and the predictions of the two models. (D) Position and velocity responses of the simple and expanded models.

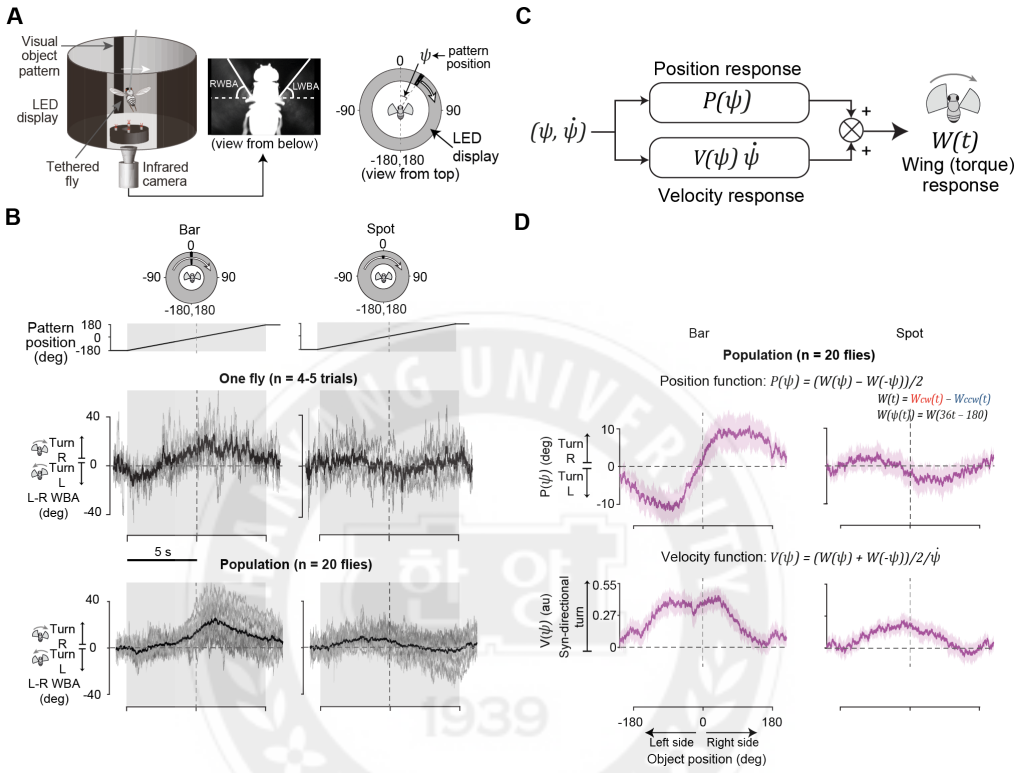


Figure 3.1–figure supplement 2. Construction of flight control models for Canton S flies. (A) Schematic of the experimental setup (left), a frame captured by the infrared camera (middle), and a simplified schematic of the experimental setup (right). The annulus surrounding the fly schematic represents the visual display viewed from above. (B) L-R WBA traces of a sample fly in response to the bar and spot visual patterns. The thick black lines indicate the average of all trials (top) or the average of all flies (bottom). Thin gray lines indicate individual trials (top) or fly averages (bottom). (C) Schematic of the position-velocity-based flight control model. (D) Position and velocity functions estimated from wing responses in C. Light purple shading indicates 95% confidence interval.

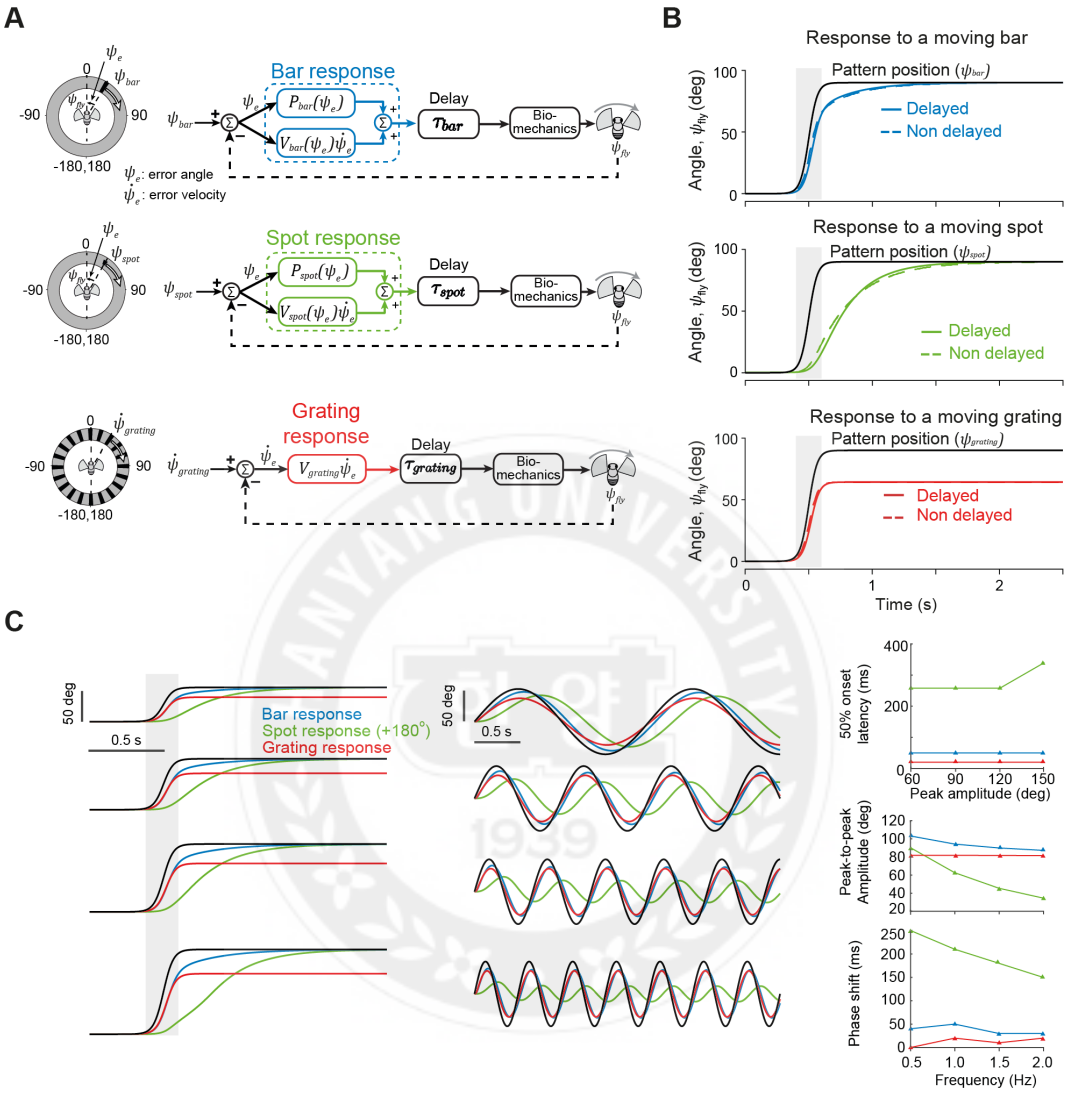


Figure 3.2–figure supplement 1. Visuomotor control models with known delays. (A) Same as in Figure 3.2A, but with an additional delay block (SPECIFY DELAY VALUES HERE). (B) Same as in Figure 3.2C comparing the delayed and non-delayed heading of the virtual fly model for each visual pattern. (C) Same as in Figure 3.2C for different peak amplitudes of the sigmoid-like pattern trajectory and different frequencies of the sine-like trajectory. Plots on the right show the change in the response latency with respect to the stimulus amplitude (top) and the frequency of the sinusoidally shifting patterns (bottom two).

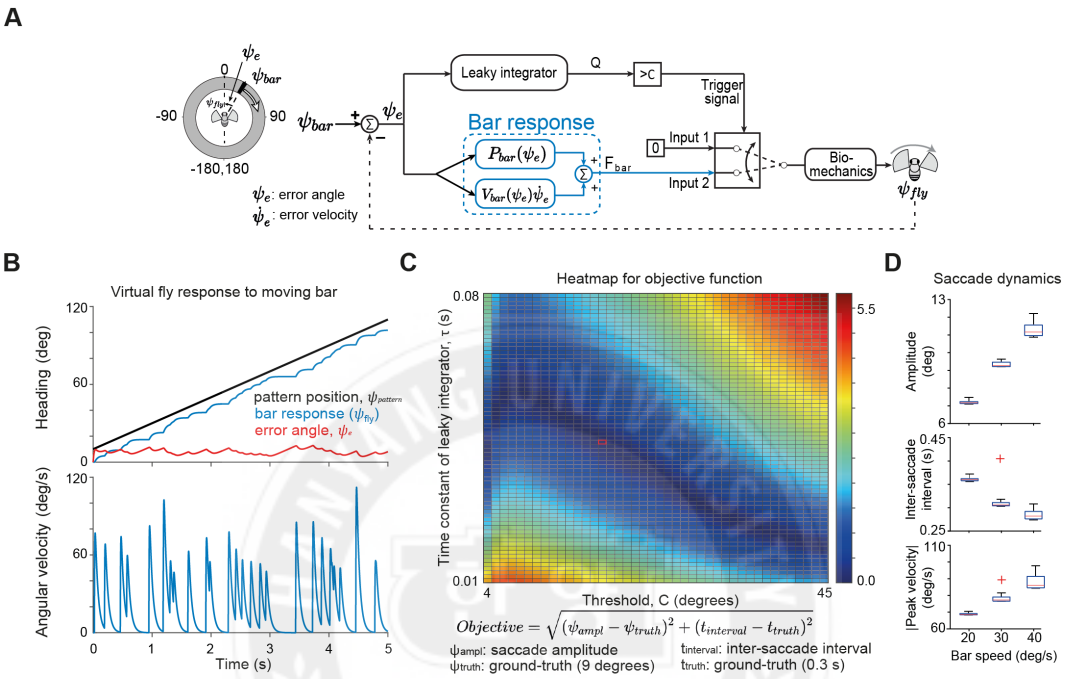


Figure 3.2-–figure supplement 2. The simple bar tracking model expanded with a saccade mechanism. (A) A schematic of the experimental display; the model diagram including a leaky error integrator and a threshold that provides a switch signal to trigger the saccades. (B) Simulation results showing the angular position of the fly in blue, the bar position in black, and the error angle in red; below the angular velocity of the fly. (C) A heatmap showing the objective function values for the pair values of τ for the time constant of the leaky integrator and C for the threshold value for the saccade initiation. The red square shows the minimum of the objective function. (D) Box plots indicating the averaged saccade amplitude, inter-saccade interval, and angular velocity peak. The box represents the interquartile range (IQR), with the median indicated by the horizontal red line. The whiskers extend to the minimum and maximum values within 2 times the IQR. Outliers are denoted by "+" marks beyond the whiskers.

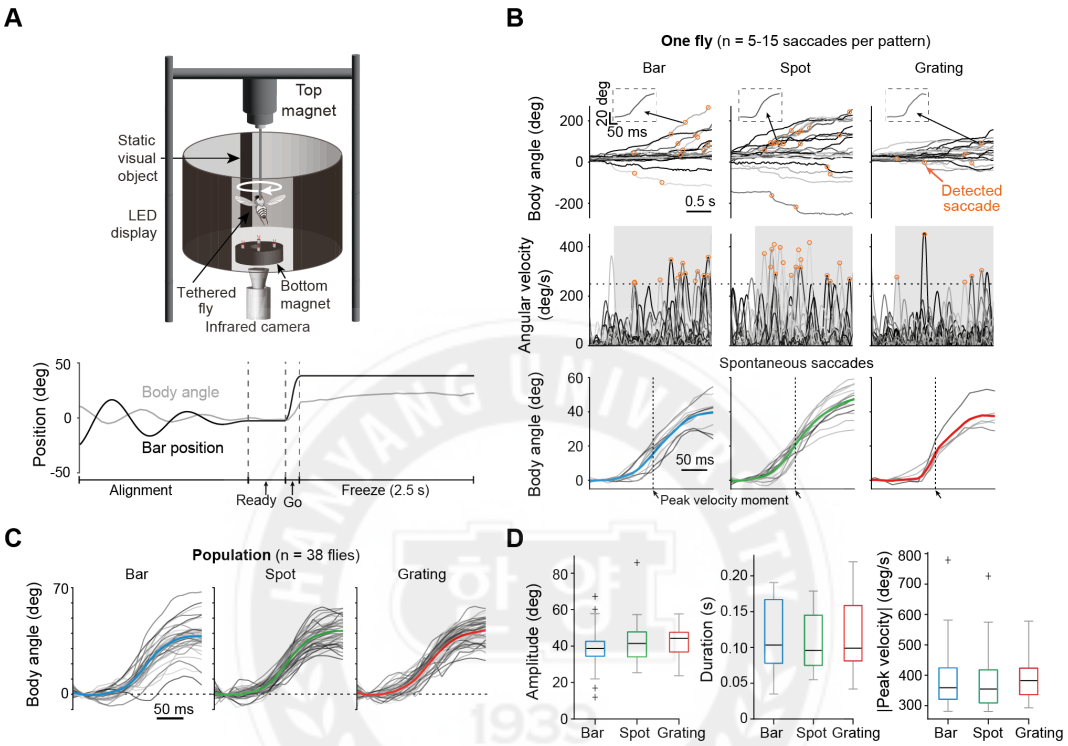


Figure 3.3–figure supplement 1. The flies exhibited spontaneous saccades when the patterns were static. (A) Same as in Figure 3.3 but with a saccade detection period of 2.5 seconds. (B) Single fly body angle traces in the presence of static bar, spot, and grating patterns showing the saccade events marked with a blue circle for bar; Angular velocity corresponding to the top traces. The dashed line indicates the threshold to detect the saccade event; Individual saccades extracted from the body angle traces on top. (C) Spontaneous saccades for a population of flies. (D) Amplitude, duration, and peak velocity boxplots of the spontaneous saccades. The box represents the interquartile range (IQR), with the median indicated by the horizontal black line. The whiskers extend to the minimum and maximum values within 2 times the IQR. Outliers are denoted by "+" marks beyond the whiskers.

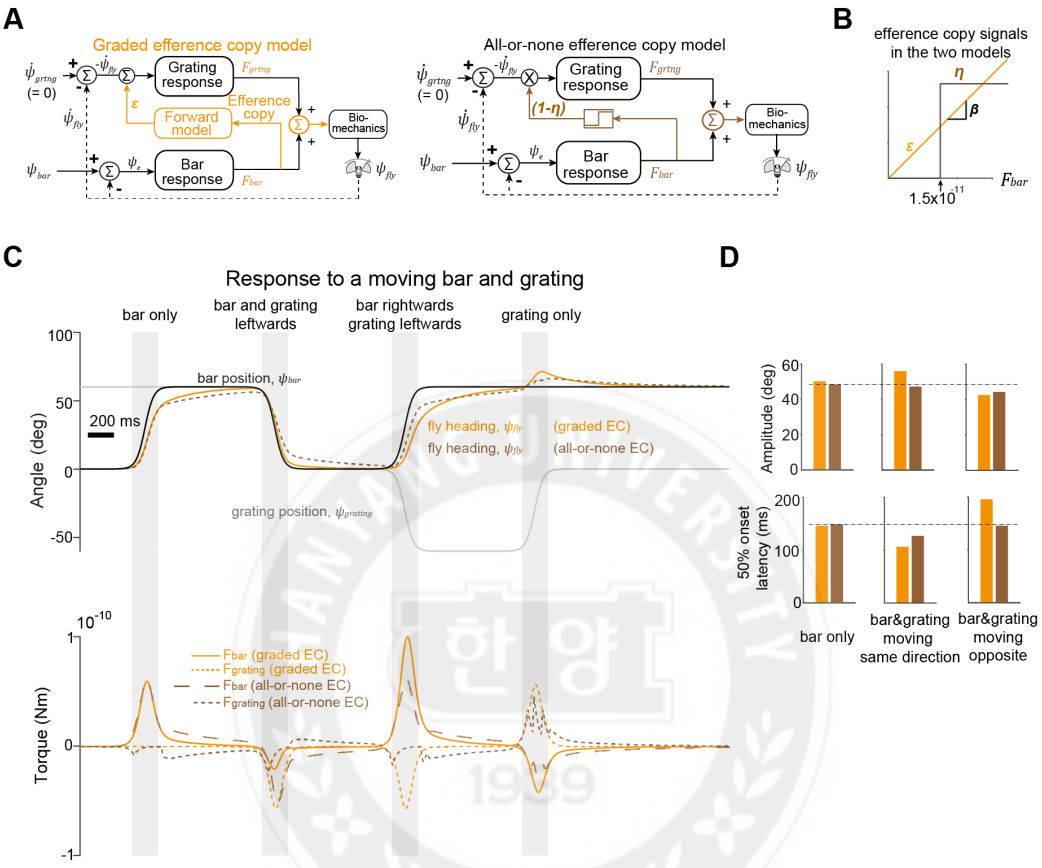


Figure 3.4–figure supplement 1. Orientation behaviors of the two EC-based models for different stimulus conditions. (A) Diagrams of the graded and all-or-none EC models. (B) The change in the EC signals with respect to the bar evoked flight torque. (C) Heading and torque responses to four different stimulus conditions: a moving bar in the first step, a moving bar and grating leftwards in the second step, a moving bar rightwards and grating leftwards in the third step, and a moving grating in the fourth step. (D) Bar plots indicating the amplitude of the response and the latency with respect to the stimulus onset, measured at the 50% point of the pattern movement for the different stimulus combinations. Dotted lines were added to facilitate the comparison across conditions. The all-or-none EC model exhibited less variability than the graded EC model.

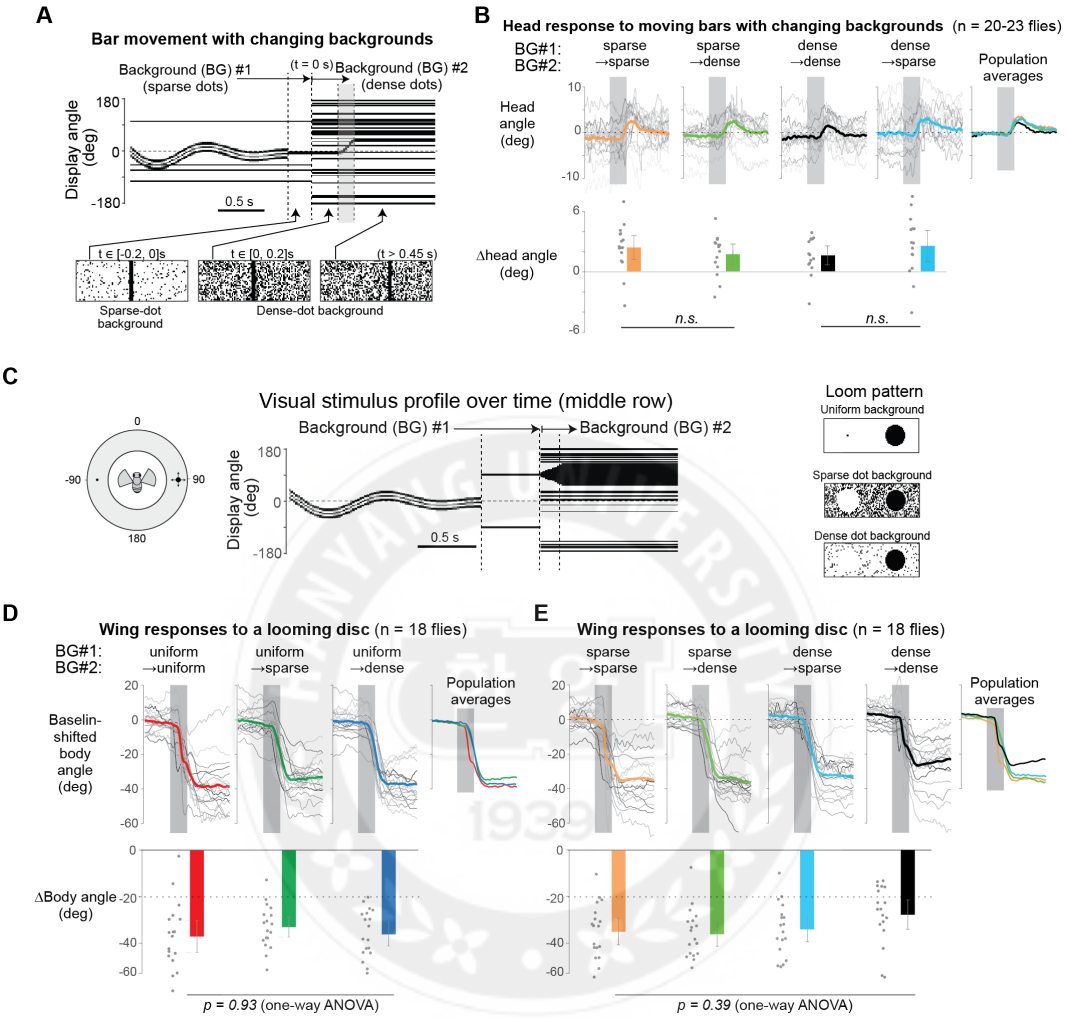


Figure 3.6–figure supplement 1. Unexpected changes of the background pattern did not affect loom-evoked behavioral responses. (A) Temporal change of the visual stimulus profile along the horizontal midline of the display. Body angle responses are shown in Figure 3.6D. (B) Head angle responses to the stimulus profile in A with four different combinations of background patterns. The head angle was calculated from DeepLabCut. (C) On the left, schematic of the experimental display showing the initial position of the loom stimulus. On the right, stimulus profile and background combinations. (D) Body angle responses to the stimulus profile in C, for the different background combinations. (E) Same as in D for different background combinations.

Chapter 4. Conclusion

4.1 Summary

This thesis presents the *Drosophila*'s brain as a baseline to develop algorithms for vision-based navigation in complex visual environments. By mimicking fly's vision, our FlyVisNet achieved higher accuracy performance than other networks of similar size in classification task. The artificial neurons exhibited selectivity to visual patterns reminiscent of those of the fly. FlyVisNet allowed a virtual agent to perform visually-guided flight in a virtual environment. The network allowed the agent to make decisions according to pattern recognitions. Furthermore we successfully deployed the network in a nano drone achieving on-board AI flight. This strategy showed faster cycle time than relying on a communication to a computer where the network was executed. FlyVisNet showed also faster cycle time than MobileNetV2. The drone running FlyVisNet on board performed successfully a visually guided flight. Together, our network provides solutions to power and memory demanding embedded systems and can further help in unveiling additional fly's visual functions.

To coordinate the different neural pathways that activate a shared actuator in complex visual environments, we built an integrative model for visuomotor behaviors. Through quantitative experimental analysis from a tethered fly essay, we derived mathematical models for simple moving visual patterns. Our models captured fly responses in a magnetically-tethered setup. We combined the models for simple patterns through 3 different strategies: addition-only, graded EC, all-or-none EC. The models with EC strategy allowed the virtual fly to respond faster to the moving patterns. We found behavioral evidence

that suggests that the fly uses most likely all-or-none EC. This could be a way to save energy and improve survival with fewer resources. Our model recapitulates fly's visuomotor behavior and EC via all-or-none mechanism. It can be used as a mathematical framework to study fly's visual reflexive behaviors and could be applied for the flight control of artificial agents such as drones.

Together, our algorithms demonstrate how *Drosophila*'s brain can be used as a baseline for engineering solutions. Our models can be used to study further fly's vision and behavior and can be applied to other domains such as robotics, autonomous driving, etc. Our models could be also expanded by incorporating additional neural circuits and pathways that would allow to perform additional functions to the artificial agent, such as color brightness learning, spatial navigation, visual learning, etc.

4.2 A preliminary study. An insect-inspired model-based framework for navigation

RL algorithms have been studied extensively in recent years to solve numerous problems mainly using model-free strategies, which try to find the solution using raw computation in a "trial and error" basis. Differently, a model-based makes use of a pre-acquired latent representation of the environment [31]. It has been found that animals like mice can use this strategy to solve mazes by acquiring an internal representation via head cells [128], place cells [129], and grid cells [130] helping them to find reward.

By having a model, an agent can simulate possible future trajectories and outcomes, enabling more optimized behavior and planning (see Figure 4.1 A), especially in scenarios where direct trial and error exploration (as in

model-free approaches) might be too costly or impractical. Thus, a model-free strategy is learned over many trials so that at each state there is an action with the greatest associated reward (see Figure 4.1 B). Whereas in the model-based strategy, the agent learns an environment model and a reward model. So it can simulate sequences of actions that provide the highest reward (see Figure 4.1 B). These findings inspired RL models that outperformed existing algorithms in navigation tasks [46].

In the case of fruit fly *Drosophila*, spatial-related internal representations take place in the central complex. The fly can perform goal-driven behaviors via visual cues, which allows to remember cooler places in a hot environment for example [28] (see Figure 4.1 C). Specifically, the visual pathway toward ellipsoid body projection gall (EPG) neurons goes from medulla (Me) layer in the optic lobe to the anterior optic tubercle (AOTU), and the bulb (BU), and connects to the ellipsoid body (EB), and the protocerebral bridge (PB) [27] (see Figure 4.1 D). EPG neurons integrate these inputs and convey spatial information to the EB, contributing to the fly's ability to navigate and maintain orientation [29, 30] (see Figure 4.1 D). The EB is organized into a series of concentric rings or layers, and its neurons encode and process visual cues relative to the fly's orientation and movement in space [32, 33] (see Figure 4.1 D).

In this part, we propose a model-based RL framework inspired by the EB, in which a Variational Autoencoder (VAE) is used to encode the frames from a virtual world environment into a two-dimensional ring-shaped representation. The internal variables are input to the policy network, updated by deep deterministic policy gradient (DDPG), that provides steering and velocity to an artificial agent. Our model allowed the agent to find a visual target converging

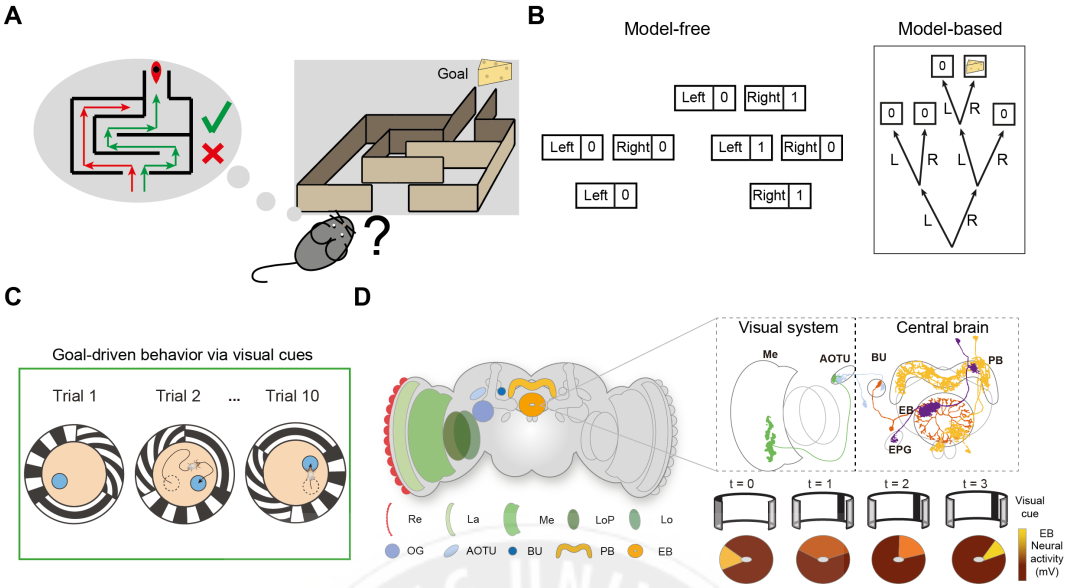


Figure 4.1: Spatial internal representations allow animals to navigate in different environments. (A) A mouse predicting possible outcomes from an internal maze simulation. (B) The model-free and model-based RL strategies for the maze in A. (C) Fruit fly learns to navigate towards a target via visual cues. (D) *Drosophila*'s neural structures of the visual system and the central brain Re: retina, La: lamina, Me: medulla, LoP: lobula plate, Lo: lobula, OG: optic glomeruli, AOTU: anterior optic tubercle, BU: bulb, PB: protocerebral bridge, EB: ellipsoid body. The neurons involved in visual goal-driven behavior and the connection between the visual system and the central brain. How EB encodes a visual cue.

to greater reward values than a standard VAE and a model-free DDPG-only counterparts in 2 different environments. Our algorithm highlights the importance of brain-inspired internal representations in navigation tasks and could be extended to other domains such as robotics, drones, autonomous driving, etc.

4.2.1 Model-based framework

In the proposed Model-Based RL algorithm the VAE encodes the state observed from the virtual environment, 64x64 frames, in a bivariate modified

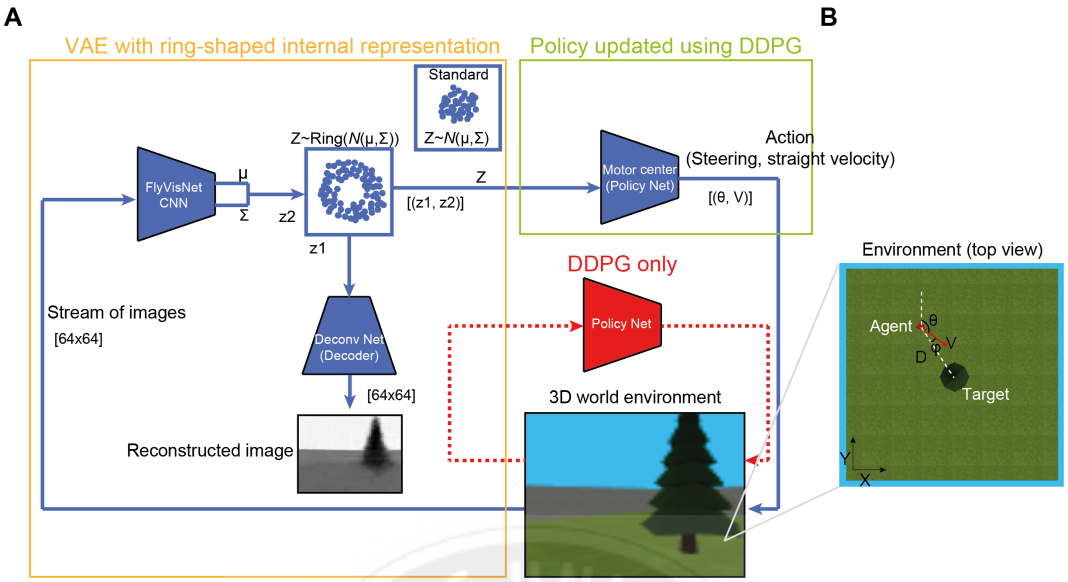


Figure 4.2: The model-based framework for navigation in a virtual environment. (A) The framework is made up of a VAE with a ring-shaped internal representation (z1,z2) and a policy network updated by using DDPG that provides steering and velocity (θ, V). A standard latent normal distribution is also shown. A DDPG-only strategy is also shown in red color. (B) The top view of the environment showing the Euclidean distance D between agent and target, the steering angle θ, the velocity vector V, and the error angle between agent and target ϕ.

normal distribution (see Algorithm 3) that provides ring-shaped latent variables (see Figure 4.2 A). The latent variables represent the relative position of the agent with respect to the target. From these variables, the decoder reconstructs the frame. Finally, this pair is fed to the policy network that provides the steering angle, and the velocity to the agent, updated by the DDPG (see Figure 4.2 A).

4.2.2 Training

The goal of the agent is to navigate to arrive at the target (a pine tree) (see Figure 4.2 B) so the received reward is higher when the agent is near the

target, and heading towards it. If the agent collides with the walls of the maze it receives a reward penalty, and if it arrives at the target receives a reward bonus (see Equation 4.1).

$$r = -0.1D - 0.1\phi^2 + n \tag{4.1}$$

where D is the Euclidean distance between the agent and the tree, ϕ is the error angle between the agent's velocity vector and the target (see Figure 4.2 B), and n is defined as follows:

$$n = \begin{cases} 100, & \text{if agent arrives at target} \\ -1, & \text{if agent collides with walls} \\ 0, & \text{otherwise} \end{cases}$$

To build and train the VAE and the policy network we used the Python library PyTorch. First, we trained the VAE isolated for 200 episodes on a supervised learning basis by using the images captured from the agent placed in the maze at random positions (see Algorithm 3). Then, we applied DDPG to train the policy network for 500 episodes with an experience buffer (see Algorithm 3).

Algorithm 3 Model-based RL framework

Initialize VAE encoder $EN_{\theta 1}$, decoder $D_{\theta 2}$, and policy networks π_ϕ with random parameters θ_1, θ_2, ϕ
Initialize experience buffer B
for $episode = 1$ **do**
  Initialize environment state
  for $t = 1, T$ **do**
    Place agent at a random position and receive image observation state s_t
    Store observation (s_t) in buffer B
    Sample random minibatch of length 40, (s_i), from B.
    Encode states from minibatch $\mu_i, \Sigma_i = EN(s_i|\theta_1)$ with $Z_i \sim N(\mu_i, \Sigma_i)$
    Decode latent variables into reconstructed images $\hat{s}_i = D(Z_i|\theta_2)$
    Compute prior mean as $\mu_{ring} = \frac{Z_i}{6} + 6\frac{XY_{irel}}{||XY_{irel}||}$ where XY_{irel} are the relative coordinates (X_i, Y_i) between the agent and the target
    Update $EN_{\theta 1}$, and $D_{\theta 2}$ via
    $L = \frac{1}{40}\sum_i^{40}(s_i - \hat{s}_i)^2 + D_{KL}(N(\mu, \Sigma)||N(\mu_{ring}, I))$
  end for
end for
for $episode = 1$ **do**
  Initialize environment state
  Initialize a random process N for action exploration
  $t \leftarrow 1$
  while $t \leq 200$ & $termination = False$ **do**
    Receive image observation state s_t
    Encode state $\mu, \Sigma = EN(s_t|\theta_1)$ with $Z \sim N(\mu, \Sigma)$
    Feed policy $a_t = \pi(Z|\phi) + N_t$
    Apply action a_t to the environment to obtain reward r_t, and next state s_{t+1}
    Store transition (s_t, a_t, r_t, s_{t+1}) in buffer B
    Sample random minibatch of length 40, (s_i, a_i, r_i, s_{i+1}), from B
    Update the policy π_ϕ with the minibatch by using DDPG
    if agent arrived at target **then**
      $termination \leftarrow True$
    end if
    $t \leftarrow t + 1$
  end while
end for

4.2.3 Results

To navigate, the agent needs to generate action from input images from the virtual environment. The straightaway approach is to apply a CNN as a policy network, that provides the steering angle and the velocity to the agent. We first used FlyVisNet as CNN policy network along with DDPG algorithm to update its weights (see Figure 4.2 A, red colored block) by following our reward function (see Equation 4.1). We trained the agent in the simple maze for 10 trials and did not obtain a significant reward increase over training episodes (see Figure 4.3 A, red trace). This is most likely due to the high dimensionality of the input. In fact, in RL one of the most challenging approaches is learning from raw pixels. Thus, policy networks applying DDPG perform better when the input is low dimensional rather than with high dimensional observation spaces such as images for most of the tasks [131]. Dimensionality reduction of the inputs can be done via autoencoders (AEs) which are a type of ANNs used for unsupervised learning tasks, particularly in dimensionality reduction, data denoising, and feature learning. They consist of an encoder and a decoder, which allow to compress input data into a lower-dimensional representation and then reconstruct the original input from this representation. AEs have been successfully applied in different RL tasks such as navigation, robotics, etc. [34, 132, 35, 36]. We used a VAE, which allows regularizing the latent variables by setting a prior distribution as a target. We set as target prior a standard normal distribution $N(0, I)$ (see Figure 4.2 A, with the standard normal distribution in the latent space) and trained the agent again in the simple maze. This time, using the VAE-standard, it achieved a higher reward over episodes (see Figure 4.3 A, green trace), showing the importance of dimension-

ality reduction from 64x64 frames to 2-dimensional latent space so that DDPG achieves a good policy. We believed that by encoding the frames to a EB-like representation would improve further the model. To do so, we set as target prior a normal distribution with a custom mean (see Algorithm 3) that would encode the frames as a relative position between the target and the agent mimicking the ring-shaped encoding representation of EB. Indeed, we found that this strategy, the VAE-ring, improved significantly the increase of reward over episodes compared to the previous approaches (see Figure 4.3 A, blue trace). Once trained, we placed the agent at random positions in the simple maze (see Figure 4.3 B, top view and first person view) and for 10 trials the agent was able to arrive at the tree (see Figure 4.3 B, agent position). Interestingly, the VAE-standard captured the salient features of the images by encoding the positions of the walls of the maze and the tree relative to the agent in the Zx variable (see Figure 4.3 B, internal representation). Thus, when the walls are left-handed respect to the agent Zx takes negative values and the other way around for the right-handed walls. If the walls are mostly in front of the agent it will be encoded with Zx values near 0. The Zy encoded the distance to the walls, being positive when the walls were far away, and becoming negative when the agent was close to them (see Figure 4.3 B, internal representation). The VAE-ring captured the spatial relative position of the agent with respect to the target given the target prior distribution we were using (see Figure 4.3 B, internal representation). For instance, when the tree was southeast from the agent, the Zx values were mostly positive and the Zy values were mostly negative (see Figure 4.3 B, internal representation).

We repeated the same process for the T-shaped maze case. Similarly, the VAE-ring approach achieved greater reward over episodes than the other

strategies (see Figure 4.3 C). We simulated again 10 trials with the agent starting at random positions on the main corridor (see Figure 4.3 D, top view and first person view). The agent arrived successfully at the tree every single trial (see Figure 4.3 D, agent position). As before, the VAE-standard captured the walls and tree relative position to the agent as internal variables (see Figure 4.3 D, internal representation). The VAE-ring captured again the relative position between the agent and the tree (see Figure 4.3 D, internal representation).

4.2.4 Discussion

We have found that by applying a model-free strategy, a virtual agent could not achieve an optimal policy in a visual navigation task (see Figure 4.3 A, and C, red trace). Similarly to biological systems, when we included a model-based approach via a VAE that previously mapped the whole maze environment to an internal representation (see Algorithm 3), the agent was able to increase the obtained reward and arrive at the target location (see Figure 4.3, green trace). Rather than more complex neural structures like the hippocampus in mice, the *Drosophila*'s navigation-related areas, which are functionally known and even detailed at cell level through the connectomes [1, 2] (see Figure 4.1 D), provide an opportunity to develop models in silico. By using an EB-like internal representation in the VAE where we modified the prior target distribution, the artificial agent outperformed the VAE-standard (see Figure 4.3, blue trace). We believe that encoding the images from the environment into a meaningful spatial representation such as the case of the fly, helped the agent to enhance the navigation capability to find the target. These findings showed the potential of brain-inspired computations in line with previously reported models in which grid and place cell representations

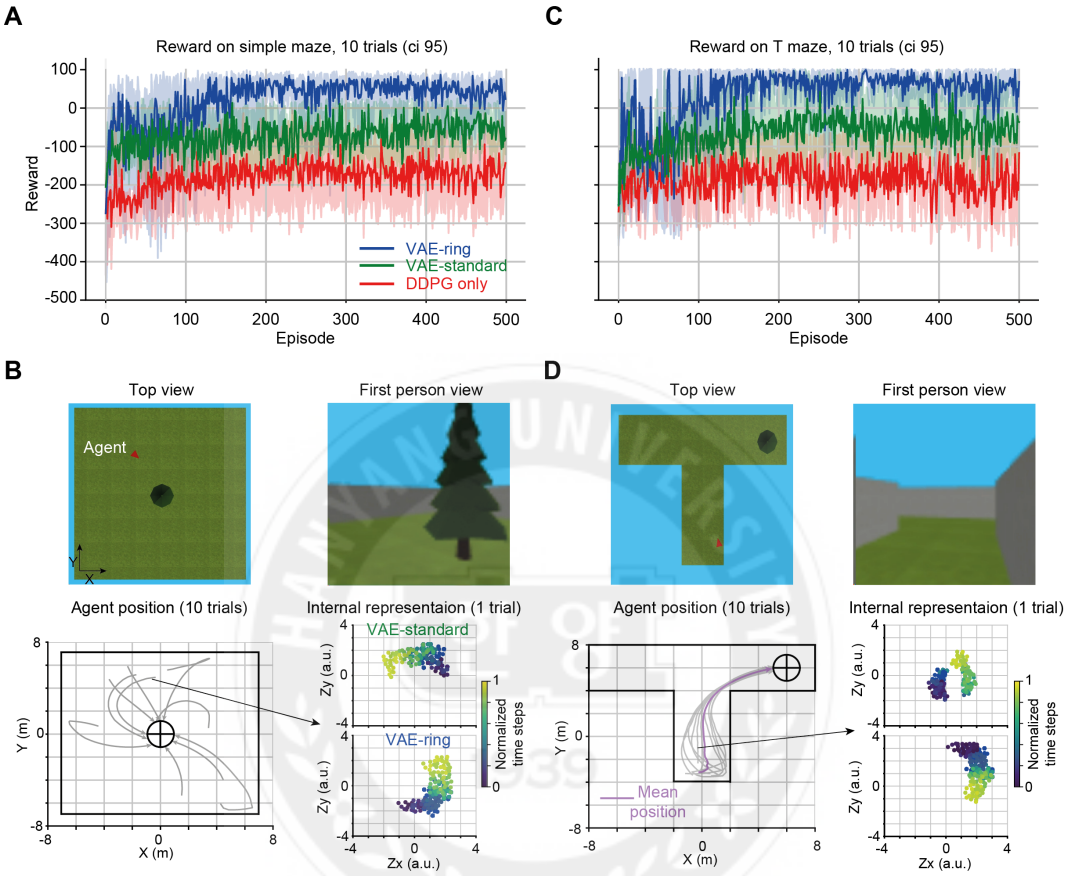


Figure 4.3: The model-based framework with ring-shaped internal representation obtained higher reward and allowed the agent to arrive at the target. (A) The obtained reward on the simple maze for the framework with the VAE with ring representation, VAE with standard representation, and with DDPG only. (B) Top and first person view of the simple maze, where the pine tree is the target. Below, the 2-dimensional position of the agent for 10 trials starting at random positions. On the right, the internal variables for one of the trials for the VAE with standard and ring-shaped representations. The colormap indicates the corresponding time step of the simulation. (C) Same as in A for the T maze. (D) Same as in B for the T maze. The magenta colored line represents the mean position of the agent over 10 trials.

improved the performance in the navigation of artificial agents [46].

Another advantage of the model-based approach is that in the case of changes in the maze, the internal map can be updated via planning. If that is the case our VAE-ring would just need a few iterations to be updated by placing the agent at random positions again in the new maze. The policy network would not need to be updated since the VAE would have already encoded the relative position between the agent and the target.

The model could be expanded with the possibility of simulating possible outcomes beforehand through a dreaming mechanism, as it has been reported [133]. This would allow reducing the number of training episodes to maximize the reward by sampling latent values and taking the value that would provide the greater reward without picking actions that would otherwise slow down the learning process. Our model could have utility not only in navigation but also in other domains such as robotics, autonomous driving, etc.

4.3 Challenges and future prospects

In future work, FlyVisNet could steadily incorporate additional neural pathways. This would be eventually possible with the continuous expansion of the connectome. Developing a whole-brain model would map sensory input to behavior through an end-to-end process and would allow agents to adapt to their environments and make intelligent decisions. In the case of vision, the model would capture visual features from the visual inputs, in the central brain decisions would be made, and at last, it would be possible to activate the right motor neurons to control actuators. The ventral cord connectome is also available so accurate motor control would be feasible. A various set

of behaviors would be executed enabling a versatile model to be applied in robotics, drones, etc.

Another challenge is the multimodal integration. A whole brain model would capture inputs from different modalities such as vision, olfaction, proprioception, etc. By integrating more than one sensory the decision would be more reliable so given the case one sensory could take priority like the somatosensory for temperature or touch to avoid damage. This could improve further the control of artificial agents in safety conditions.

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국문요지

초파리의 뇌는 최근 몇 년 동안 단일 뉴런 수준에서 구조와 기능 모두에 대해 광범위하게 연구되었다. 이 모델 동물은 정교한 시각적 행동과 함께 가상의 에이전트를 위한 수학적 프레임워크를 개발하는데 이상적인 플랫폼으로 활용할 수 있으며, 이때 가상의 에이전트는 초파리의 시각을 기반으로 한 탐색을 수행한다. 이 연구에서는 초파리의 시각적 처리를 기반으로 한 수학적 모델을 개발하여 인공 에이전트가 주변 환경에서 복잡한 시각적 특징을 감지하고 그에 따라 탐색할 수 있도록 하였다.

우리는 초파리의 시각 시스템 아키텍처를 기반으로 하는 뉴로모픽 컨볼루션 신경망(CNN)인 FlyVisNet을 개발하였다. FlyVisNet은 패턴 인식을 가능하게 하는 신경 경로를 구현하여 다양한 모양과 흐릿하게 보이는 물체를 감지할 수 있습니다. 우리는 FlyVisNet이 다양한 복잡성을 가진 물체의 분류 정확도에서 동일한 크기를 가진 세그먼트의 다른 아키텍처보다 성능이 뛰어나다는 것을 발견하였다. 나노 드론과 같이 가장 적은 메모리와 전력을 요구하는 임베디드 시스템에 배포할 수 있다. FlyVisNet은 가상 에이전트와 실제 에이전트가 시각적으로 유도된 비행 및 충돌 회피를 위한 결정을 내릴 수 있도록 하였다.

지능적인 결정을 내리기 위해서는 초파리의 날개 근육과 같은 모터 액츄에이터를 공유할 때 서로 다른 신경 경로를 조정해야 한다. 시각 반사 행동에서 초파리는 시각적 흐름을 억제하기 위해 efference copy (EC) 신호를 사용하는 것으로 밝혀졌다. 이 메커니즘을 통해 초파리는 비행을 안정화하고 특정 시각적 입력에 우선 순위를 매길 수 있다. 우리는 EC 메

커니즘을 통해 여러 시각적 패턴이 있는 복잡한 시각적 환경에서 초파리의 결정을 조정할 수 있는 반사적 시각 행동에 대한 통합 모델을 제공한다. EC를 사용할 때 가상 초파리가 시각적 패턴에 더 빠르게 반응한다는 것을 발견하였다. 또한 초파리가 어수선한 복잡한 시각적 환경에서 탐색할 때 EC의 행동에 대한 증거를 발견하였다.

목표 지향적인 행동에서 초파리는 시각적 단서를 통해 특정 장소를 찾을 수 있다. 예비 연구를 통해 autoencoder (AE)가 시각적 입력을 초파리의 Central complex (CX)와 같은 잠재된 표현으로 매핑하는 모델 기반 강화 학습(RL) 프레임워크를 구축하였다. FlyVisNet을 인코더로 사용함으로써 에이전트는 가상 환경에서 목표를 찾기 위해 탐색할 수 있다.

요약하자면, 초파리의 뇌가 시각 기반 내비게이션 알고리즘을 개발하기 위한 기준선으로 사용될 수 있음을 입증하였다. 우리의 알고리즘은 비슷한 크기의 다른 아키텍처보다 성능이 뛰어났고 인공 가상의 에이전트가 복잡한 환경에서 시각적으로 유도된 비행과 의사 결정을 수행할 수 있도록 하였다. 우리의 결과는 다른 영역에 적용할 수 있는 잠재적인 가능성과 함께 생물학적 대응물에 대한 더 깊은 이해를 위해 뇌에서 영감을 얻은 계산의 중요성을 강조한다.

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Declaration of Ethical Conduct in Research

I, as a graduate student of Hanyang University, hereby declare that I have abided by the following Code of Research Ethics while writing this dissertation thesis, during my degree program.

"First, I have strived to be honest in my conduct, to produce valid and reliable research conforming with the guidance of my thesis supervisor, and I affirm that my thesis contains honest, fair and reasonable conclusions based on my own careful research under the guidance of my thesis supervisor.

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JUNE 07, 2024

Degree : Doctor

Department : DEPARTMENT OF ELECTRONIC ENGINEERING

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연구 윤리 서약서

본인은 한양대학교 대학원생으로서 이 학위논문 작성 과정에서 다음과 같이 연구 윤리의 기본 원칙을 준수하였음을 서약합니다.

첫째, 지도교수의 지도를 받아 정직하고 엄정한 연구를 수행하여 학위논문을 작성한다.

둘째, 논문 작성시 위조, 변조, 표절 등 학문적 진실성을 훼손하는 어떤 연구 부정행위도 하지 않는다.

셋째, 논문 작성시 논문유사도 검증시스템 "카피킬러"등을 거쳐야 한다.

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